

22050 Continuous-Time Signals and Linear Systems

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L06

Impulse trains

Sampling

Applications of Fourier transformation

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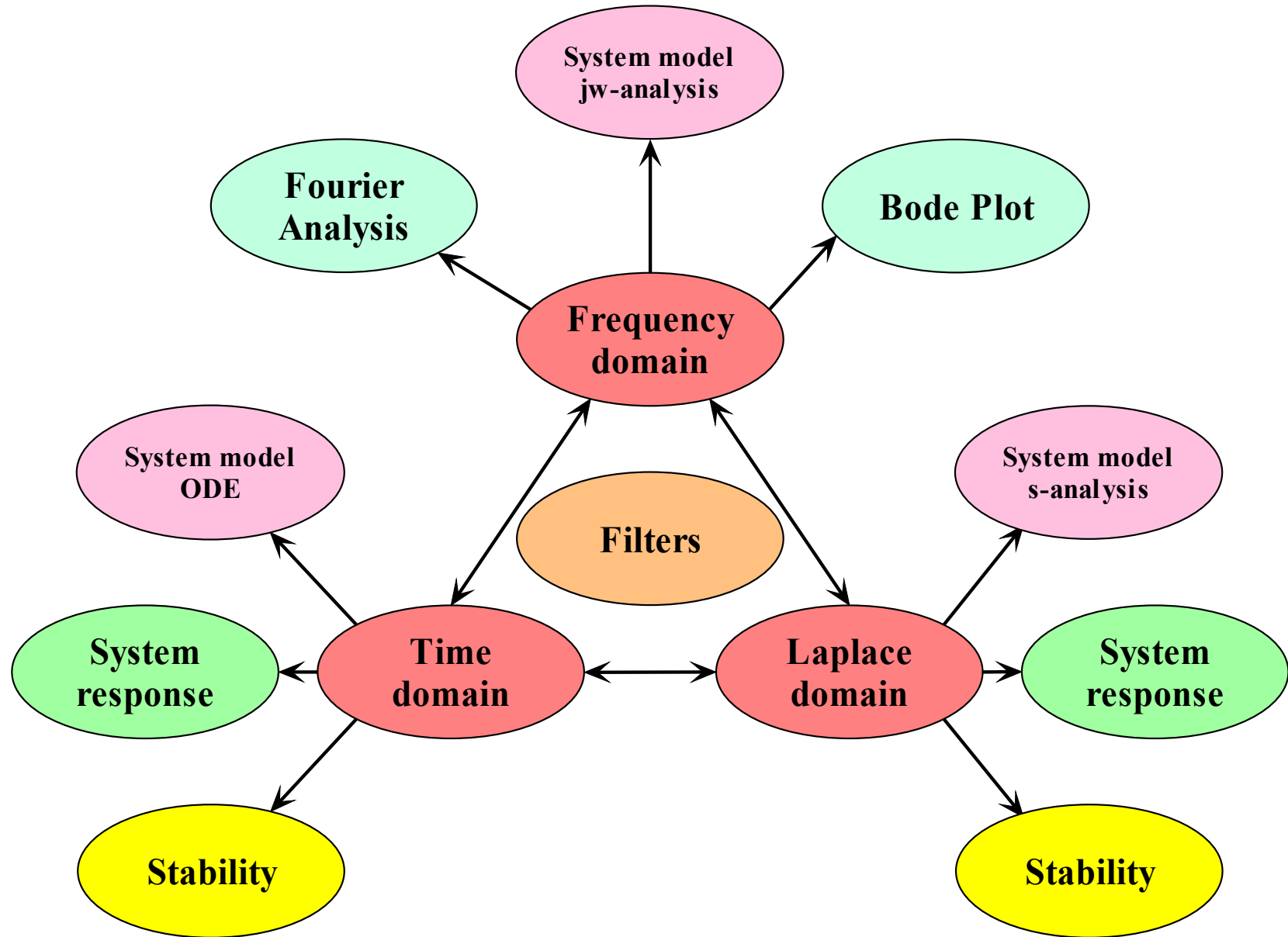
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- System
 - Model
 - Response
 - Performance
- Domains
 - Time
 - Frequency
 - s-domain



- Review of Fourier transformation
- Fourier transform of the impulse train
- Convolution with an impulse train
- Multiplication with an impulse train (sampling)
- Ideal and practical sampling
- Ideal and practical reconstruction from samples
- Amplitude and phase spectra
- Fourier analysis of filter circuits
 - First order lowpass filter
 - Nodal equations in frequency domain
 - Asymptote analysis
 - Frequency scaling
 - Impedance scaling
 - AC sweep
 - First order highpass
 - Nodal equations in frequency domain
 - Asymptote analysis
 - Frequency scaling
 - Impedance scaling
 - AC sweep

With this set of transforms we can go back and forth between time domain ($x(t)$) and frequency domain ($X(\omega)$).

We say that $X(\omega)$ is the Fourier transform of $x(t)$, and that $x(t)$ is the inverse Fourier transform of $X(\omega)$.

As $X(\omega)$ is a continuous density distribution of sinusoidal basis functions, $X(\omega)$ is also called a **frequency spectrum**.

The spectrum is a complex valued function of frequency with modulus and angle:

Forward Transform:	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$	Analysis
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Inverse Transform:	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$	Synthesis
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$X(\omega) = X(\omega) e^{j\angle X(\omega)}$	: Frequency spectrum
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$ X(\omega) $	: Amplitude spectrum
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$\angle X(\omega)$	: Phase spectrum
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Time scaling:

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

Time shift:

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$$

Frequency shift:

$$x(t)e^{j\omega_0 t} \leftrightarrow X(\omega - \omega_0)$$

Time domain convolution:

$$x(t) * h(t) \leftrightarrow H(\omega) X(\omega)$$

Frequency domain convolution:

$$\frac{1}{2\pi} X(\omega) * H(\omega) \leftrightarrow h(t)x(t)$$

Time domain integration:

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \pi X(0)\delta(\omega) + \frac{X(\omega)}{j\omega}$$

Time domain differentiation:

$$\frac{dx}{dt} \leftrightarrow j\omega X(\omega)$$

Review: Temporal Differentiation

Application of the transform pair

$$\frac{d}{dt} \leftrightarrow j\omega$$

$$\ddot{y} + a_1\dot{y} + a_0y = b_2\ddot{x} + b_1\dot{x} + b_0x$$

$$[(j\omega)^2 + a_1 \cdot (j\omega) + a_0]Y(\omega) = [b_2 \cdot (j\omega)^2 + b_1 \cdot (j\omega) + b_0]X(\omega)$$

System frequency
characteristic:

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{b_2 \cdot (j\omega)^2 + b_1 \cdot (j\omega) + b_0}{(j\omega)^2 + a_1 \cdot (j\omega) + a_0} = \frac{P(j\omega)}{Q(j\omega)}$$

We now have the means to Fourier transform a differential equation in order to obtain the frequency characteristic of the system.

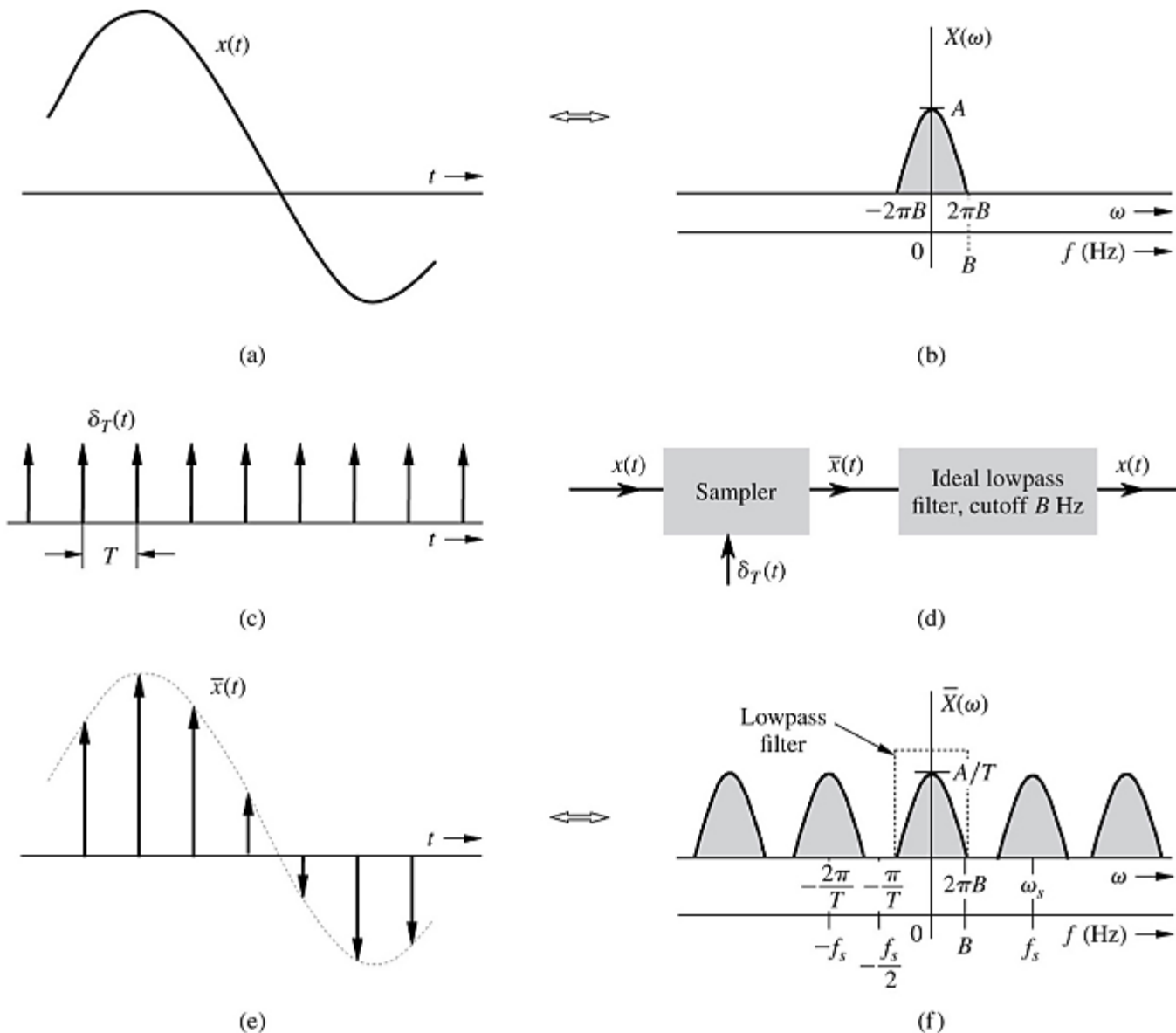
Conversely, if we have the frequency characteristic, we can translate it into the differential equation for the system. In many cases it might be faster to derive $H(\omega)$ from an electric circuit diagram first and then obtain the differential equation from $H(\omega)$.

In cases where we are only interested in $H(\omega)$, we do not need to derive the differential equation.

Fig. 5.1 in the textbook shows that if we multiply a time signal $x(t)$ by a train of impulse functions $\delta_T(t)$, a new signal is obtained $\bar{x}(t)$ which is zero everywhere except for impulses located at discrete times nT . The strength of each impulse equals the amplitude of $x(nT)$.

We will call $\bar{x}(t)$ a **sampled version** of $x(t)$.

The amplitude spectrum of the original signal is shown at the top on the right-hand side. The amplitude spectrum of $\bar{x}(t)$ is shown at the bottom on the right-hand side.



Fourier Transformation of Periodic Signals

Fourier Series
synthesis of periodic
signal:

Fourier
Transformation:

$$x(t) = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

$$X(\omega) = \mathcal{F}\{x(t)\} = \mathcal{F}\left\{\sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}\right\} = \sum_{n=-\infty}^{\infty} \mathcal{F}\{X_n e^{jn\omega_0 t}\} = \sum_{n=-\infty}^{\infty} 2\pi X_n \delta(\omega - n\omega_0)$$

Frequency shift

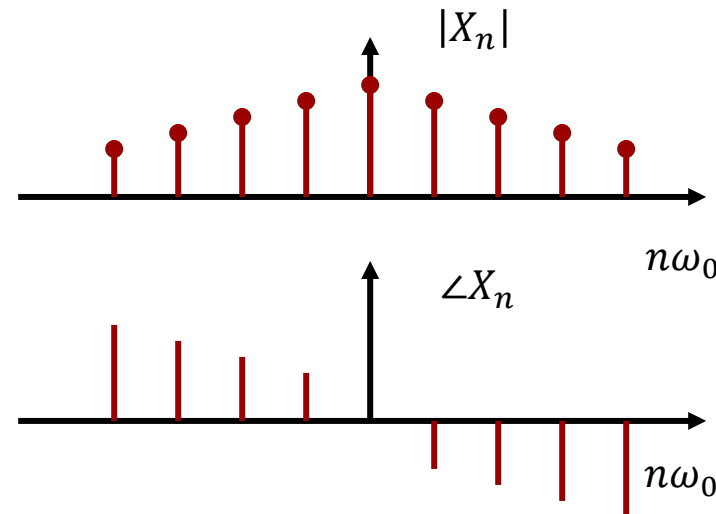
$$A \leftrightarrow A2\pi\delta(\omega)$$

$$X_n \leftrightarrow X_n 2\pi\delta(\omega)$$

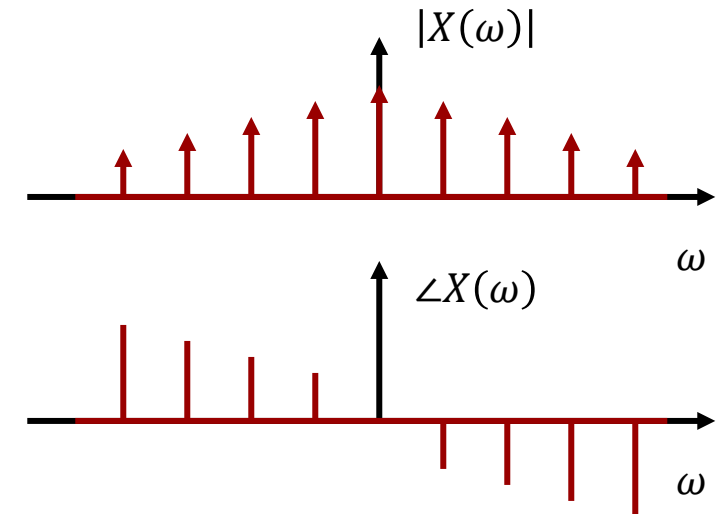
$$X_n e^{jn\omega_0 t} \leftrightarrow 2\pi X_n \delta(\omega - n\omega_0)$$

Discrete and continuous
functions of frequency:
The former is not
defined between stems.
The latter is defined
everywhere.

Discrete $n\omega_0$



Continuous



Fourier Series synthesis of periodic signal:

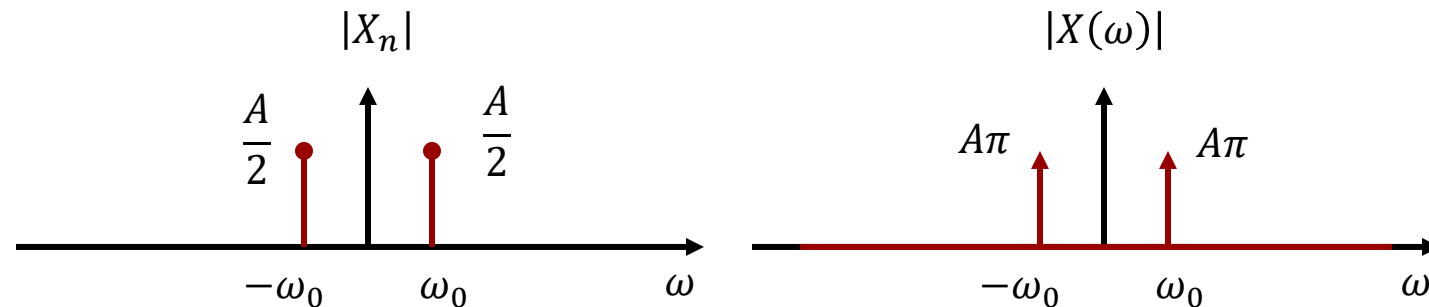
$$A \cos(\omega_0 t) = A \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right] = \sum_{n=-\infty}^{\infty} X_n e^{jn\omega_0 t}$$

$$X_n = \begin{cases} A/2 & n = \pm 1 \\ 0 & n \neq \pm 1 \end{cases}$$

Fourier Transformation:

$$X(\omega) = \sum_{n=-\infty}^{\infty} 2\pi X_n \delta(\omega - n\omega_0) = A\pi \delta(\omega + \omega_0) + A\pi \delta(\omega - \omega_0)$$

Discrete and continuous functions of frequency:

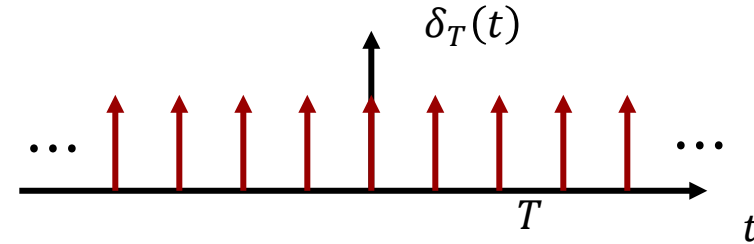


Why is there no phase?

Cos(x) is real and even

Impulse train

An impulse train is a periodic repetition of impulses, hence a periodic signal.



We can expand it into a complex exponential Fourier series:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \qquad \omega_s = \frac{2\pi}{T}$$

Fourier Series coefficients:

$$D_n = \frac{1}{T} \int_{-T/2}^{T/2} \delta_T(t) e^{-jn\omega_s t} dt = \frac{1}{T} \int_{-T/2}^{T/2} \delta_T(t) e^{-jn\omega_s 0} dt = \frac{1}{T}$$

Fourier Series expansion:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_s t} = \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn\omega_s t}$$

Fourier Series expansion:

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn\omega_s t} \quad \omega_s = \frac{2\pi}{T}$$

Fourier Transformation:

$$\Delta_T(\omega) = \int_{-\infty}^{\infty} \delta_T(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{T} e^{jn\omega_s t} e^{-j\omega t} dt = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{1}{T} e^{jn\omega_s t} \right) e^{-j\omega t} dt$$

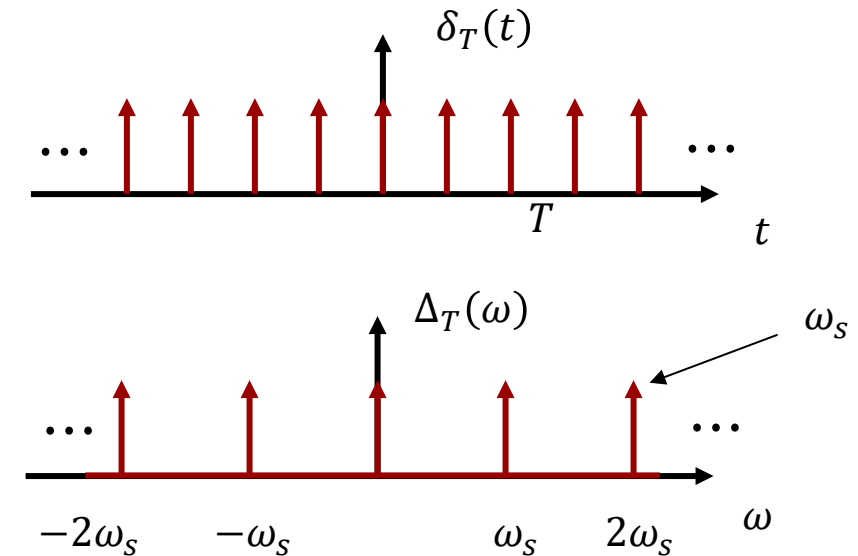
$$= \sum_{n=-\infty}^{\infty} \frac{2\pi}{T} \delta(\omega - n\omega_s) = \sum_{n=-\infty}^{\infty} \omega_s \delta(\omega - n\omega_s)$$

$$A \leftrightarrow 2\pi A \delta(\omega)$$

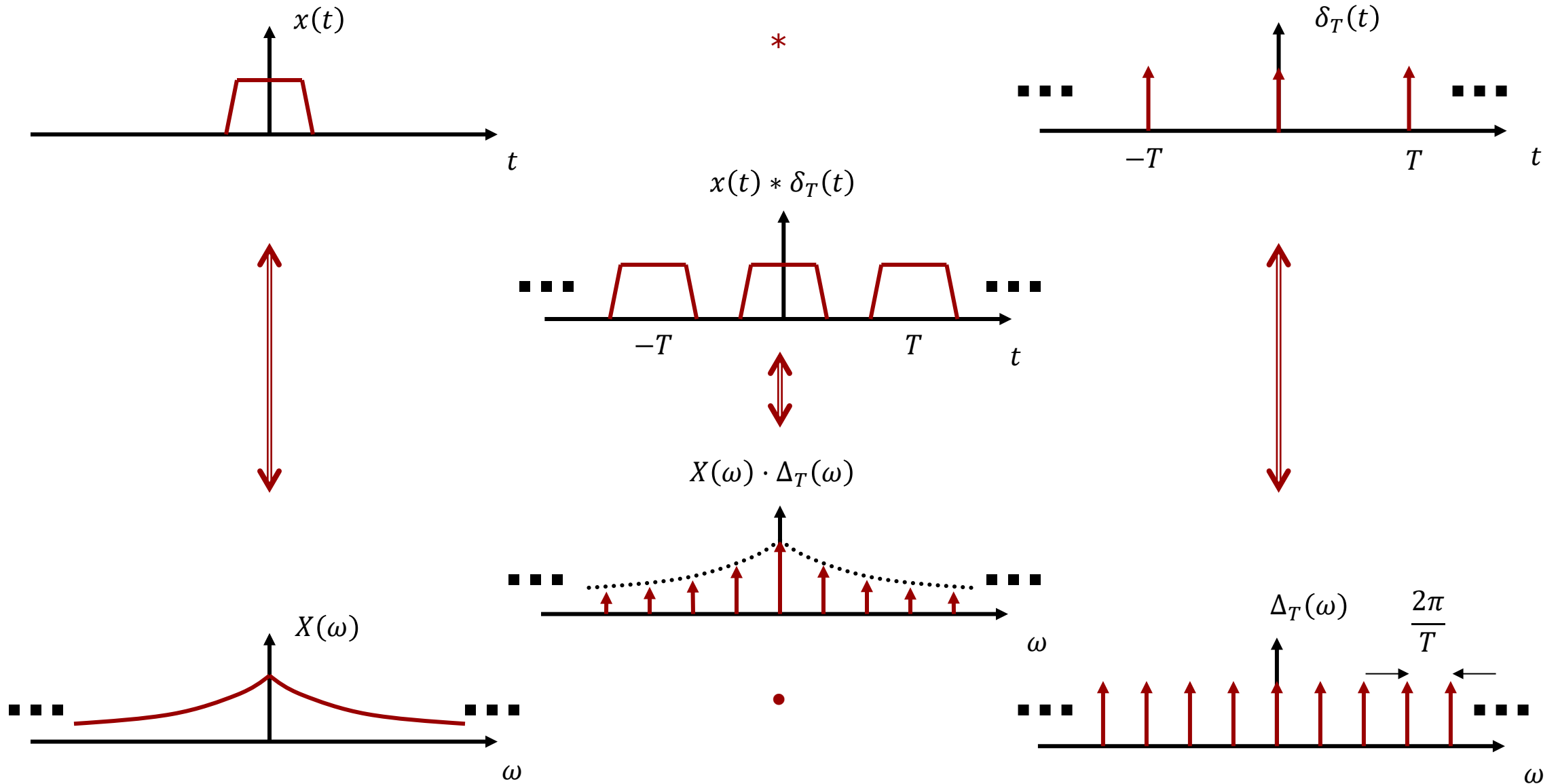
$$A e^{j\omega_0 t} \leftrightarrow 2\pi A \delta(\omega - \omega_0)$$

$$\frac{1}{T} e^{jn\omega_0 t} \leftrightarrow \frac{2\pi}{T} \delta(\omega - n\omega_0)$$

The spectrum of an impulse train is periodic in ω with period ω_s :



Impulse Train Convolution with Aperiodic Signal



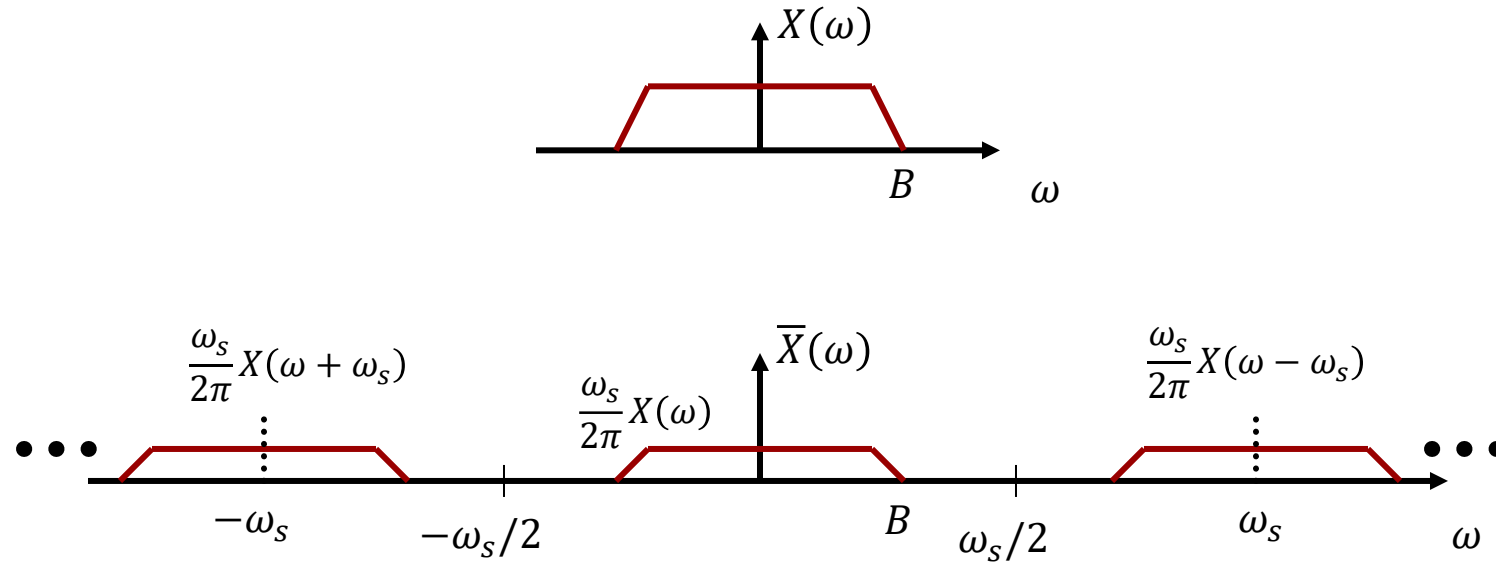
Aperiodic signal multiplied by impulse train

Multiplication in
time:

$$\bar{x}(t) = x(t) \cdot \delta_T(t)$$

Convolution in
frequency
domain:

$$\bar{X}(\omega) = \frac{1}{2\pi} X(\omega) * \Delta_T(\omega) = \frac{1}{2\pi} X(\omega) * \sum_{n=-\infty}^{\infty} \omega_s \delta(\omega - n\omega_s) = \frac{\omega_s}{2\pi} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

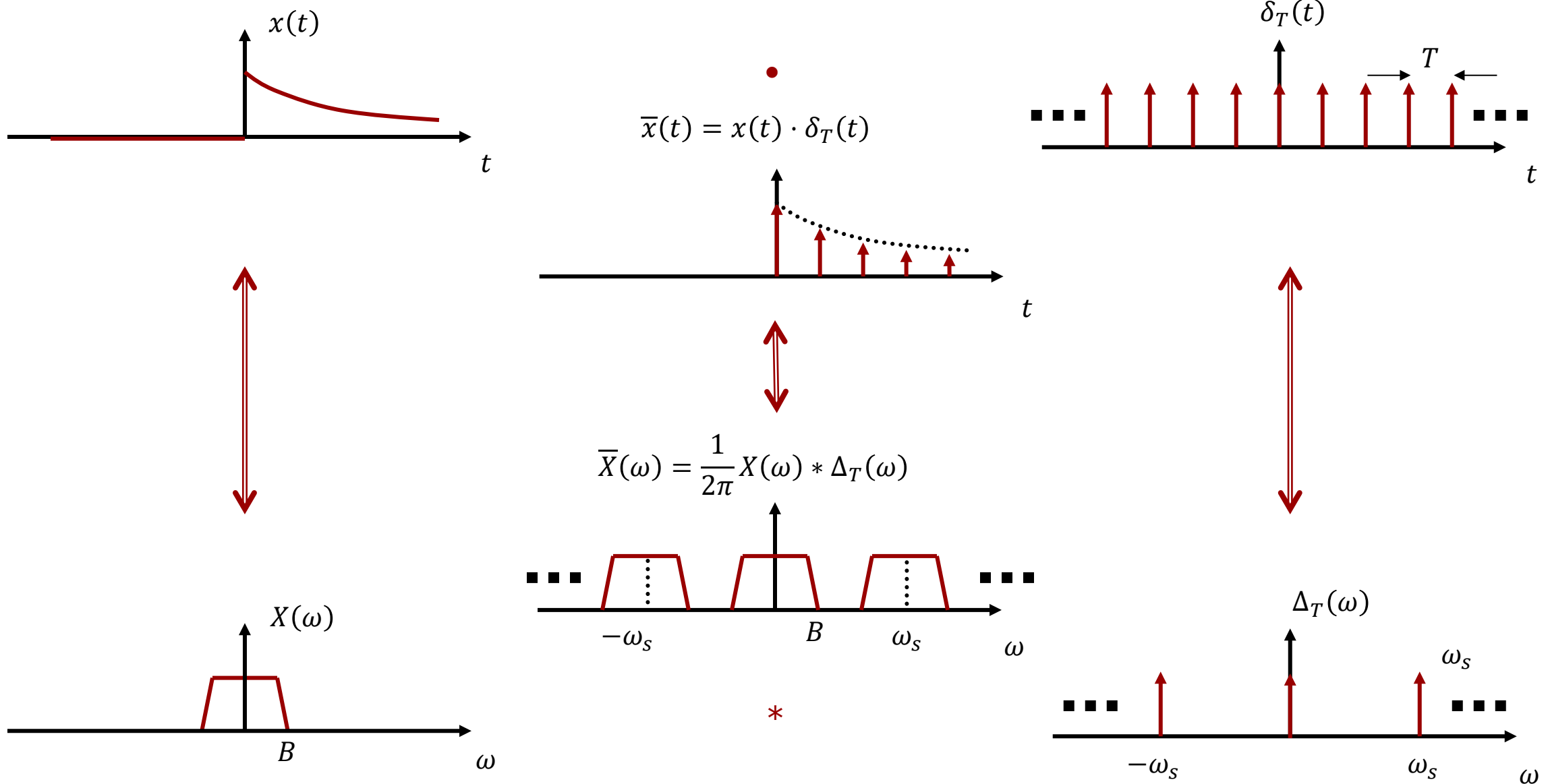


**Nyquist
theorem:**

$$\omega_s > 2 \cdot B$$

If the sampling frequency is too small, adjacent extensions of the frequency spectrum overlap and "irreversible" aliasing occurs.

Mathematical sampling of Continuous-Time Signals

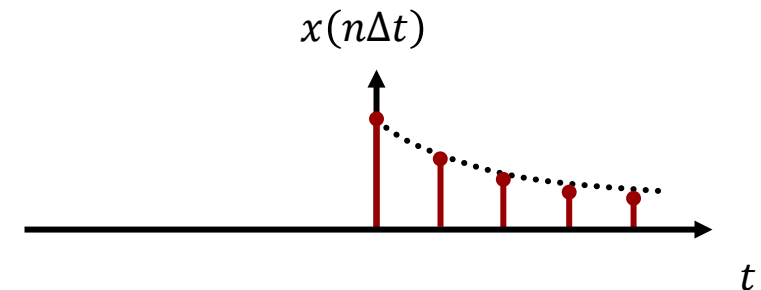
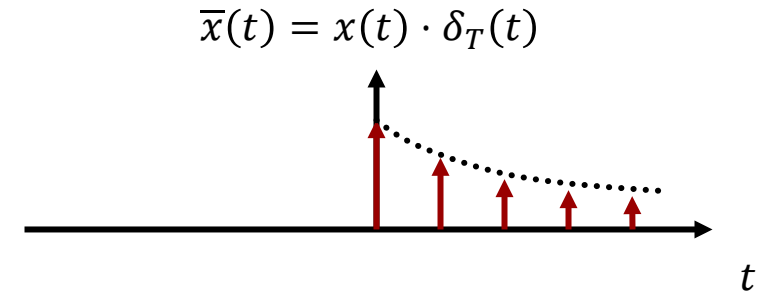


A sampled continuous-time signal is a series of equally spaced **impulses** with a strength determined by its amplitude.

This type of sampling is a mathematical process, not a process done by an analog-digital converter (ADC).

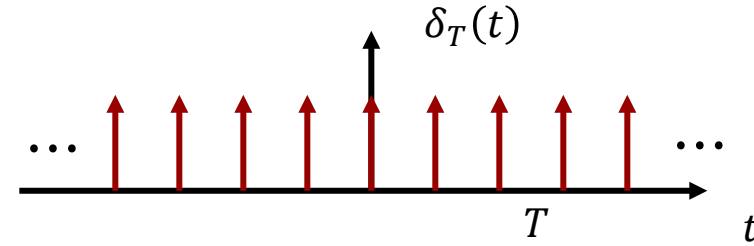
A discrete-time signal is a series of equally spaced **numbers** representing the same information.

This sequence of numbers is produced by an ADC.



We just obtained the frequency characteristic of the mathematically sampled CT signal $\bar{x}(t)$ using convolution in the frequency domain.

An alternative approach is to use the **modulation theorem**.



$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad \omega_s = \frac{2\pi}{T}$$

The trigonometric Fourier series:

$$\delta_T(t) = \frac{1}{T} (1 + 2 \cos \omega_s t + 2 \cos 2\omega_s t + 2 \cos 3\omega_s t + \dots)$$

$$\bar{x}(t) = x(t)\delta_T(t) = \frac{1}{T} (x(t) + 2x(t) \cos \omega_s t + 2x(t) \cos 2\omega_s t + 2x(t) \cos 3\omega_s t + \dots)$$

$$\bar{X}(\omega) = \frac{1}{T} (X(\omega) + X(\omega - \omega_s) + X(\omega + \omega_s) + X(\omega - 2\omega_s) + X(\omega + 2\omega_s) + \dots)$$

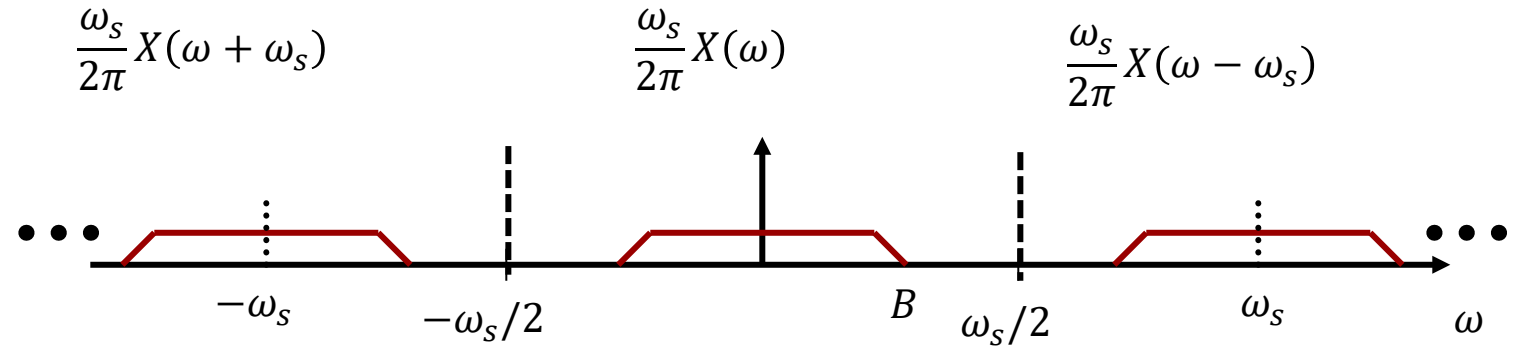
$$\bar{X}(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

Nyquist sampling theorem

The frequency spectrum $\bar{X}(\omega)$ is periodic, with periodicity ω_s .

We can reconstruct $X(\omega)$ from $\bar{X}(\omega)$ only if there is no overlap between the periodic repetitions of $X(\omega)$ in $\bar{X}(\omega)$.

The conditions for reconstructing $X(\omega)$ from $\bar{X}(\omega)$ has been formulated by several scientists and forgotten again. It was rediscovered by Nyquist and is now known as the **Nyquist sampling theorem**.

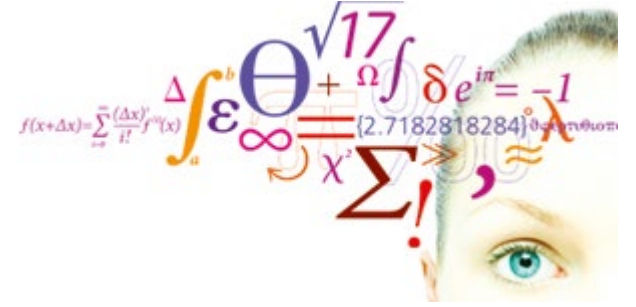


Expressed in radians/s	$B \leq \frac{\omega_s}{2}$	$\omega_s \geq 2B$
------------------------	-----------------------------	--------------------

Expressed in Hz	$B \leq \frac{f_s}{2}$	$f_s \geq 2B$	$T = \frac{1}{f_s}$
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Nyquist sampling rate:	$f_s = 2B$
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Some signals require:	$f_s > 2B$
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Practical sampling

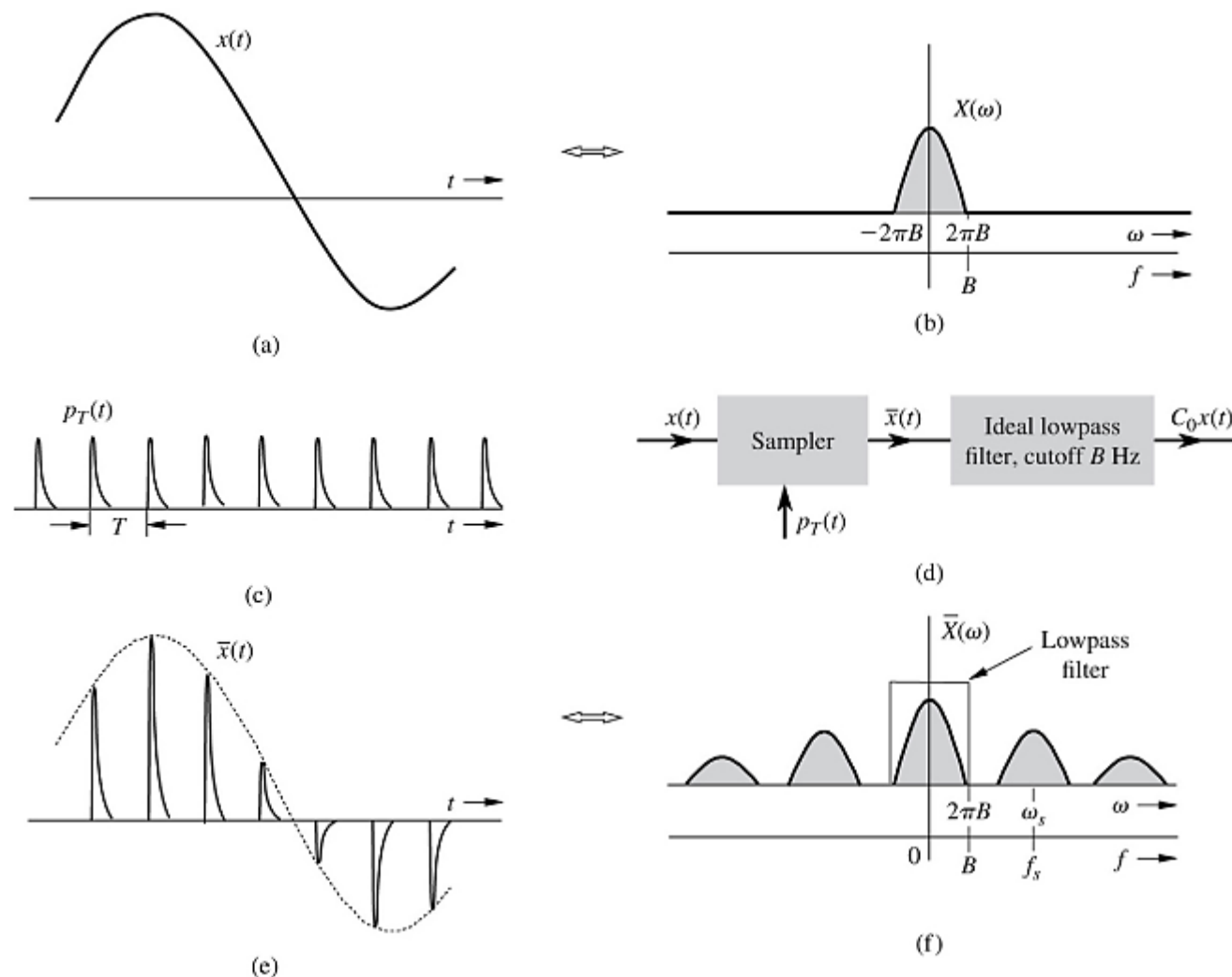
Video 2

Fig. 5.3 in the textbook illustrates the practical sampling process where it is not possible to produce an impulse. Instead, a practically realizable pulse $p_T(t)$ is multiplied onto the signal to be sampled.

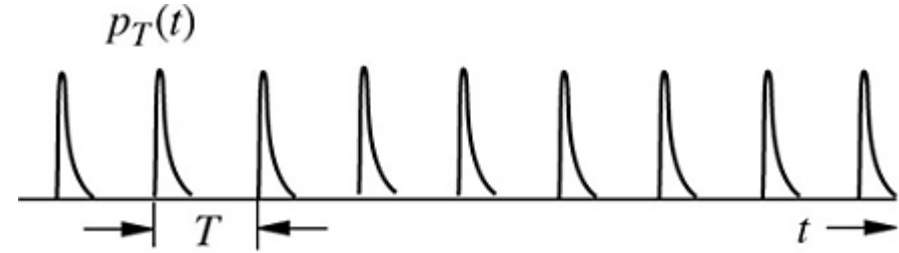
The pulse must be periodic with period T and the duration of each pulse must be less than T .

It is shown that the resulting frequency spectrum still has periodic repetitions of $X(\omega)$ but that each repetition now is scaled in magnitude. The exact scaling depends on the shape of the pulse.

When the sampling pulse was an ideal impulse, all spectral repetition had the same scaling.



The train of practically realizable pulses can be expressed as a compact Fourier series. Remember that the Fourier series is a periodic function, hence the Fourier series represent the entire train of pulses, not just one pulse.



$$p_T(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_s t + \theta_n)$$

The practically sampled signal $\bar{x}(t)$ can now be expressed as a Fourier series expansion:

$$\bar{x}(t) = x(t)p_T(t) = x(t) \left(C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_s t + \theta_n) \right)$$

Each term in the series is a modulation of the CT signal $x(t)$, but each repetition of the spectrum is scaled by $C_n/2$.

$$= C_0 x(t) + \sum_{n=1}^{\infty} C_n x(t) \cos(n\omega_s t + \theta_n)$$

$$\bar{X}(\omega) = \frac{1}{T} \left(C_0 X(\omega) + \frac{C_1}{2} (X(\omega - \omega_s) + X(\omega + \omega_s)) + \frac{C_2}{2} (X(\omega - 2\omega_s) + X(\omega + 2\omega_s)) + \dots \right)$$

Practical sampling

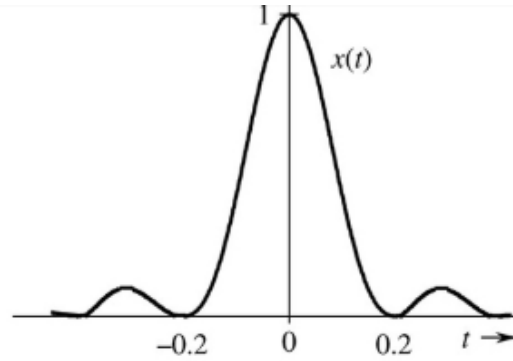
Now we would like to consider a sampling pulse train for which we can derive its Fourier transform.

The sampling pulse train is a train of square pulses. They are spaced apart by T , i.e. the time interval between samples.

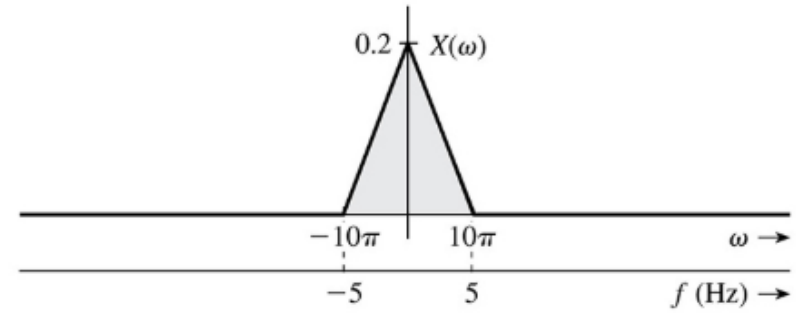
Each pulse has a height of 1 and a width of T_1 .

We will first approach this as done in the textbook, ch. 5.1.

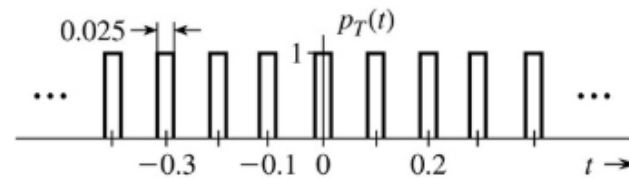
Then we will use an alternative approach.



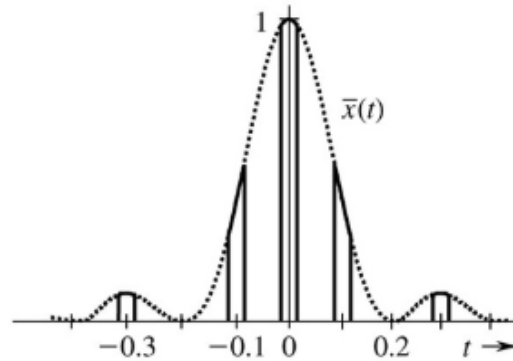
(a)



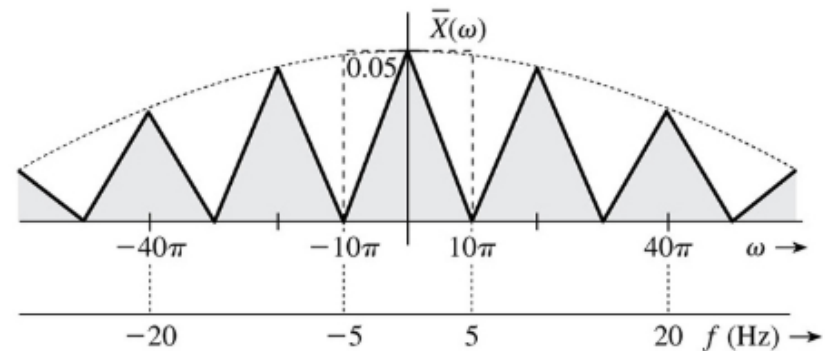
(b)



(c)



(d)

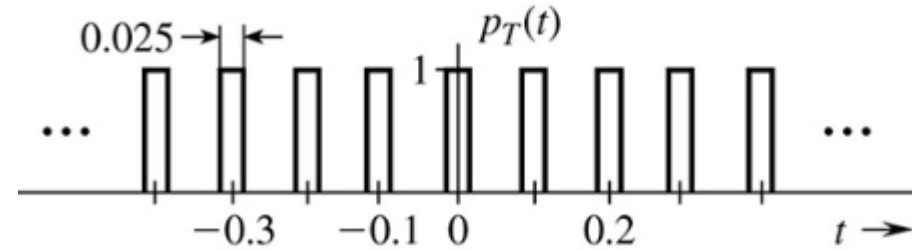


(e)

Fourier series of square pulse

We will assume that each pulse has a duration T_1 and a periodicity of T .

$$\omega_s = \frac{2\pi}{T}$$



$$p_T(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_s t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_s t)$$

$$a_0 = \frac{1}{T} \int_{-T_1/2}^{T_1/2} p_T(t) dt = \frac{T_1}{T}$$

$$a_n = \frac{2}{T} \int_{-T_1/2}^{T_1/2} p_T(t) \cos n\omega_s t dt = \frac{2}{T} \int_{-T_1/2}^{T_1/2} \cos n\omega_s t dt = \frac{2}{Tn\omega_s} [\sin n\omega_s t]_{-T_1/2}^{T_1/2} = \frac{2}{Tn\omega_s} \left(\sin \frac{n\omega_s T_1}{2} - \sin \frac{-n\omega_s T_1}{2} \right)$$

$$a_n = \frac{2T_1}{T} \frac{2}{n\omega_s T_1} \sin \frac{n\omega_s T_1}{2} = \frac{2T_1}{T} \operatorname{sinc} \left(\frac{n\omega_s T_1}{2} \right) = \frac{2T_1}{T} \operatorname{sinc} \left(\frac{n\pi T_1}{T} \right)$$

$$b_n = \frac{2}{T} \int_{-T_1/2}^{T_1/2} p_T(t) \sin n\omega_s t dt = 0, p_T \text{ even}$$

Inserting the Fourier series coefficients, we obtain:

$$\bar{x}(t) = x(t)p_T(t) = c_0 x(t) + \sum_{n=1}^{\infty} c_n x(t) \cos(n\omega_s t + \theta_n)$$

$$\begin{aligned} \bar{x}(t) = x(t)p_T(t) &= x(t) \left(\frac{T_1}{T} + \sum_{n=1}^{\infty} \frac{2T_1}{T} \operatorname{sinc}\left(\frac{n\pi T_1}{T}\right) \cos(n\omega_s t) \right) \\ &= \frac{T_1}{T} x(t) + \frac{2T_1}{T} \sum_{n=1}^{\infty} \operatorname{sinc}\left(\frac{n\pi T_1}{T}\right) x(t) \cos(n\omega_s t) \end{aligned}$$

Modulation theorem:

$$x(t) \cos(\omega_0 t) \leftrightarrow \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$

$$\begin{aligned} \bar{X}(\omega) &= \frac{T_1}{T} X(\omega) \\ &\quad + \frac{T_1}{T} \operatorname{sinc}\left(\frac{1\pi T_1}{T}\right) (X(\omega - \omega_s) + X(\omega + \omega_s)) \\ &\quad + \frac{T_1}{T} \operatorname{sinc}\left(\frac{2\pi T_1}{T}\right) (X(\omega - 2\omega_s) + X(\omega + 2\omega_s)) \\ &\quad + \dots \end{aligned}$$

Let us approach the same analysis using a different approach where we use the Fourier Transform theorems.

The main idea here is to consider the practical sampling pulse to be the result of convolving a train of ideal impulse functions $\delta_T(t)$ with a single square pulse $p(t)$.

We make the Fourier Transform of the time-domain expression and then find the Fourier Transform of all three functions.

We can then get the product:

$$\bar{x}(t) = x(t)p_T(t) = x(t)(\delta_T(t) * p(t))$$

$$\bar{X}(\omega) = \frac{1}{2\pi} X(\omega) * (\Delta_T(\omega)P(\omega))$$

$$x(t) \Leftrightarrow X(\omega)$$

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \Leftrightarrow \Delta_T(\omega) = \frac{2\pi}{T} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

$$p(t) = \text{rect}\left(\frac{t}{T_1}\right) \Leftrightarrow P(\omega) = T_1 \text{sinc}\left(\frac{\omega T_1}{2}\right)$$

$$\Delta_T(\omega)P(\omega) = \frac{2\pi T_1}{T} \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n\omega_s T_1}{2}\right) \delta(\omega - n\omega_s)$$

From the previous slide we have:

$$\bar{X}(\omega) = \frac{1}{2\pi} X(\omega) * (\Delta_T(\omega) P_T(\omega))$$

$$\Delta_T(\omega) P_T(\omega) = \frac{2\pi T_1}{T} \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n\omega_s T_1}{2}\right) \delta(\omega - n\omega_s)$$

Next, we calculate the convolution:

$$\bar{X}(\omega) = \frac{1}{2\pi} X(\omega) * \frac{2\pi T_1}{T} \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n\omega_s T_1}{2}\right) \delta(\omega - n\omega_s)$$

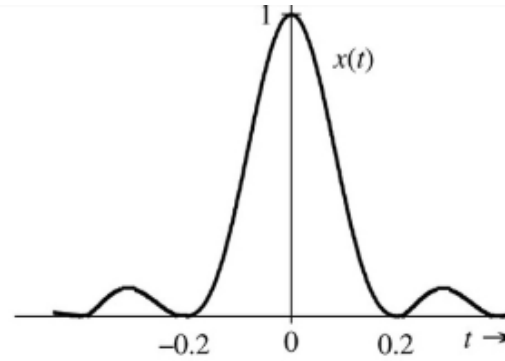
$$\bar{X}(\omega) = \frac{T_1}{T} \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n\omega_s T_1}{2}\right) X(\omega - n\omega_s)$$

Observations:

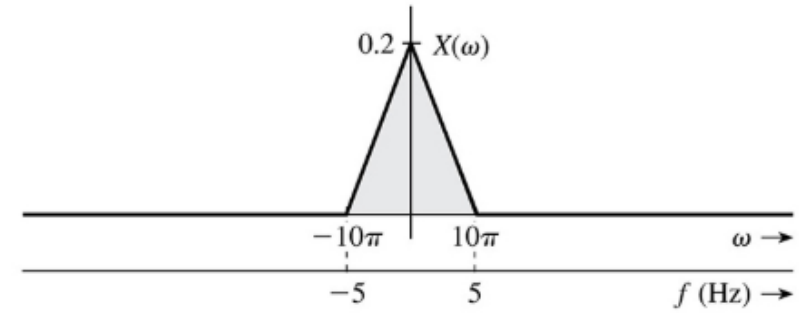
Although we replaced the ideal sampling function (the impulse) with a square pulse, the frequency spectrum $X(\omega)$ of the CT signal is still contained within the periodic spectrum $\bar{X}(\omega)$.

It is scaled in amplitude depending on the width of the sampling pulse, but provided the Nyquist sampling theorem is not violated, we can recover the spectrum $X(\omega)$ by lowpass filtering $\bar{X}(\omega)$.

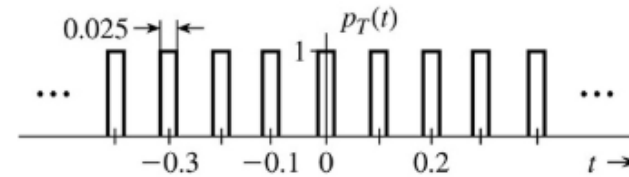
We see that spectrum scaling is determined by the pulse width T_1 . Letting $T_1 \rightarrow 0$ would not work.



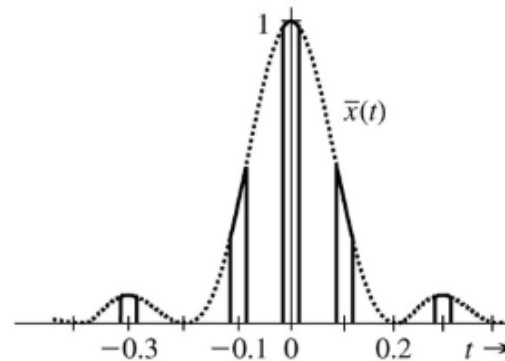
(a)



(b)

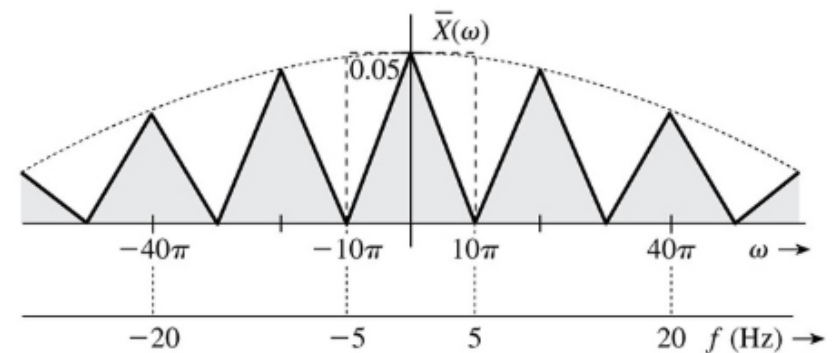


(c)

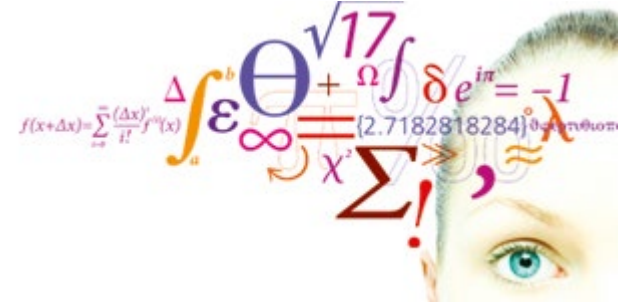


(d)

$$\bar{X}(\omega) = \frac{T_1}{T} \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n\omega_s T_1}{2}\right) X(\omega - n\omega_s)$$



(e)



Lathi 5.2

Reconstruction of CT signal

Video 2

Reconstruction of Continuous-Time Signal

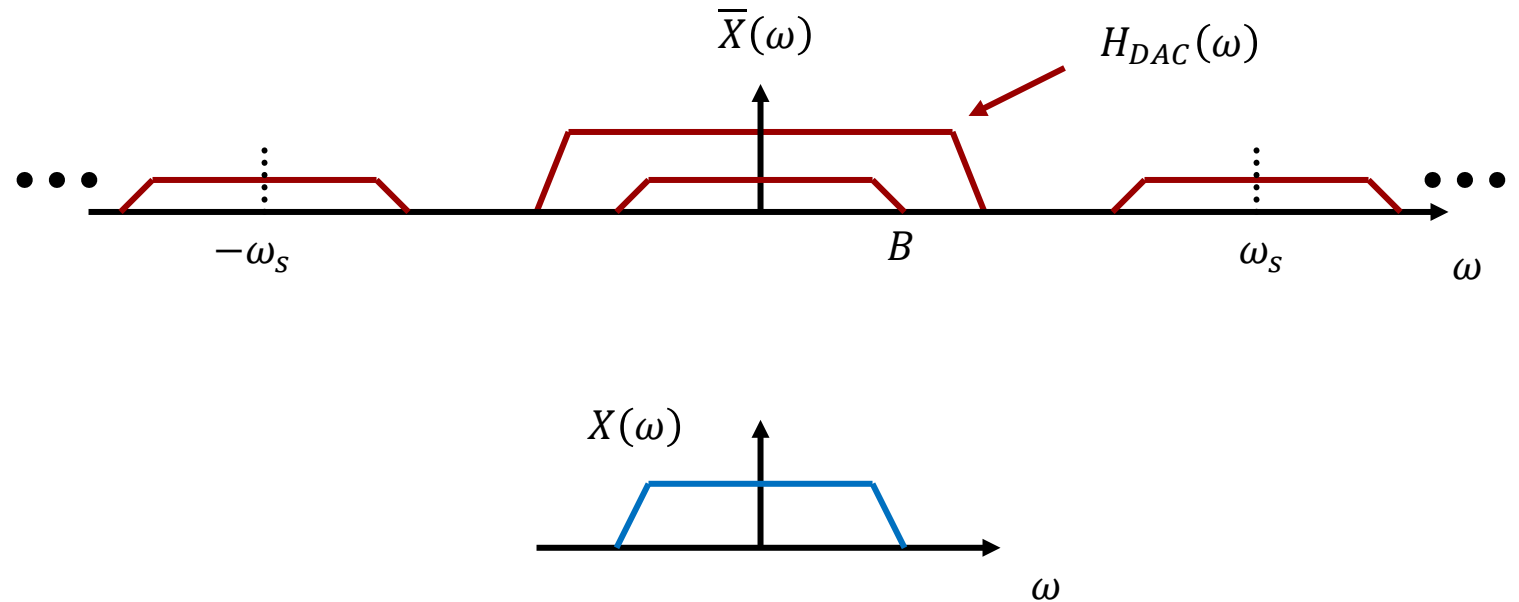
We have just learned that we can reconstruct the spectrum $X(\omega)$ by lowpass filtering $\bar{X}(\omega)$.

Ideal reconstruction:

We call this an ideal reconstruction because the periodic repetitions of spectra do not overlap.

This is due to the Nyquist theorem not being violated.

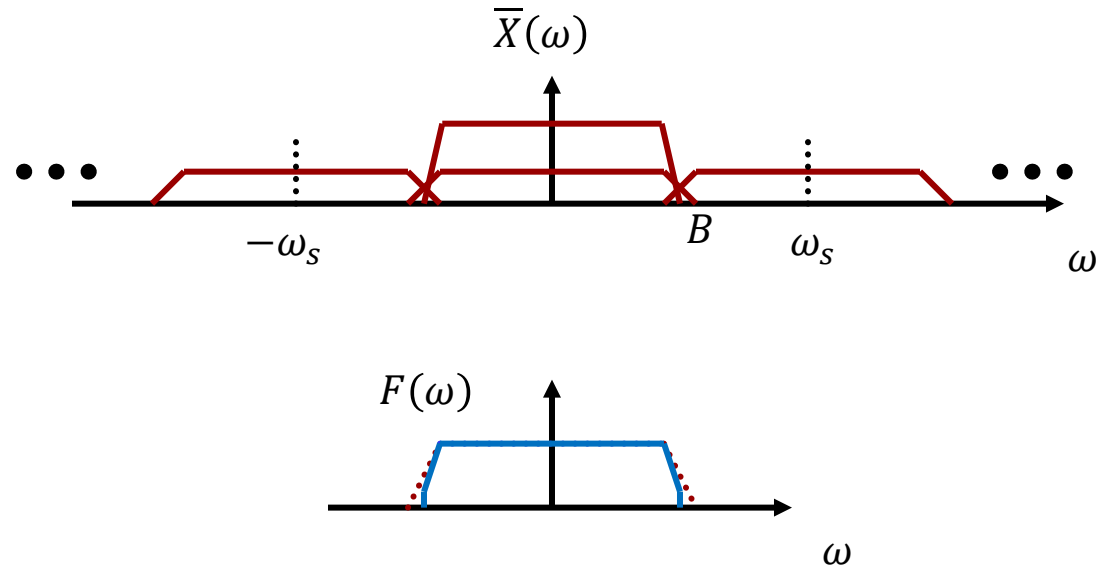
If the spectrum $X(\omega)$ is preserved by the sampling, it means that it is possible to reconstruct $x(t)$ from its samples.



Reconstruction of Continuous-Time Signal

Here the Nyquist sampling theorem has been violated ($\omega_s < 2B$).

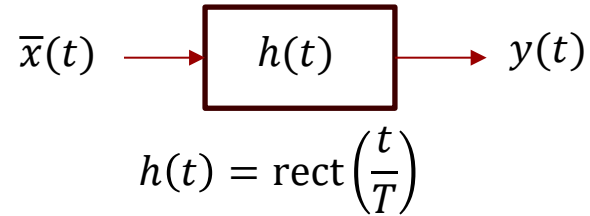
Imperfect
reconstruction:



In this case, the spectra overlap and in fact their overlapping parts add up and modify the spectrum in a region centered around $\omega_s/2$. This error is called **aliasing**.

Aliasing is the irreversible consequence of violating the Nyquist sampling theorem.

Reconstruction of Continuous-Time signal from its samples



$$\bar{x}(t) = x(t) \cdot \delta_T(t) \quad \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

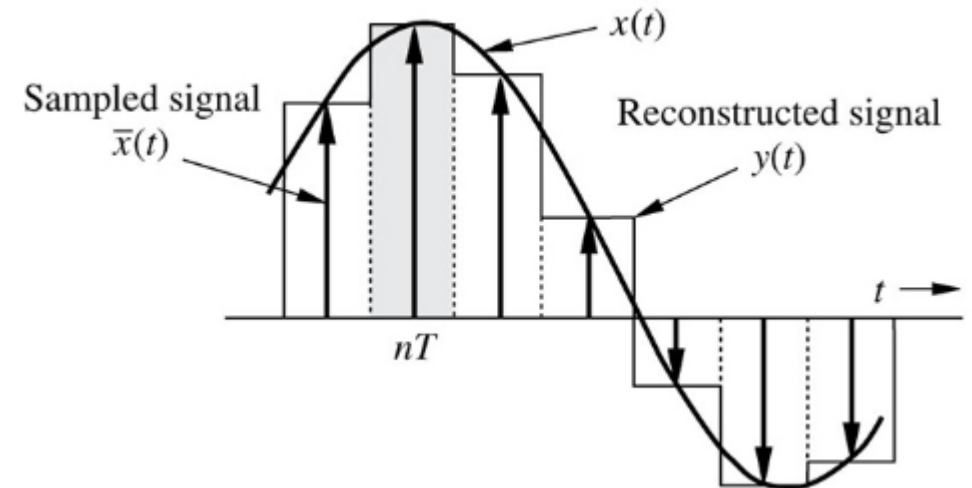
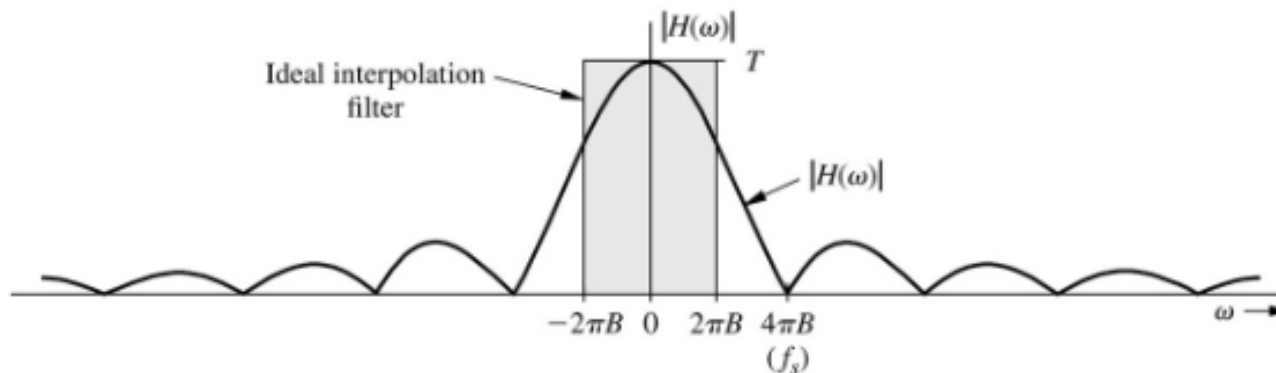
$$y(t) = \bar{x}(t) * h(t) = x(t) \cdot (\delta_T(t) * h(t))$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \text{rect}\left(\frac{t - nT}{T}\right)$$

To create a CT signal from a set of samples, interpolation must be used.

If we use a filter with a pulse as its impulse response to interpolate between samples, we get a staircase approximation $y(t)$.

The reason for the poor interpolation is that we are multiplying the spectrum with a sinc.



Reconstruction of Continuous-Time signal from its samples by ideal interpolation

To isolate the spectrum $X(\omega)$ from $\bar{X}(\omega)$ we could multiply with a rect function in the frequency domain, i.e. an ideal filter.

$$X(\omega) = \bar{X}(\omega) H(\omega)$$

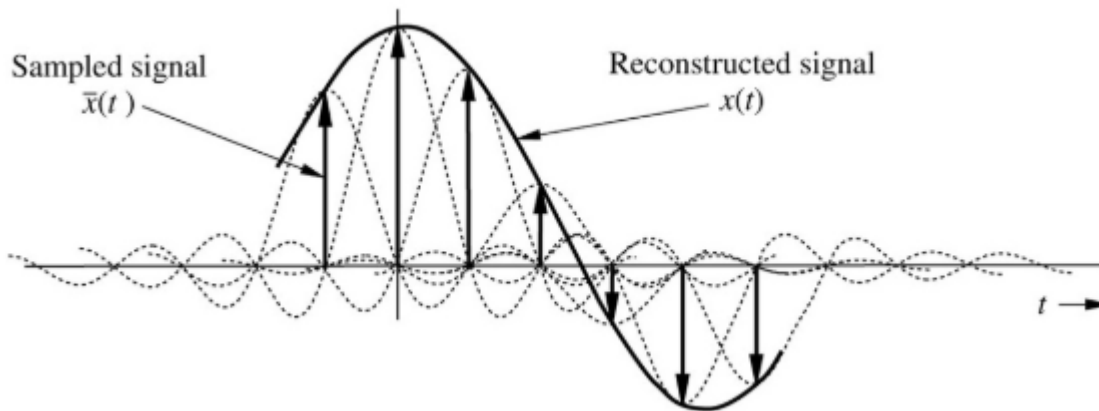
$$H(\omega) = T \operatorname{rect}\left(\frac{\omega}{4\pi B}\right)$$

$$h(t) = \operatorname{sinc}(2\pi B t)$$

$$y(t) = \sum_{n=-\infty}^{\infty} x(nT) h(t - nT)$$

$$y(t) = \sum_{n=-\infty}^{\infty} x(nT) \operatorname{sinc}(2\pi B(t - nT))$$

$$y(nT) = x(nT)$$



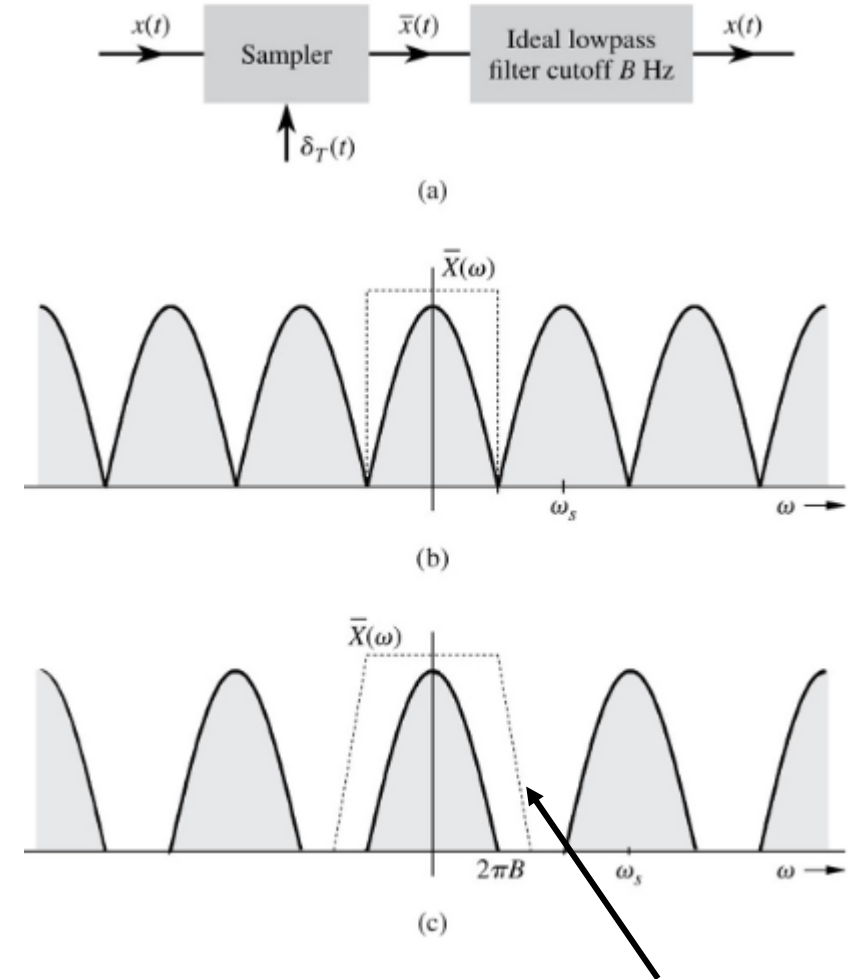
The reconstructed signal $y(t) = x(t)$ because they are both equal to $\mathcal{F}^{-1}\{\bar{X}(\omega) H(\omega)\}$

The ideal interpolation requires a non-causal filter:

$$h(t) = \text{sinc}(2\pi B t)$$

If the sampling frequency is increased so $\frac{\omega_s}{2} > 2\pi B$, or $\frac{f_s}{2} > B$, then we can use a lowpass filter with a finite slope to attenuate all periodically repeated cycles.

Ideally such a filter should have zero gain for all frequencies above $f_s/2$. No practical realizable filter can achieve this. The best we can aim for is some feasible compromise between increasing the sampling frequency and increasing the slope of the filter cut-off.



Time-limited signals have infinite bandwidth.

Let us make a signal $y(t)$ time-limited by multiplying it with a pulse of duration T_1 .

This means that the frequency spectrum of $y(t)$ will be the spectrum of $x(t)$ convolved with a never ending sinc function.

All practically realizable signals are time-limited. Hence, all practically realizable signals have infinite bandwidth.

$$y(t) \stackrel{\text{def}}{=} x(t)p(t)$$

$$p(t) = \text{rect}\left(\frac{t}{T_1}\right) \Leftrightarrow P(\omega) = T_1 \text{sinc}\left(\frac{\omega T_1}{2}\right)$$

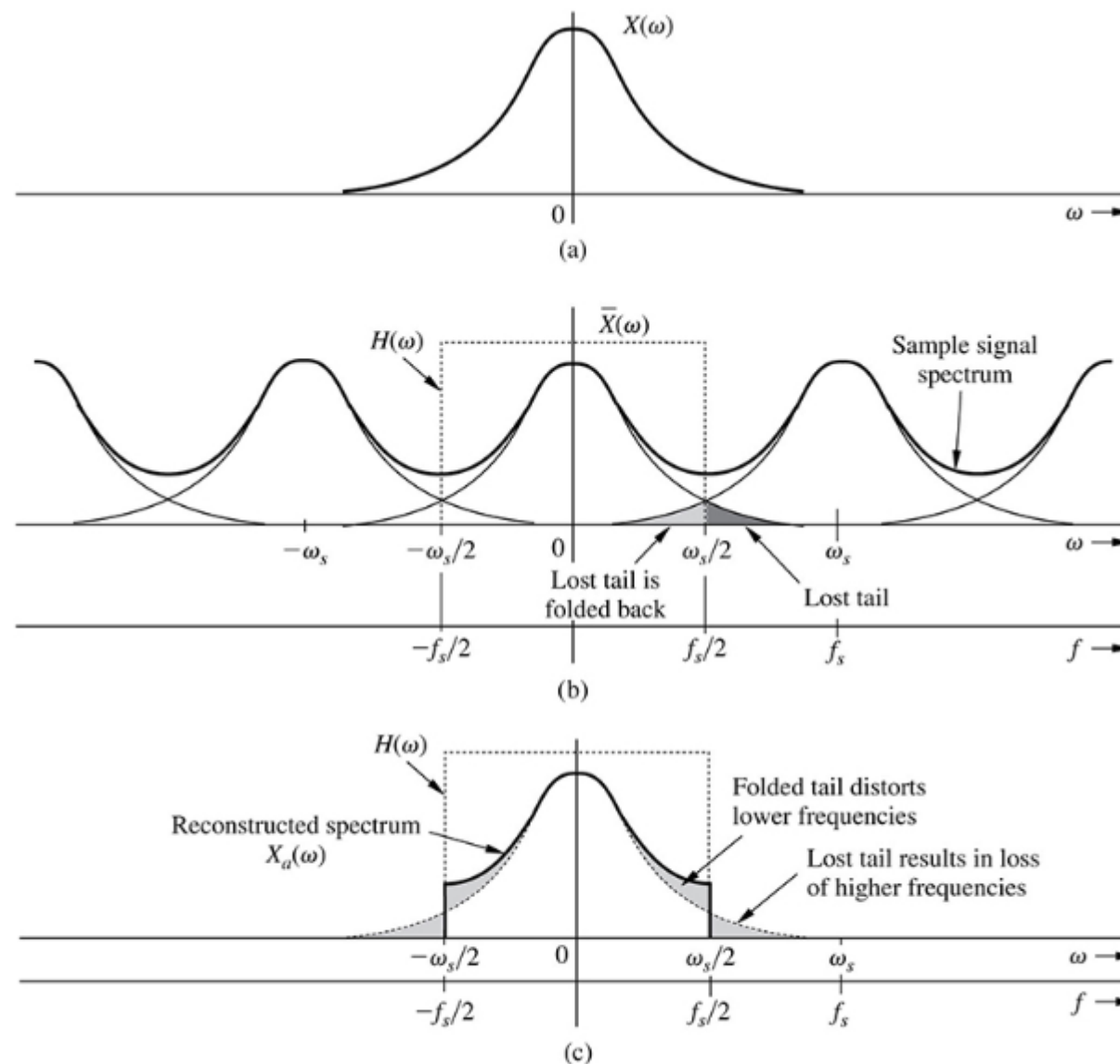
$$Y(\omega) = X(\omega) * T_1 \text{sinc}\left(\frac{\omega T_1}{2}\right)$$

Time-limited signals have infinite bandwidth.

When sampled, the frequency spectrum $\bar{X}(\omega)$ will show overlapping tails, as shown on the sketch. The tail above $f_s/2$ is also found reflected below $f_s/2$.

It looks as if the upper tail has been folded about $f_s/2$, which is therefore termed the **folding frequency**.

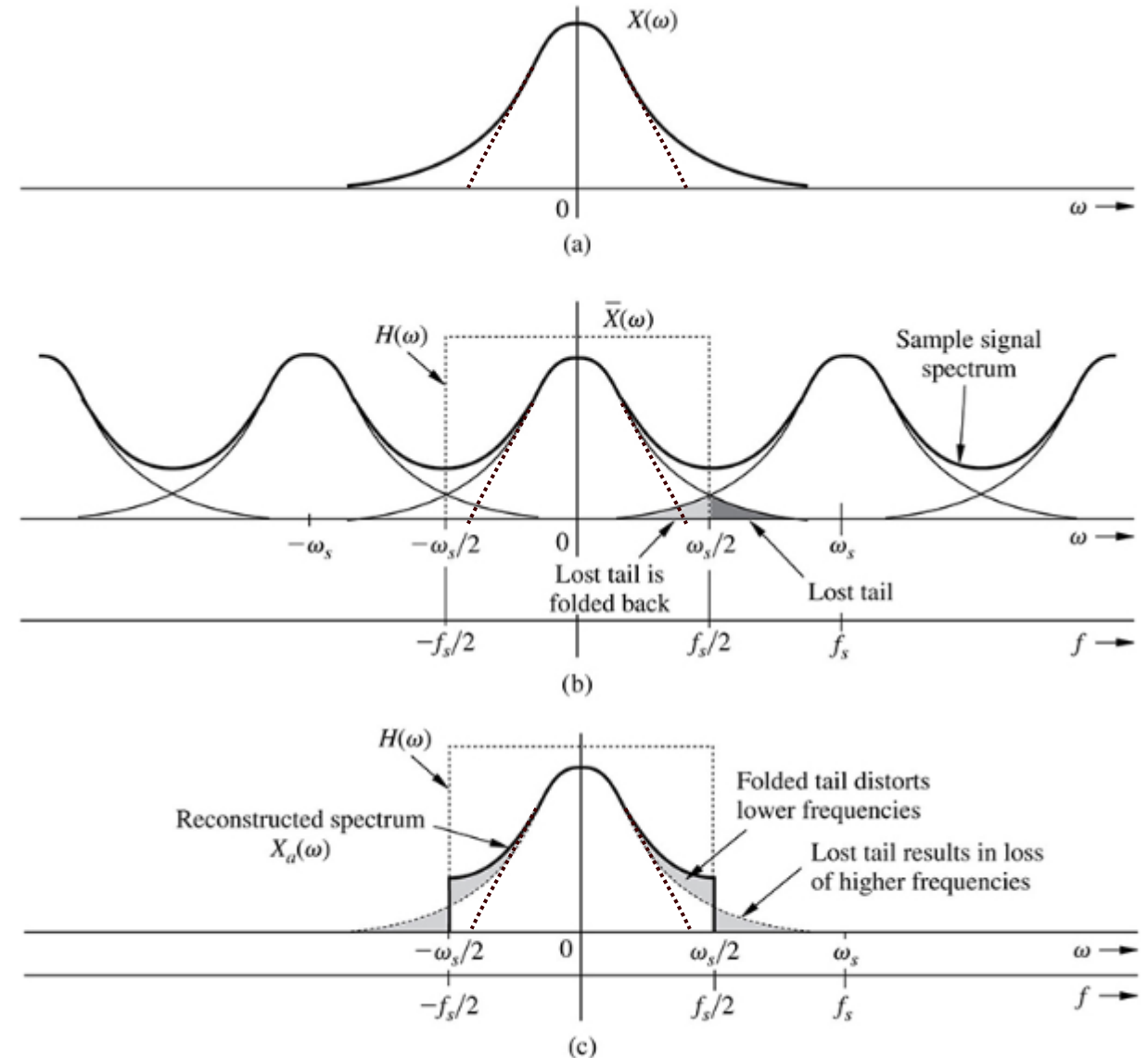
The folded tail adds to the spectrum and distorts it irreversibly.

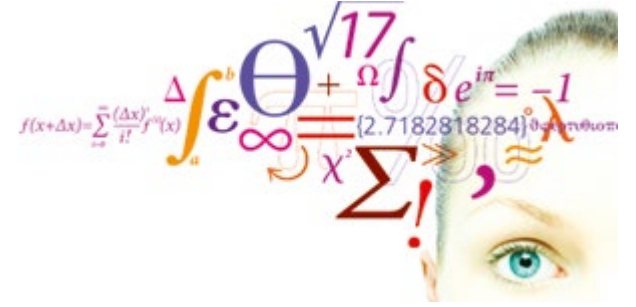


Strictly speaking Nyquist sampling theorem can never be satisfied for practically realizable signals because the sampling frequency would need to be infinite.

If the frequencies near the folding frequency originates from noise, we can use a lowpass filter to reduce its magnitude to an insignificant level. In such cases the distortion of the frequency spectrum is minimal at frequencies originating from noise.

Such a lowpass filter is called an **anti-aliasing filter**. It is always placed in front of an ADC to minimize the effect of frequencies folding across the folding frequency.





Fourier analysis of filter circuits

Video 2

We will often derive the frequency characteristic $H(\omega)$ of a filter. It is in general a complex-valued function, and to visualize it, we need to plot its modulus (amplitude) and argument (phase).

Here we show that the modulus of $H(\omega)$ is the modulus of the numerator divided by the modulus of the denominator.

Comparing the exponentials on the left and right sides shows that the phase angle of $H(\omega)$ is the phase angle of the numerator minus the phase angle of the denominator:

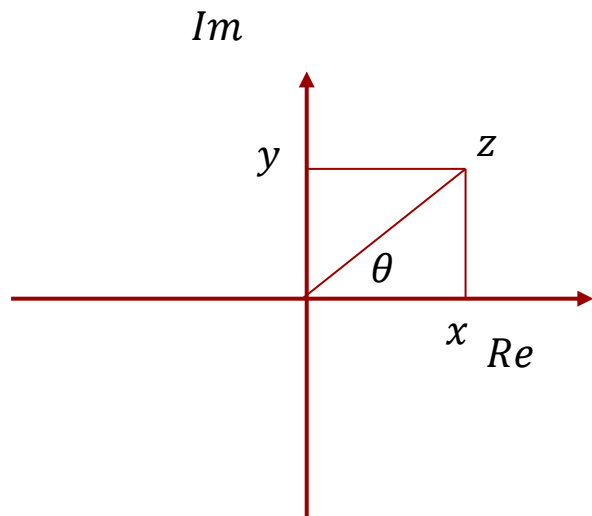
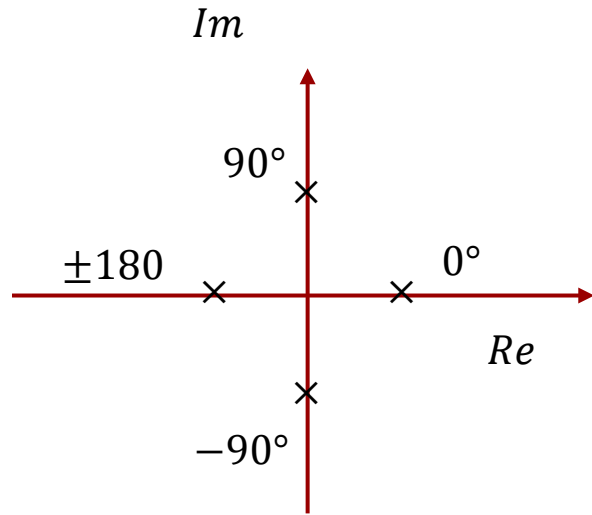
$$H(\omega) = \frac{P(j\omega)}{Q(j\omega)}$$

$$H(\omega) = \underbrace{|H(\omega)|e^{j\angle H(\omega)}}_{\text{Polar form}} = \frac{P(j\omega)}{Q(j\omega)} = \frac{|P(\omega)|e^{j\angle P(\omega)}}{\underbrace{|Q(\omega)|e^{j\angle Q(\omega)}}_{\text{Polar form}}}$$

$$|H(\omega)| = \frac{|P(\omega)|}{|Q(\omega)|}$$

$$e^{j\angle H(\omega)} = \frac{e^{j\angle P(\omega)}}{e^{j\angle Q(\omega)}} = e^{j\angle P(\omega)} e^{-j\angle Q(\omega)} = e^{j(\angle P(\omega) - \angle Q(\omega))}$$

$$\angle H(\omega) = \underbrace{\angle P(\omega)}_{\text{numerator angle}} - \underbrace{\angle Q(\omega)}_{\text{denominator angle}}$$



$$z = x + jy$$

$$|z| = |x + jy| = \sqrt{x^2 + y^2}$$

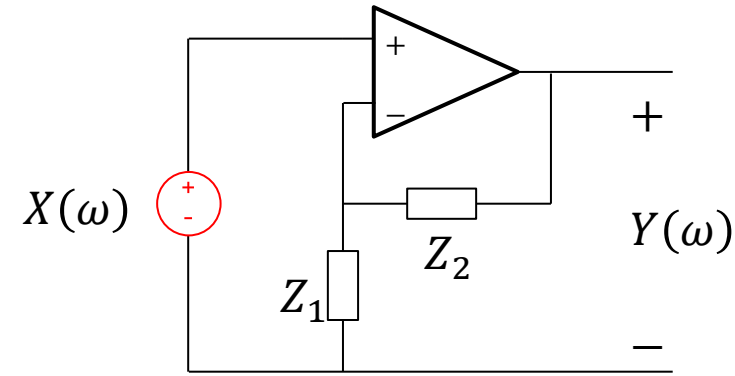
$$y = x \tan \theta$$

$$\angle z = \theta = \tan^{-1} \frac{y}{x} = \tan^{-1} \frac{\Im(z)}{\Re(z)}$$

A non-inverting 1st order filter:

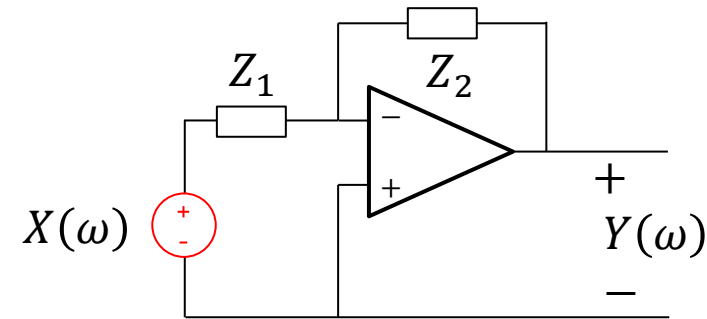
$$Y(\omega) = \left(1 + \frac{Z_2(\omega)}{Z_1(\omega)}\right) X(\omega)$$

We simply replace resistors with impedances in the gain equations.



An inverting 1st order filter:

$$Y(\omega) = -\frac{Z_2(\omega)}{Z_1(\omega)} X(\omega)$$



It is the frequency dependent nature of the impedances that determines, if the two circuits are lowpass, highpass, bandpass or bandstop filters.

First order active low pass filter

Now that we have seen an interesting example of using an inverting amplifier as a lowpass filter, let us look at how we design such a filter.

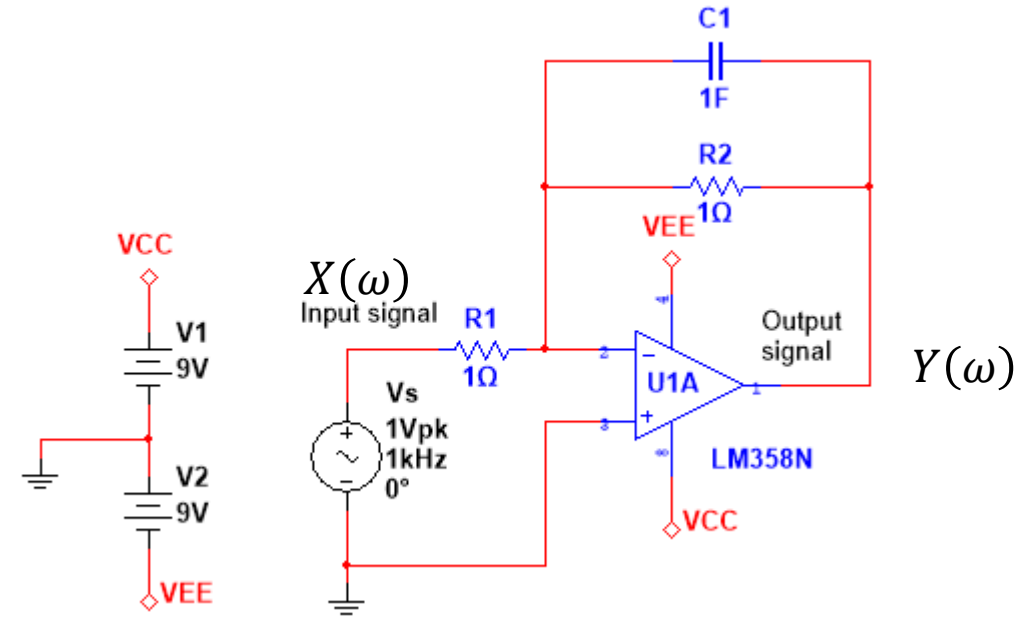
Gain equation for inverting amplifier.

$$H(\omega) = \underbrace{\frac{Y(\omega)}{X(\omega)}}_{\text{phasors}} = -\frac{Z_2(\omega)}{Z_1(\omega)}$$

Component impedances:

$$Z_1(\omega) = R_1$$

$$Z_2(\omega) = R_2 || Z_{C_1}(\omega) = \frac{R_2 Z_{C_1}(\omega)}{R_2 + Z_{C_1}(\omega)} = \underbrace{\frac{R_2 \cdot \frac{1}{j\omega C_1}}{R_2 + \frac{1}{j\omega C_1}}}_{\substack{C_1 \text{ and } R_2 \\ \text{in parallel}}} = \underbrace{\frac{R_2}{1 + j\omega R_2 C_1}}_{\substack{\text{polynomial} \\ \text{form}}}$$



Substitute impedances into gain equation:

Impedances

$$ZC1 := \omega \rightarrow \frac{1}{j \omega C1} :$$

Transfer function

$$H := \omega \rightarrow - \frac{Par(R2, ZC1(\omega))}{R1} :$$

$$\text{simplify}(\text{evalf}(H(\omega)))$$

$$\frac{1 R2}{(R2 \omega C1 - 1. I) R1}$$

Choice of components

$$C1 := 1 ;; R1 := 1 ;; R2 := 1 ;;$$

$$\frac{Y(\omega)}{X(\omega)} = - \frac{Z_2(\omega)}{Z_1(\omega)} = \frac{j R_2}{(R_2 \omega C_1 - j) R_1} = - \frac{\frac{R_2}{R_1}}{\underbrace{1 + j \omega R_2 C_1}_{\text{polynomial form}}}$$

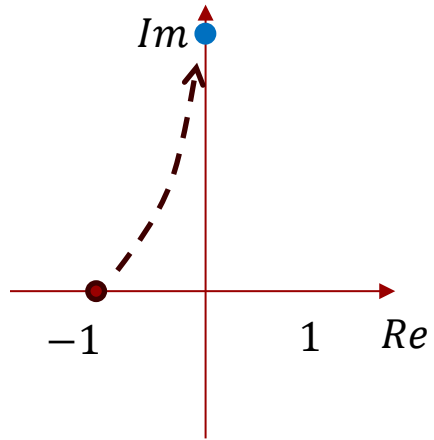
$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = - \frac{1}{1 + j \omega} \quad \text{Component values inserted}$$

$$H_{LFasymptote}(\omega) = - \frac{1}{1} = -1 = 1 e^{\pm j \pi}$$

$$H_{HFasymptote}(\omega) = - \frac{1}{j \omega} = \frac{j}{\omega} = \frac{1}{\omega} e^{j \pi/2}$$

$$\underbrace{|H(\omega)| = \frac{1}{\sqrt{1 + \omega^2}} = \frac{1}{\sqrt{2}}}_{\text{cut-off magnitude}} \Rightarrow \underbrace{\omega_c = 1 \frac{\text{rad}}{\text{s}} \Rightarrow f_c = \frac{1}{2\pi} = 0.159 \text{Hz}}_{\text{cut-off frequency}}$$

First order active low pass filter



$$\angle H(\omega) = \underbrace{\angle P(\omega)}_{\text{numerator angle}} - \underbrace{\angle Q(\omega)}_{\text{denominator angle}}$$

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = -\frac{1}{1 + j\omega}$$

$$\theta(\omega) = \angle H(\omega) = 180^\circ - \tan^{-1} \omega$$

$$\theta(0) = \angle H(0) = 180^\circ - \tan^{-1} 0 = 180^\circ$$

$$\theta(\infty) = \angle H(\infty) = 180^\circ - \tan^{-1} \infty = 180^\circ - 90^\circ = 90^\circ$$

Low frequency asymptote

High frequency asymptote

A filter will let frequencies pass through in some frequency interval (**pass band**) and attenuate the output signal in other frequency intervals (**stop band**). The frequency that marks the transition between pass band and stop band is called the **cut-off frequency** ω_c or f_c .

A quantitative definition for the cut-off frequency is that frequency where the power of the output signal is 50% of the power of the input signal. Power is $V \cdot I = V^2/R$.

It follows that the cut-off frequency is that frequency, where the amplitude characteristic is $1/\sqrt{2}$:

$$\underbrace{|H(\omega_c)|^2 = \frac{|Y(\omega_c)|^2}{|X(\omega_c)|^2}}_{\text{cut-off magnitude}} = \frac{1}{2}$$

For some amplitude characteristics, this is equivalent to a 3 dB attenuation. The cut-off frequency is therefore also called the **3-dB frequency**.

$$\underbrace{|H(\omega_c)| = \frac{1}{\sqrt{2}}}_{\text{cut-off magnitude}}$$

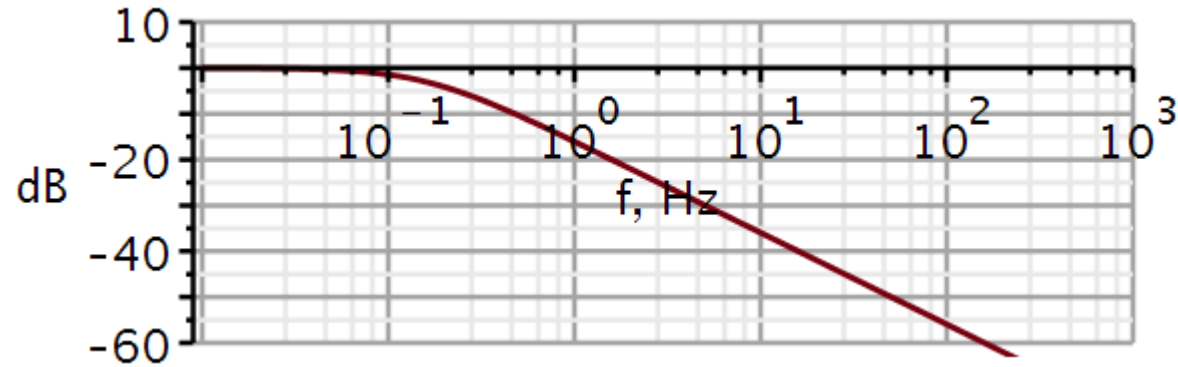
Observation:

The equations often use radians ω , but in graphs, f is used. Design specifications are usually given in Hertz. This must then be translated to radians before deriving component values.

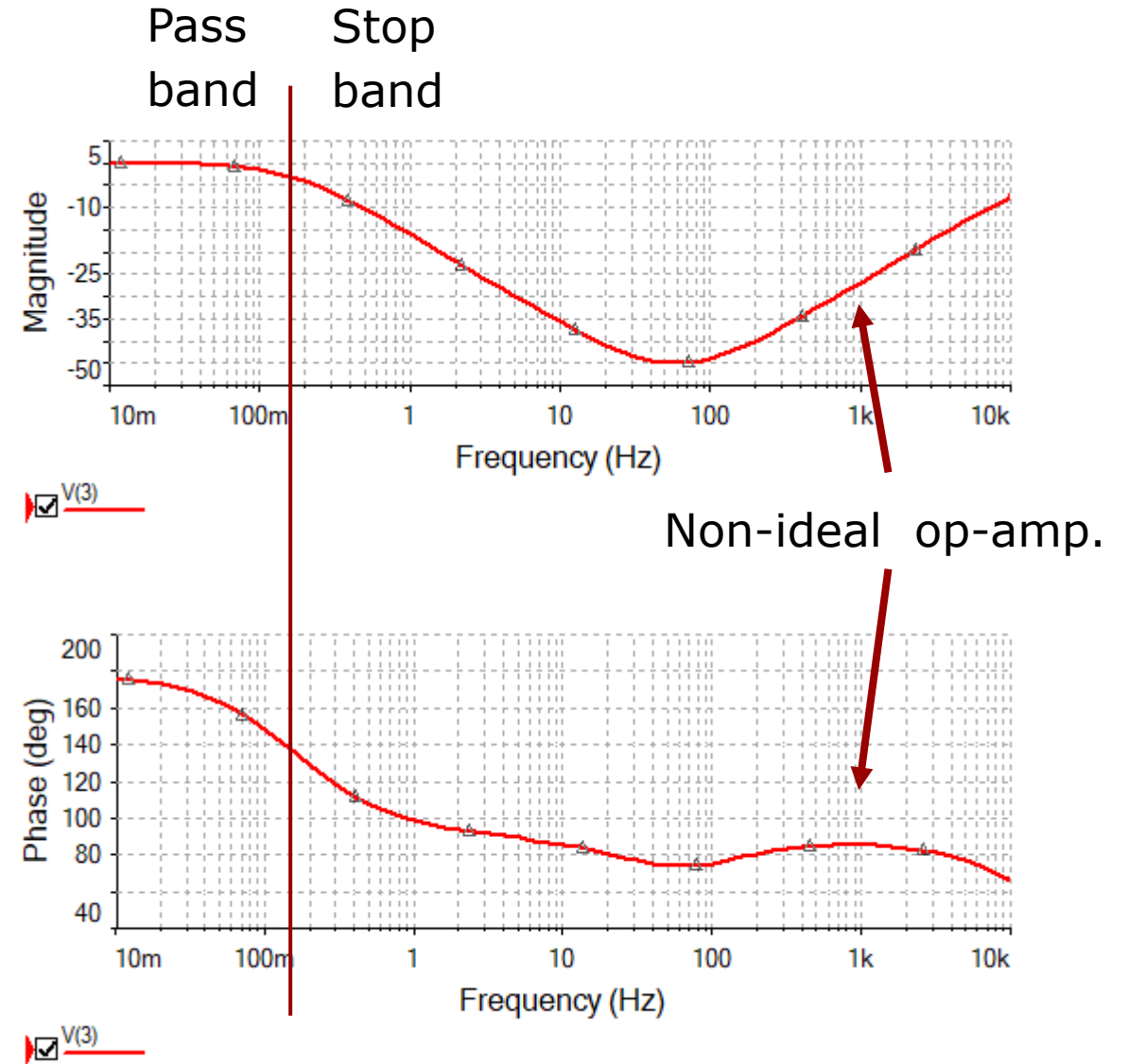
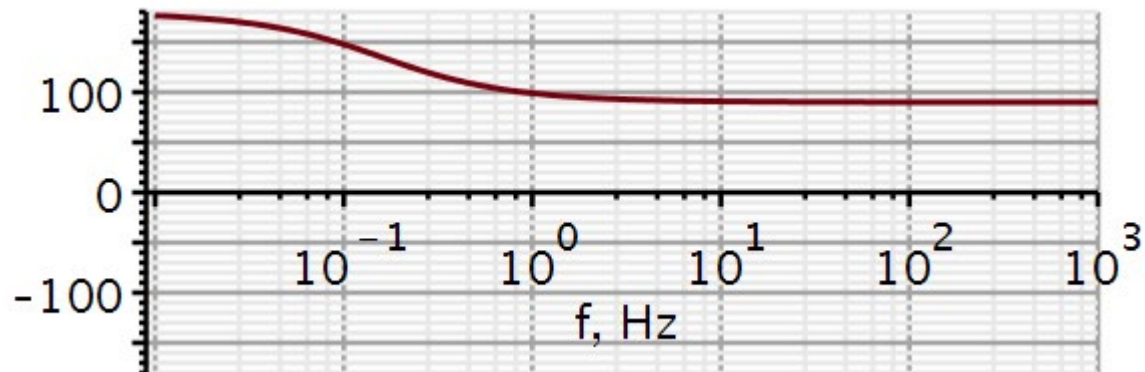
$$\underbrace{|H(\omega_c)|_{dB} = 20 \cdot \log_{10} \frac{1}{\sqrt{2}}}_{\text{cut-off magnitude}} = -3 \text{ dB}$$

First order active low pass filter

Amplitude characteristic



Phase characteristic



Let us now change the cut-off frequency from 0.159 Hz to 200 Hz. We do **frequency scaling** as follows:

$$\omega = K_F \cdot \omega_1$$

$$K_F = \frac{200\text{Hz}}{0.159\text{Hz}} = 1258 \Rightarrow C_1 = \frac{1\text{F}}{1258} = 795\mu\text{F}$$

We can also do an **impedance scaling**, where we scale all component with the same factor K_Z , but without changing time constants.

If we desire a compact capacitor, we would like it to be smaller than $1\mu\text{F}$.

$$\frac{Y(\omega)}{X(\omega)} = - \frac{\frac{R_2}{R_1}}{1 + j(K_F \cdot \omega)R_2 \frac{C_1}{K_F}}$$

$$\tau = R \cdot C = K_Z R \cdot C / K_Z$$

$$\frac{Y(\omega)}{X(\omega)} = - \frac{\frac{K_Z R_2}{K_Z R_1}}{1 + j(K_F \cdot \omega)K_Z R_2 \frac{C_1}{K_F K_Z}}$$

Let us aim for a capacitor of 47nF.

So here we let the impedance scaling factor K_Z be decided from a desired magnitude of the capacitor value.

Then the resistor values are determined based on this.

Capacitors are usually available in the E6 series. We may then have to fit odd values of resistors or use E48 or E96 resistors. It is much cheaper to get a high precision resistor than a high precision capacitor.

$$C_1'' = \frac{C_1'}{K_Z} = \frac{C_1}{K_F K_Z}$$

$$K_Z = \frac{795 \mu\text{F}}{47 \text{ nF}} = 16915$$

$$R_1 = 16915 \cdot 1\Omega \approx 15\text{k}\Omega$$

$$R_2 = 16915 \cdot 1\Omega \approx 15\text{k}\Omega$$

$$C_1 = 47\text{nF}$$

$$R_1 C_1 = 1\Omega \cdot 795\mu\text{F} = 0.795\text{ms}$$

$$R_1 C_1 \approx 15\text{k}\Omega \cdot 47\text{nF} = 0.705\text{ms}$$

C values

E6:

10

15

22

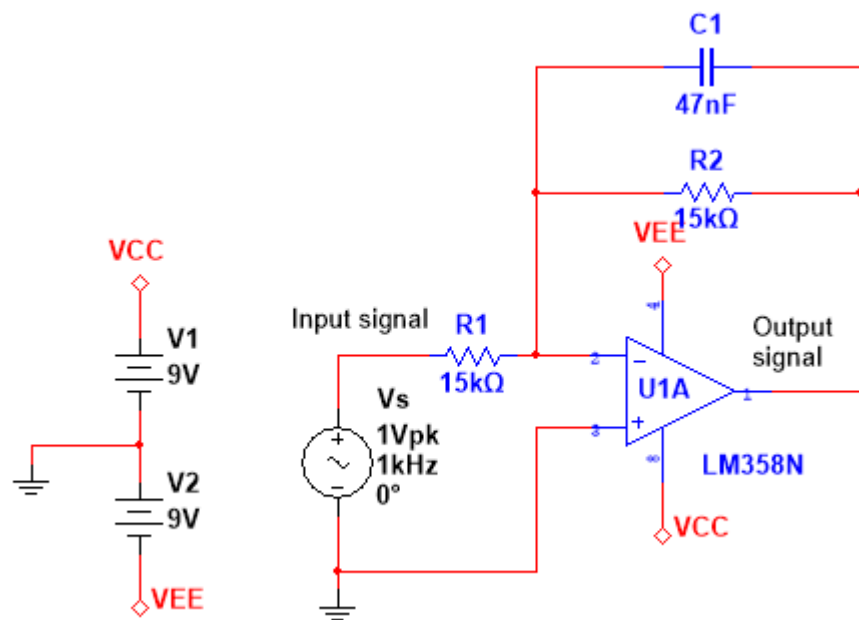
33

47

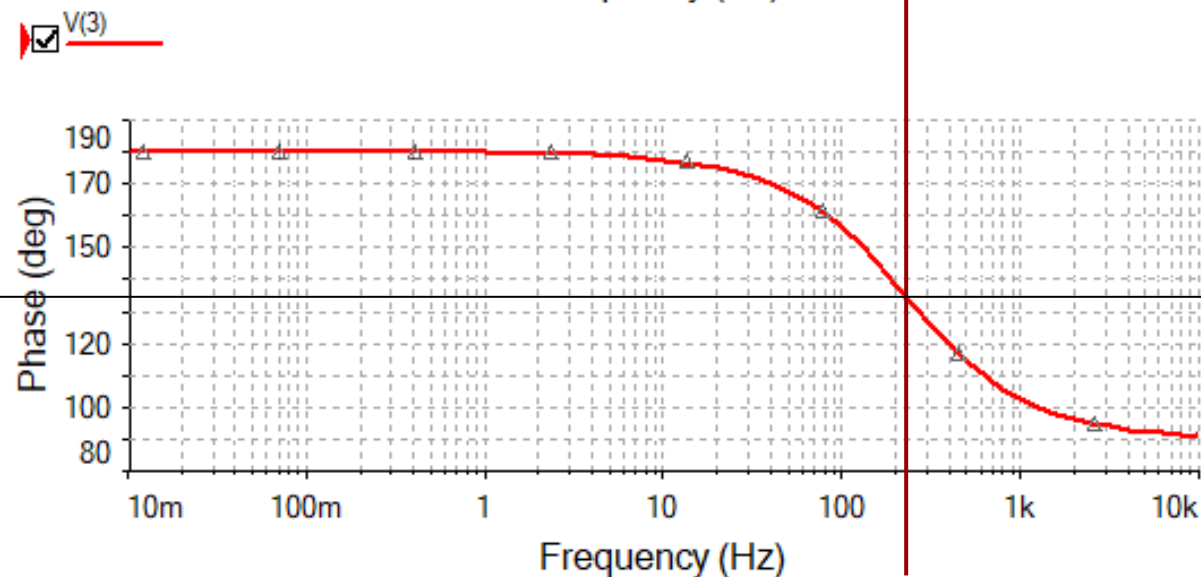
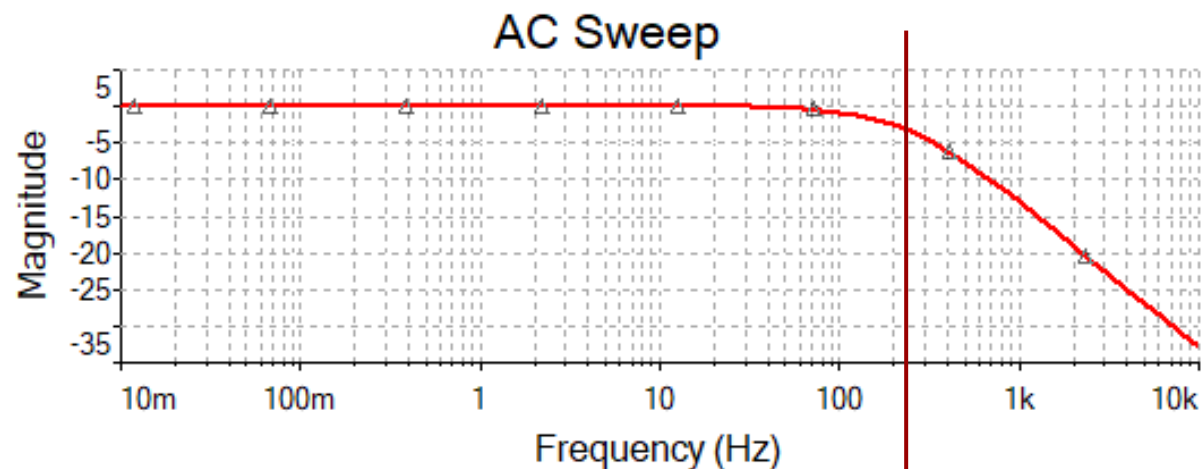
68

$$f_c = 200\text{Hz}$$

$$f_c = 226\text{Hz}$$



We have now a finished lowpass filter. We can read the cut-off frequency at -3dB on the magnitude plot or at half the phase angle change (here 135°) on the phase plot.



Using a lookup table

We just arbitrarily chose a capacitor value of 47nF. We then had to round-off the resistor value.

Using a look-up table, we can spot the best combination.

Here 8.160 is the number closest to 7.95.

$$\tau = R_1 C_1 = 1\Omega \cdot 795\mu\text{F} = 0.795\text{ms (ideal)}$$

$$\tau = R_1 C_1 \approx 15\text{k}\Omega \cdot 47\text{nF} = 0.705\text{ms}$$

$$R_1 C_1 = 1.2 \times 10^x \cdot 6.8 \times 10^y = 0.816\text{ms}$$

$$f_c = 195\text{Hz}$$

This look-up table is made in Excel.

The numbers of interest are often found in a narrow diagonal.

	$\tau = R \cdot C$												
	E6	E12											
R\C	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10
1	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000
1,2	1,200	1,440	1,800	2,160	2,640	3,240	3,960	4,680	5,640	6,720	8,160	9,840	12,000
1,5	1,500	1,800	2,250	2,700	3,300	4,050	4,950	5,850	7,050	8,400	10,200	12,300	15,000
1,8	1,800	2,160	2,700	3,240	3,960	4,860	5,940	7,020	8,460	10,080	12,240	14,760	18,000
2,2	2,200	2,640	3,300	3,960	4,840	5,940	7,260	8,580	10,340	12,320	14,960	18,040	22,000
2,7	2,700	3,240	4,050	4,860	5,940	7,290	8,910	10,530	12,690	15,120	18,360	22,140	27,000
3,3	3,300	3,960	4,950	5,940	7,260	8,910	10,890	12,870	15,510	18,480	22,440	27,060	33,000
3,9	3,900	4,680	5,850	7,020	8,580	10,530	12,870	15,210	18,330	21,840	26,520	31,980	39,000
4,7	4,700	5,640	7,050	8,460	10,340	12,690	15,510	18,330	22,090	26,320	31,960	38,540	47,000
5,6	5,600	6,720	8,400	10,080	12,320	15,120	18,480	21,840	26,320	31,360	38,080	45,920	56,000
6,8	6,800	8,160	10,200	12,240	14,960	18,360	22,440	26,520	31,960	38,080	46,240	55,760	68,000
8,2	8,200	9,840	12,300	14,760	18,040	22,140	27,060	31,980	38,540	45,920	55,760	67,240	82,000

The look-up table gives the base value, but not the scale. We need to determine the multipliers for R and C .

We get one equation and two scale parameters. We need to choose one and calculate the other:

We can aim for $C_1 = 6.8\text{nF}$, i.e., $y = -9$:

In this case we will get a resistor
 $R_1 = 1.2 \times 10^5 \Omega = 120\text{k}\Omega$

Alternatively, we could use:

$$R_1 C_1 = 1.2 \times 10^x \cdot 6.8 \times 10^y = 0.816\text{ms}$$

$$1.2 \times 10^x \cdot 6.8 \times 10^y = 8.16 \times 10^{x+y} = 8.16 \times 10^{-4}$$

$$x + y = -4$$

$$x = -y - 4 = 9 - 4 = 5$$

$$R_1 = 120\text{k}\Omega, C_1 = 6.8\text{nF}$$

$$R_1 = 12\text{k}\Omega, C_1 = 68\text{nF}$$

Observation:

Picking an arbitrary value for C and then calculating the resistor value will not give the best approximation to the time constant and cut-off frequency. Using a look-up table is an easy approach to spot the best combination of R and C values. Look-up tables are easily constructed in Excel.

First order high pass filter

Gain equation for inverting amplifier.

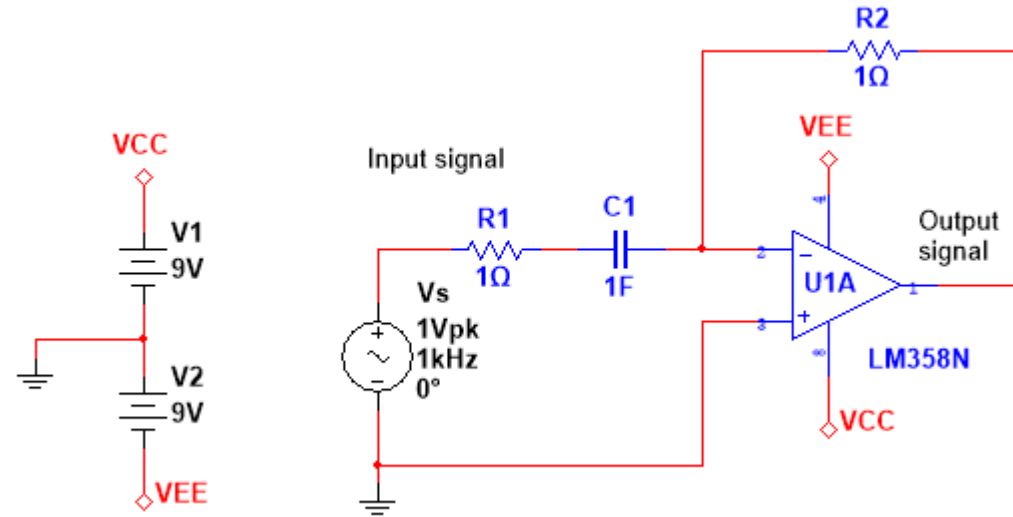
$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = -\frac{Z_2(\omega)}{Z_1(\omega)}$$

Component impedances:

$$Z_1(\omega) = R_1 + \frac{1}{j\omega C_1}$$

$$Z_2(\omega) = R_2$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = -\frac{R_2}{R_1 + \frac{1}{j\omega C_1}} = -\frac{j\omega R_2 C_1}{\underbrace{1 + j\omega R_1 C_1}_{\text{polynomial form}}}$$



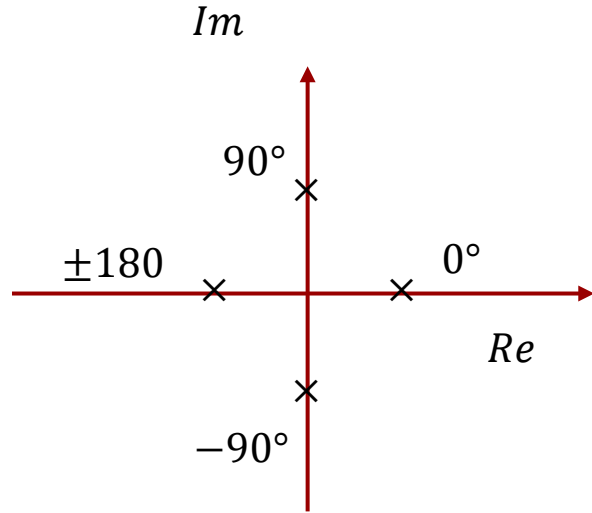
$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = -\frac{j\omega}{1 + j\omega} \quad \text{Component values inserted}$$

$$H_{LF\text{asymptote}}(\omega) = -\frac{j\omega}{1} = -j\omega = \omega e^{-j\pi/2}$$

$$H_{HF\text{asymptote}}(\omega) = -\frac{j\omega}{j\omega} = 1e^{-j\pi}$$

A highpass filter with a phase starting in -90 degrees and ending in -180 degrees.

First order high pass filter



$$z = x + jy$$

$$|z| = |x + jy| = \sqrt{x^2 + y^2}$$

$$\angle z = \tan^{-1} \frac{y}{x}$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = -\frac{j\omega}{1 + j\omega}$$

$$\angle H(\omega) = \underbrace{\angle P(\omega)}_{\text{numerator angle}} - \underbrace{\angle Q(\omega)}_{\text{denominator angle}}$$

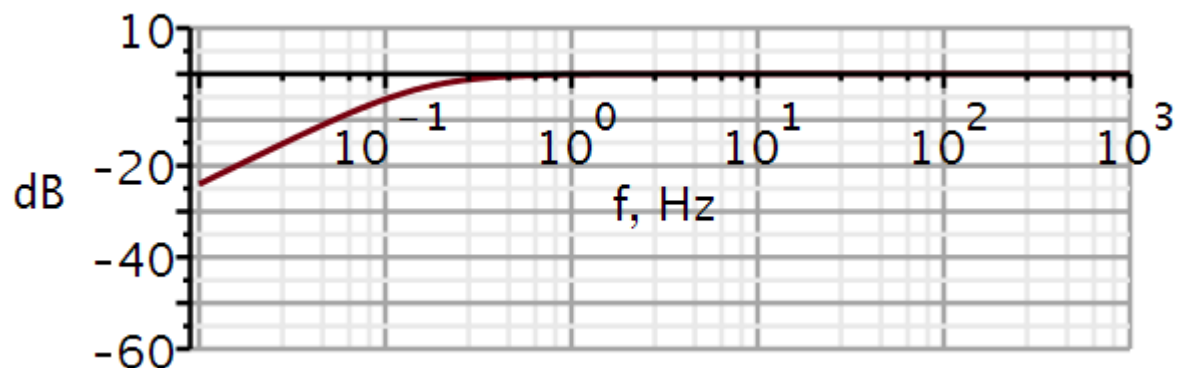
$$\theta(\omega) = \angle H(\omega) = -180^\circ + 90^\circ - \tan^{-1} \omega$$

$$\theta(0) = \angle H(0) = -90^\circ - \tan^{-1} 0 = -90^\circ$$

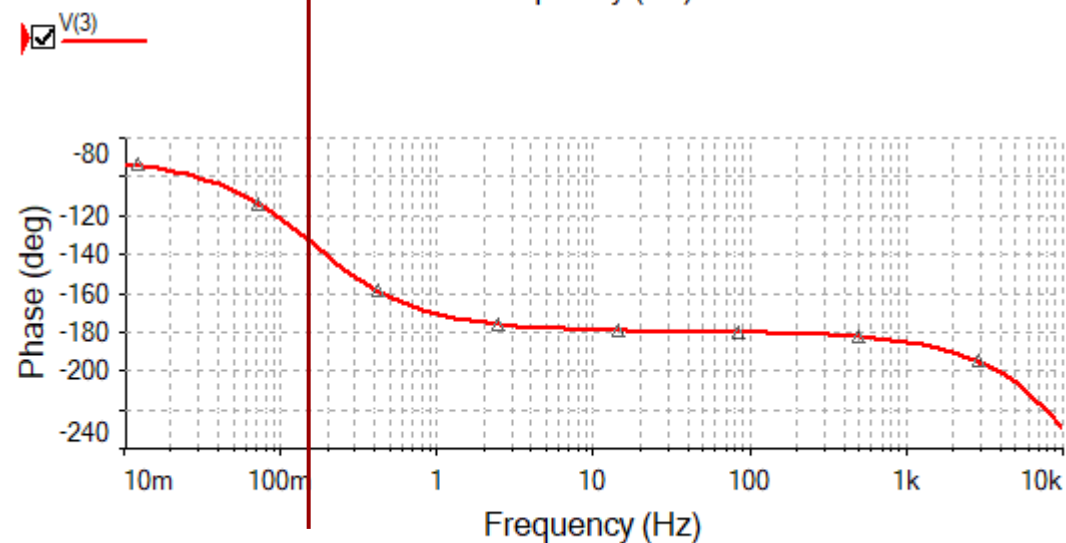
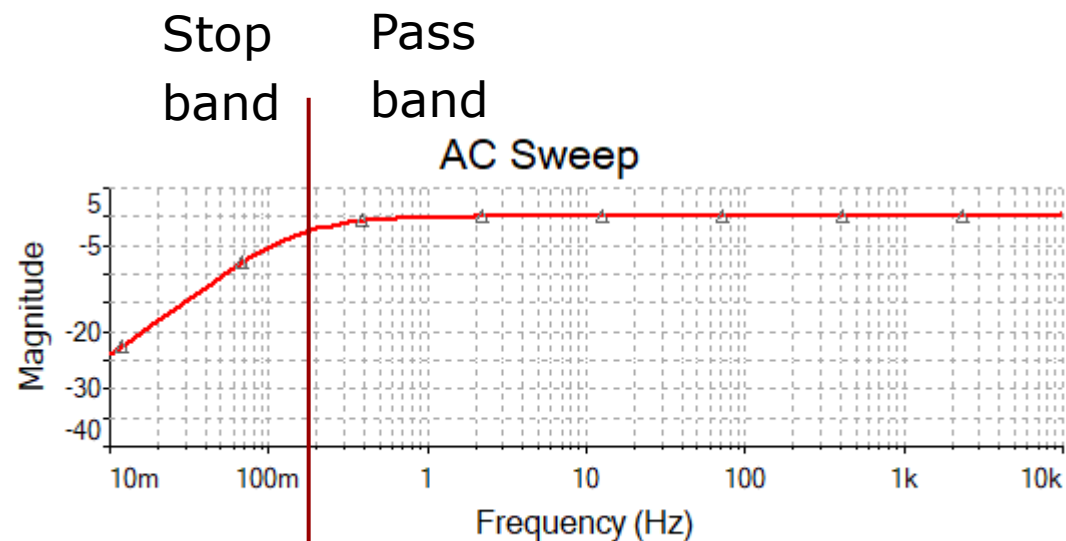
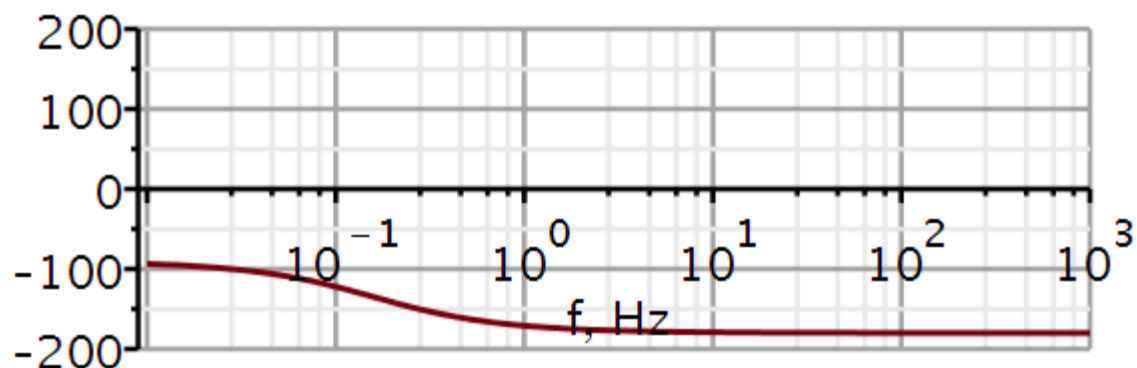
$$\theta(\infty) = \angle H(\infty) = -90^\circ - \tan^{-1} \infty = -90^\circ - 90^\circ = -180^\circ$$

High pass filter with normalized cut-off frequency

Amplitude characteristic



Phase characteristic



Let us now scale the cut-off frequency from 0.159 to 20 Hz. We can **scale the frequency characteristic** as follows:

$$\omega = K_F \cdot \omega_1$$

$$K_F = \frac{20\text{Hz}}{0.159\text{Hz}} = 125.8 \Rightarrow C_1 = \frac{1\text{F}}{125.8} = 7950 \mu\text{F}$$

We can also do an **impedance scaling**, where we scale all component with the same factor K_Z , but without changing time constants.

If we desire a compact capacitor, we would like it to be smaller than $1\mu\text{F}$.

$$H(\omega) = - \frac{j K_F \cdot \omega R_2 \frac{C_1}{K_F}}{1 + j K_F \cdot \omega R_1 \frac{C_1}{K_F}}$$

$$\tau = R \cdot C = K_Z R \cdot \frac{C}{K_Z} = 1\Omega \cdot 7950\mu\text{F} = 7.95\text{ms}$$

$$H(\omega) = - \frac{j (K_F \cdot \omega) (K_Z \cdot R_2) \frac{C_1}{K_F K_Z}}{1 + j (K_F \cdot \omega) (K_Z \cdot R_1) \frac{C_1}{K_F K_Z}}$$

Using a lookup table

Using a look-up table, we can spot the best combination of R_1 and C_1 . Here 8.160 is the number closest to 7.95.

$$\tau = R_1 C_1 = 1\Omega \cdot 7950\mu\text{F} = 7.95\text{ms (ideal)}$$

$$R_1 C_1 = 1.2 \times 10^x \cdot 6.8 \times 10^y = 8.16\text{ms}$$

This look-up table is made in Excel.

The numbers of interest are of-ten found in a narrow diago-nal.

$\tau = R \cdot C$

	E6	E12											
R\C	1	1,2	1,5	1,8	2,2	2,7	3,3	3,9	4,7	5,6	6,8	8,2	10
1	1,000	1,200	1,500	1,800	2,200	2,700	3,300	3,900	4,700	5,600	6,800	8,200	10,000
1,2	1,200	1,440	1,800	2,160	2,640	3,240	3,960	4,680	5,640	6,720	8,160	9,840	12,000
1,5	1,500	1,800	2,250	2,700	3,300	4,050	4,950	5,850	7,050	8,400	10,200	12,300	15,000
1,8	1,800	2,160	2,700	3,240	3,960	4,860	5,940	7,020	8,460	10,080	12,240	14,760	18,000
2,2	2,200	2,640	3,300	3,960	4,840	5,940	7,260	8,580	10,340	12,320	14,960	18,040	22,000
2,7	2,700	3,240	4,050	4,860	5,940	7,290	8,910	10,530	12,690	15,120	18,360	22,140	27,000
3,3	3,300	3,960	4,950	5,940	7,260	8,910	10,890	12,870	15,510	18,480	22,440	27,060	33,000
3,9	3,900	4,680	5,850	7,020	8,580	10,530	12,870	15,210	18,330	21,840	26,520	31,980	39,000
4,7	4,700	5,640	7,050	8,460	10,340	12,690	15,510	18,330	22,090	26,320	31,960	38,540	47,000
5,6	5,600	6,720	8,400	10,080	12,320	15,120	18,480	21,840	26,320	31,360	38,080	45,920	56,000
6,8	6,800	8,160	10,200	12,240	14,960	18,360	22,440	26,520	31,960	38,080	46,240	55,760	68,000
8,2	8,200	9,840	12,300	14,760	18,040	22,140	27,060	31,980	38,540	45,920	55,760	67,240	82,000

Let us aim for a capacitor of 680nF.

So again, we let the impedance scaling factor K_Z be decided from a desired magnitude of the capacitor value.

Then the resistor values are determined based on this.

$$C_1'' = \frac{C_1'}{K_Z} = \frac{C_1}{K_F K_Z}$$

$$K_Z = \frac{7950 \mu\text{F}}{680 \text{ nF}} = 11691$$

$$R_1 = 11691 \cdot 1\Omega \approx 12\text{k}\Omega$$

$$R_2 = 11691 \cdot 1\Omega \approx 12\text{k}\Omega$$

$$C_1 = 680\text{nF}$$

C values

E6:

10

15

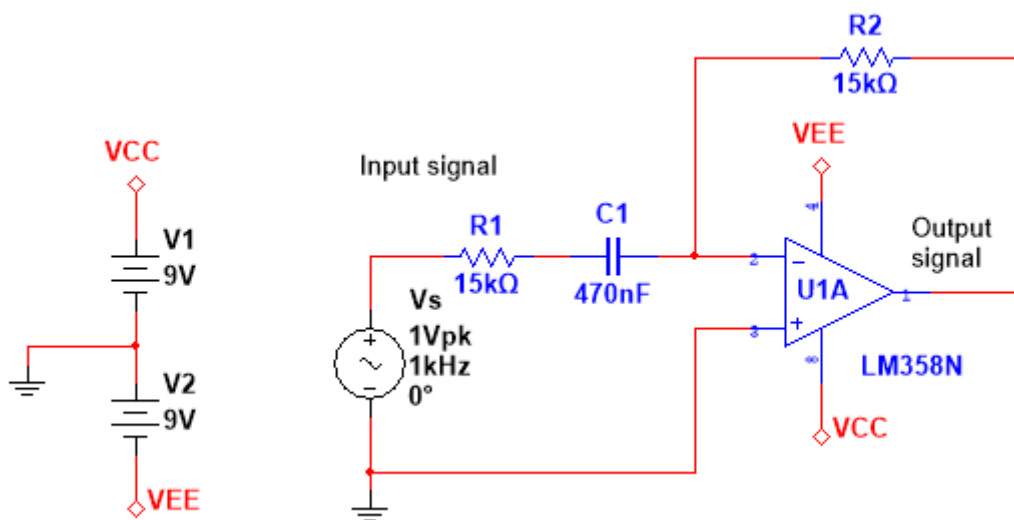
22

33

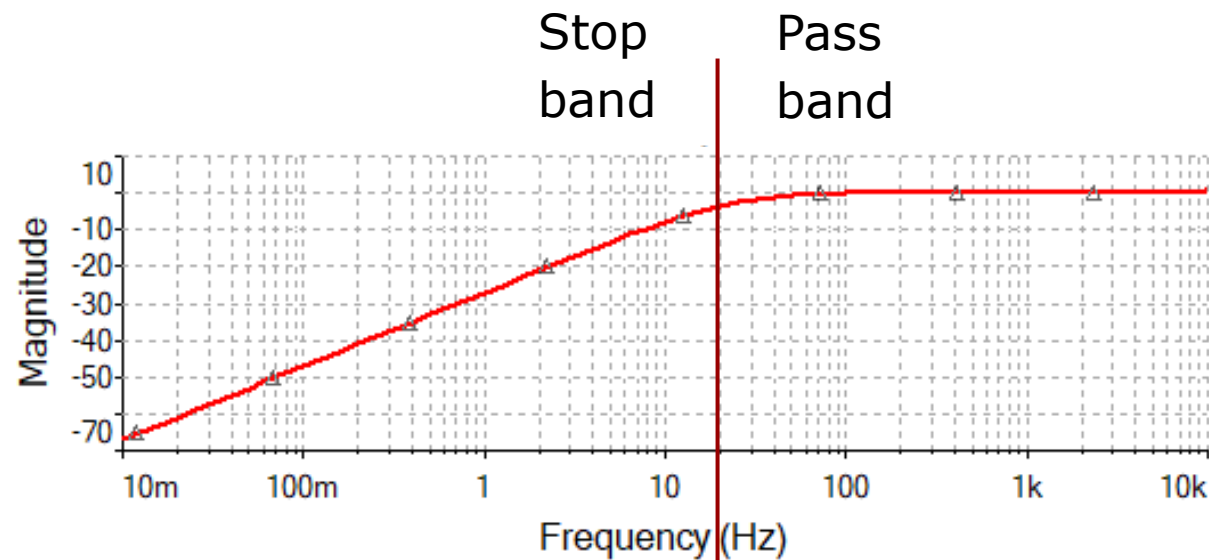
47

68

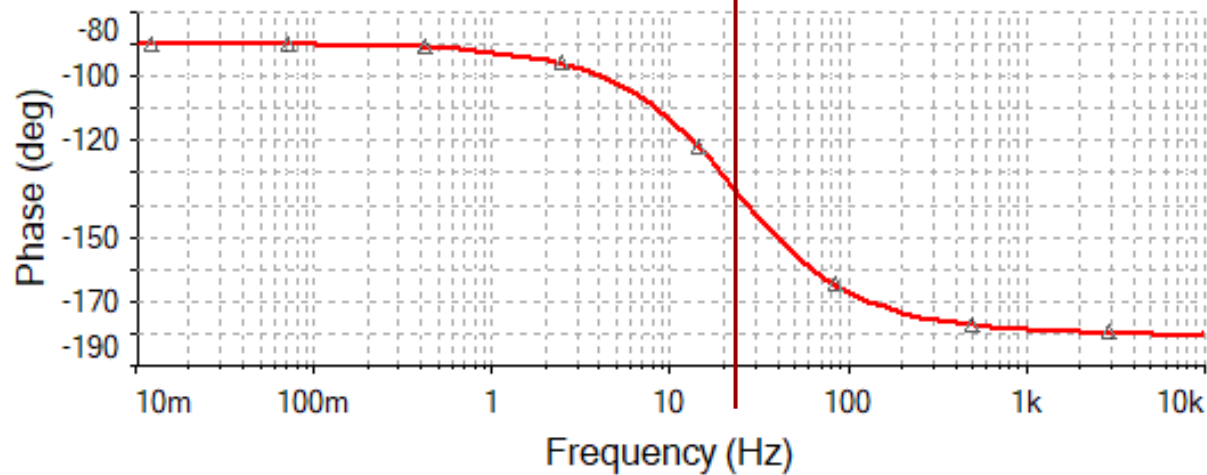
Scaled high pass filter



Simulation in Multisim

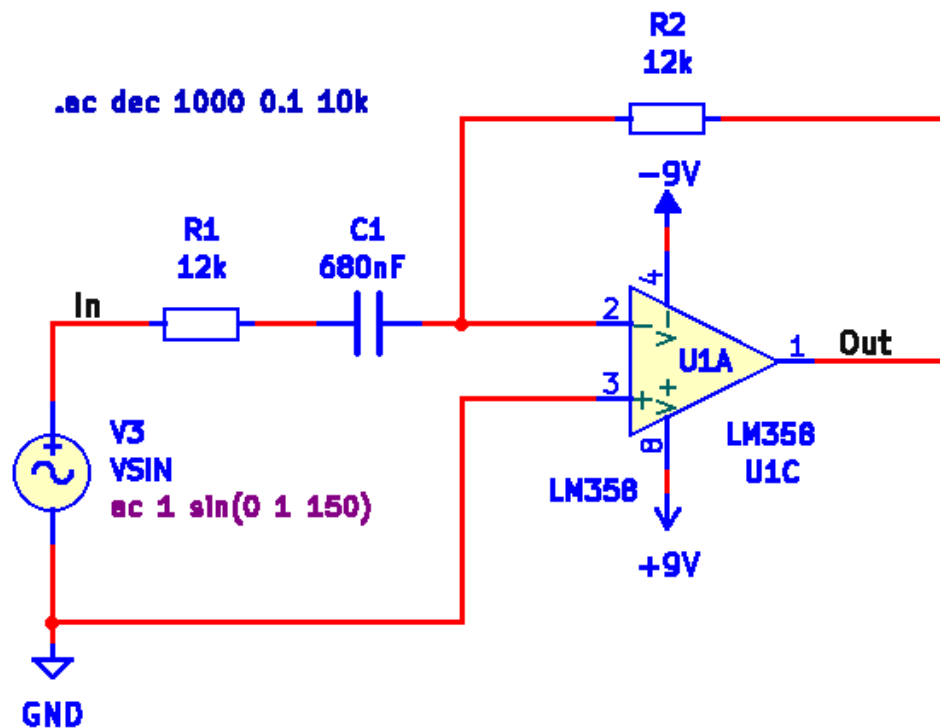


V(3)

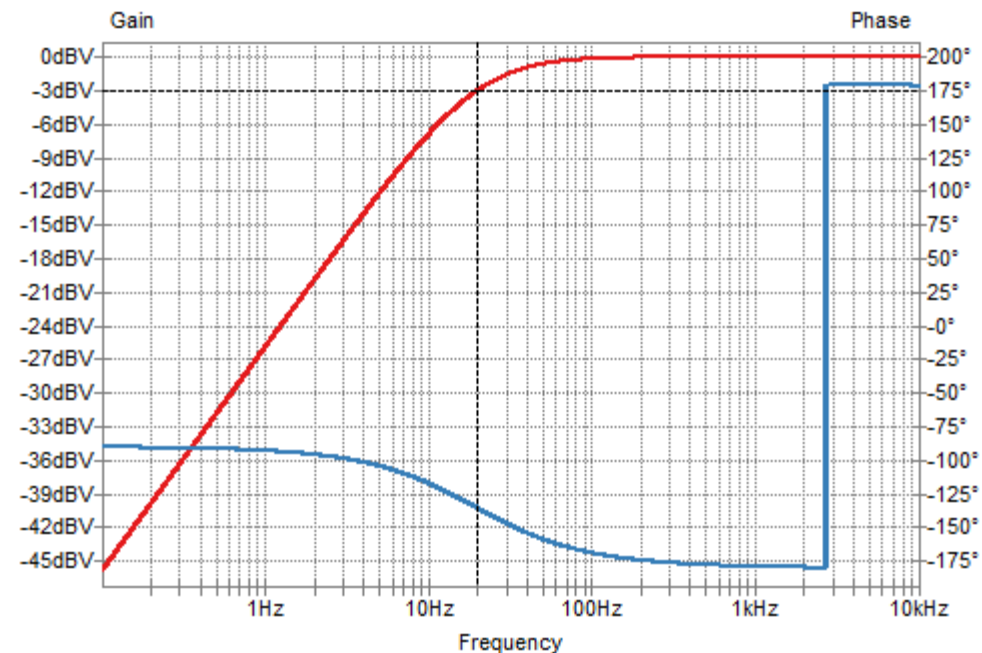


V(3)

Scaled high pass filter



Simulation in KiCad

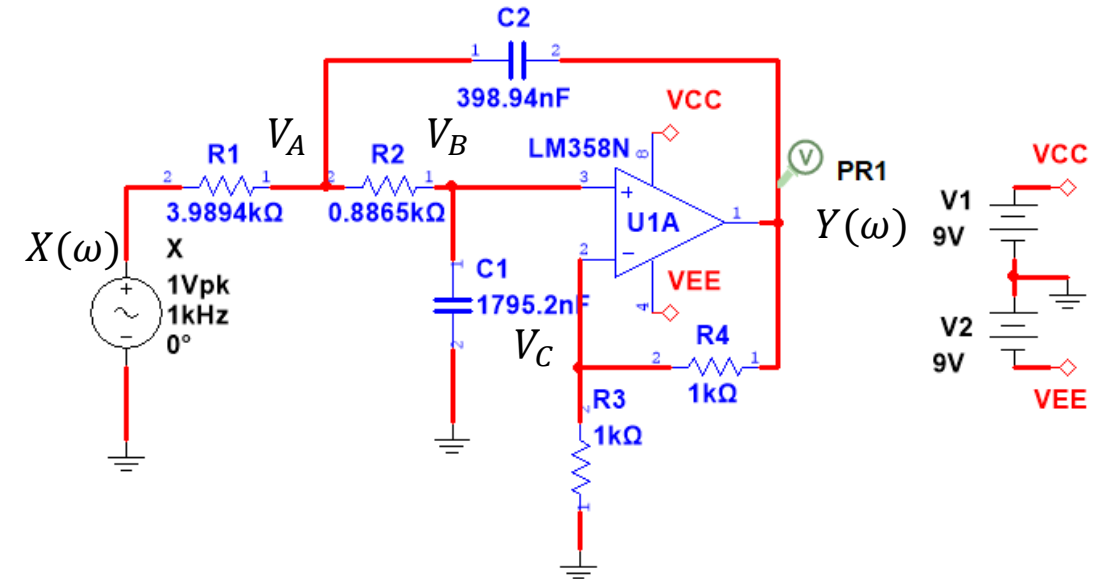


Problems

Video 3

A Sallen-Key filter is shown here. You are asked to analyse this circuit and:

1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.



C values

E6:

10

15

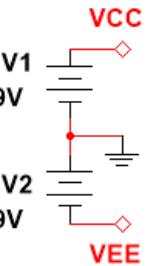
22

33

47

68

1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.

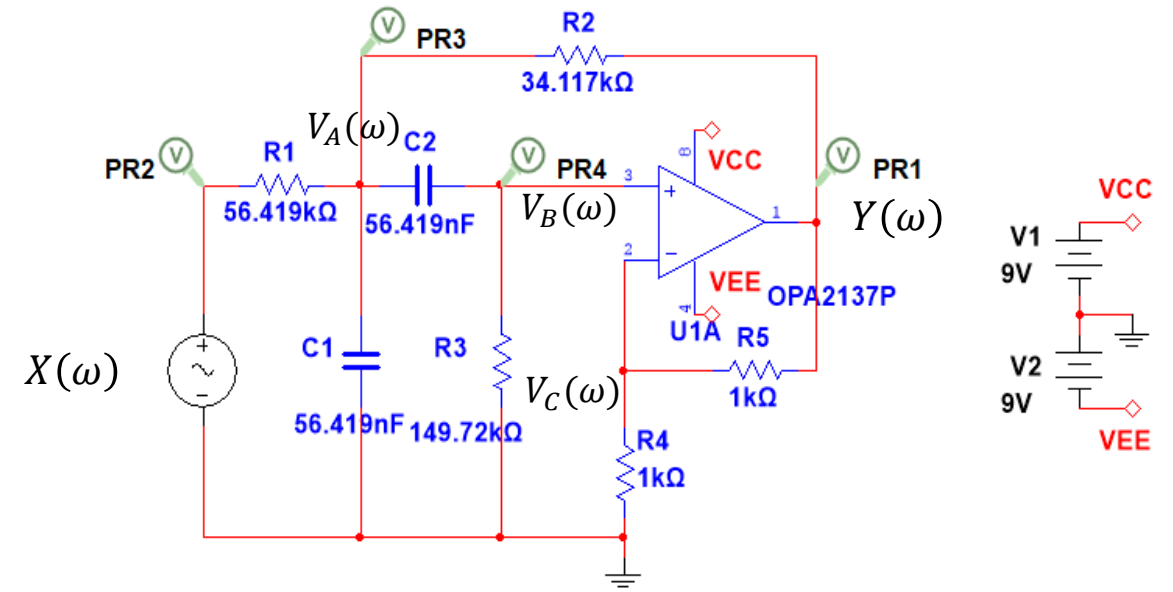


68

Filter 12 – Bandpass filter

A Sallen-Key filter is shown here. You are asked to analyse this circuit and:

1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.

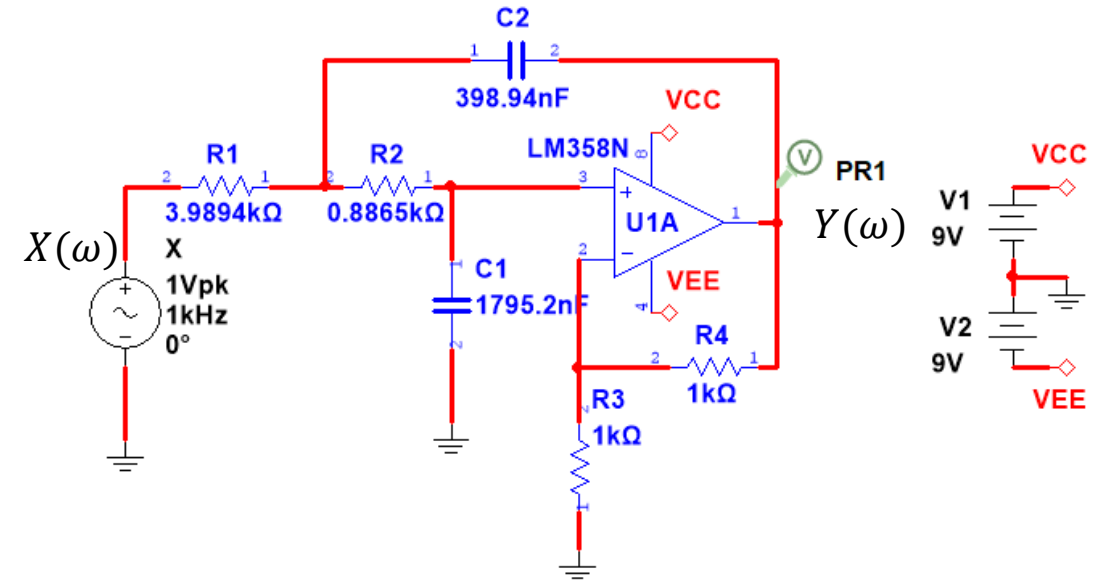


C values
E6:
10
15
22
33
47
68

Solutions

A Sallen-Key filter is shown here. You are asked to analyse this circuit and:

1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.



C values

E6:

10

15

22

33

47

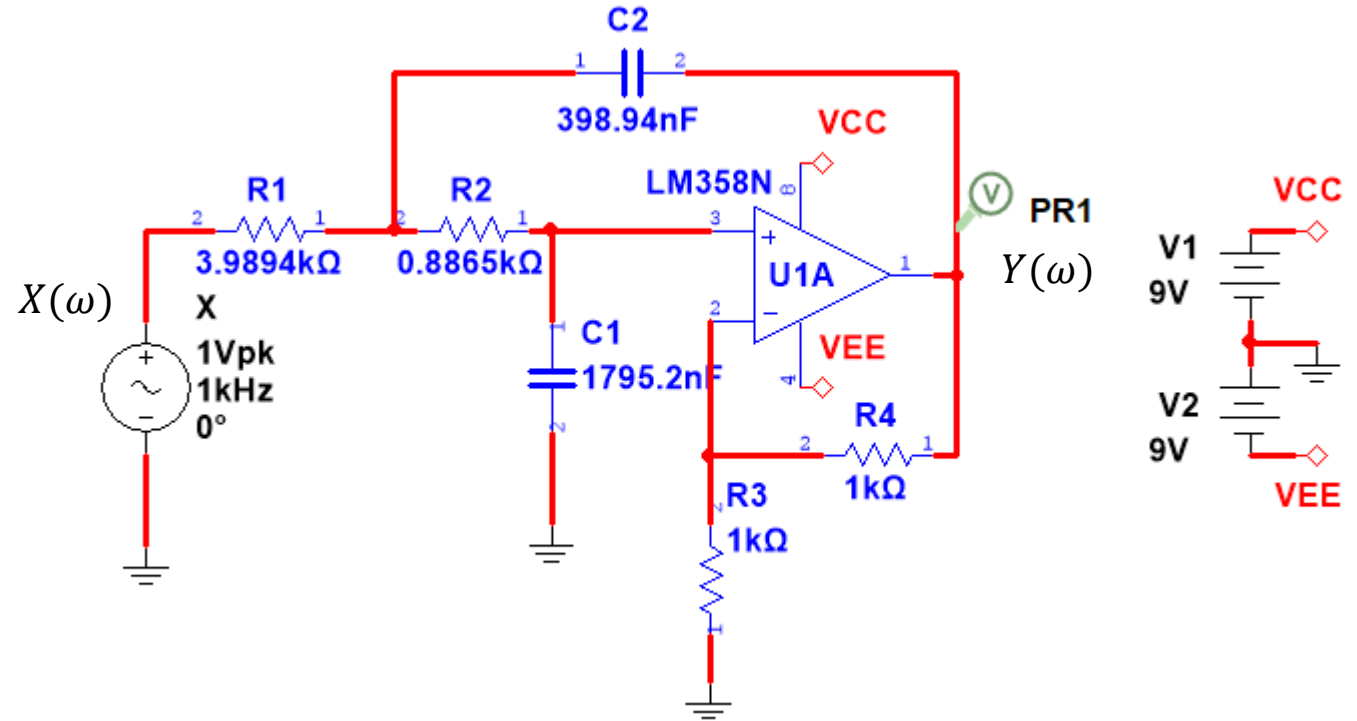
68

Filter 10 – Lowpass filter (sol)

We now want to analyse the Sallen-Key filter from a frequency domain perspective.

We want to:

1. Obtain the equation for the frequency characteristic.
2. Check the values of the equation coefficients.
3. Obtain the differential equation from the frequency characteristic
4. Predict the main features of the amplitude and phase characteristics
5. Plot the amplitude and phase characteristics in Maple/Python.



Filter 10 – Lowpass filter (sol)

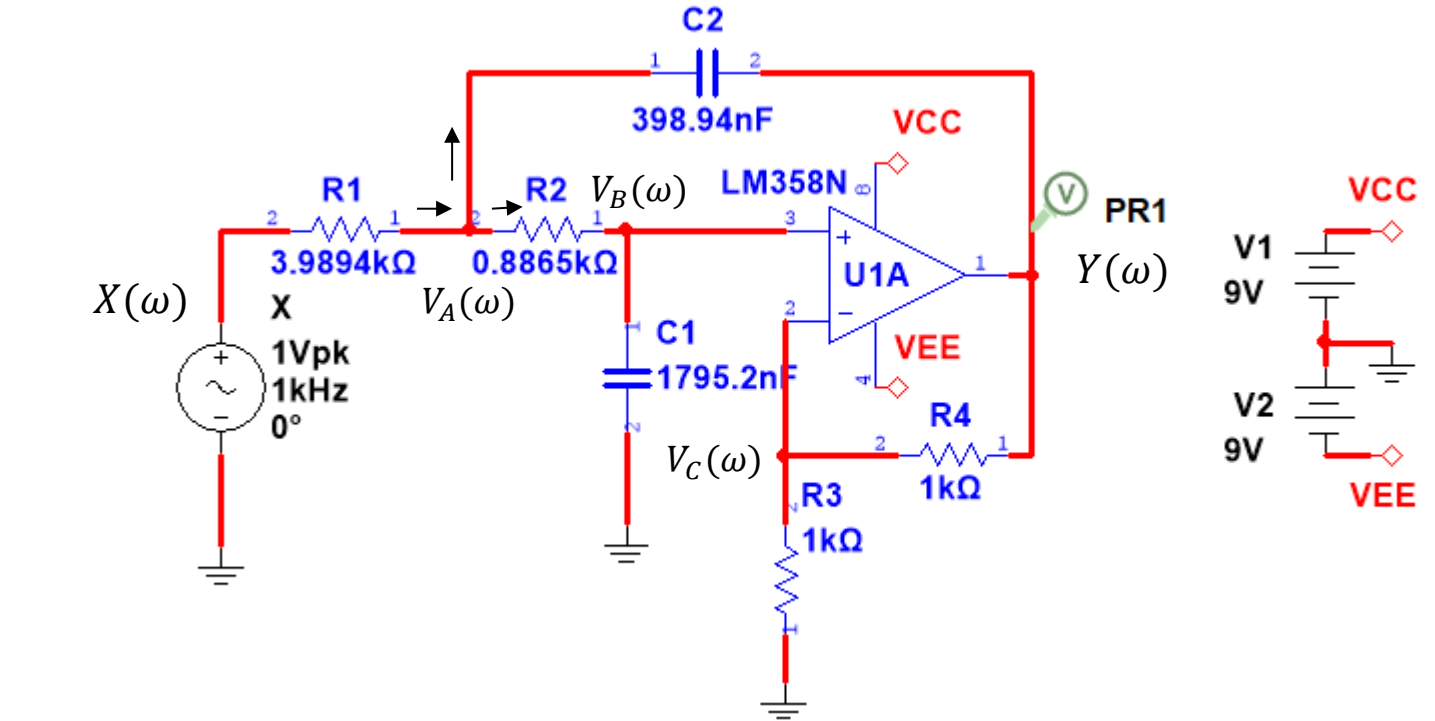
Nodal symbols have been added.
We will write the **frequency domain** equations using impedances and nodal analysis.

$$A: -\frac{(X - V_A)}{R_1} + \frac{(V_A - Y)}{Z_{C2}} + \frac{(V_A - V_B)}{R_2} = 0$$

$$B: -\frac{(V_A - V_B)}{R_2} + \frac{V_B}{Z_{C1}} = 0$$

Constraint:

$$V_B = V_C = \frac{R_3}{R_3 + R_4} \cdot Y = \frac{1}{K} \cdot Y$$



All variables are Fourier transformed with ω as their argument.
Hence, we use capital letters.

Constraint:

$$V_B = \frac{1}{K} \cdot Y$$

Equations:

$$A: -\frac{(X - V_A)}{R_1} + \frac{(V_A - Y)}{Z_{C2}} + \frac{(V_A - V_B)}{R_2} = 0$$

$$B: -\frac{(V_A - V_B)}{R_2} + \frac{V_B}{Z_{C1}} = 0$$

Writing the frequency domain equations in Maple:

Constraints

$$VB := \frac{1}{K} \cdot Y :$$

Equations

$$eqA1 := -\frac{(X - VA)}{R1} + \frac{(VA - Y)}{ZC2} + \frac{(VA - VB)}{R2} = 0 :$$

$$eqB1 := -\frac{(VA - VB)}{R2} + \frac{VB}{ZC1} = 0 :$$

Solving for a symbolic expression for Y :

Solve

```
solutions := simplify(solve( {eqA1, eqB1}, [VA, Y], symbolic = true)) :
assign(solutions)
simplify(Y)
```

$$- \frac{K X ZC1 ZC2}{((K - 1) ZC1 - ZC2 - R2) R1 - ZC2 (R2 + ZC1)}$$

Impedances

Defining the impedances:

$$ZC1 := \frac{1}{j \cdot \omega \cdot C1} \quad ; \quad ZC2 := \frac{1}{j \cdot \omega \cdot C2} \quad ;$$

Obtaining V_o expressed in terms of the components:

```
simplify(Y)
```

$$- \frac{K X}{-1 + C1 C2 R1 R2 \omega^2 + ((-1 C1 + (1 K - 1) C2) R1 - 1 C1 R2) \omega}$$

Constructing a function for $Y(\omega)$:

$$Y := \omega \rightarrow \frac{K}{R1 \cdot R2 \cdot C1 \cdot C2 (j\omega)^2 + (j\omega) (R1 \cdot C1 + R2 \cdot C1 + (1 - K) R1 \cdot C2) + 1} \cdot X :$$

Observation:

The gain K is part of the coefficient to $j\omega$. This means that the gain interferes with the cut-off frequency.

Changing the gain resistors will also change the cut-off frequency.

For this reason, the Sallen-Key filter is often used with $K = 1$. In such cases, we build the filter around a unity gain op-amp circuit.

Transfer function

$$H := \omega \rightarrow \frac{\frac{K}{R1 \cdot R2 \cdot C1 \cdot C2}}{(j\omega)^2 + (j\omega) \cdot \left(\frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} + \frac{(1-K)}{R2 \cdot C1} \right) + \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}} :$$

Choice of components

$$\begin{aligned} C1 &:= 1795.2\text{E}-9 ;; C2 := 398.94\text{E}-9 ;; R1 := 3.9894\text{E}3 ;; R2 := 0.8865\text{E}3 ;; R3 := 1\text{E}3 ;; \\ R4 &:= 1\text{E}3 ;; \\ K &:= 1 + \frac{R4}{R3} : \end{aligned}$$

Alternative form:

$$H(\omega) = \frac{K}{(j\omega)^2 R_1 R_2 C_1 C_2 + j\omega (C_1 (R_1 + R_2) + R_1 C_2 (1 - K)) + 1}$$

Filter table

		Lowpass	Highpass	Bandpass
R_1	$k\Omega$	3.9894	1784.1	56.419
R_2	$k\Omega$	0.8865	892.06	34.117
R_3	$k\Omega$	1.0	1.0	149.72
R_4	$k\Omega$	1.0	1.0	1.0
R_5	$k\Omega$	–	–	1.0
C_1	nF	1795.2	1784.1	56.419
C_2	nF	398.94	3568.2	56.419
a_1		$2.83 \cdot 10^3$	$6.283 \cdot 10^{-1}$	$3.141 \cdot 10^1$
a_0		$3.95 \cdot 10^5$	$9.87 \cdot 10^{-2}$	$9.87 \cdot 10^4$
b_2		0	2	0
b_1		0	0	$6.283 \cdot 10^2$
b_0		$7.89 \cdot 10^5$	0	0

Coefficient values

$$a1 := \frac{1}{R2 \cdot C2} + \frac{1}{R1 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1}$$

$$a1 := 2827.537902$$

$$a0 := \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$a0 := 394814.6451$$

$$b0 := \frac{K}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$b0 := 789629.2901$$

The coefficients agree with what was obtained in the time domain.

We readily obtain the differential equation from the frequency characteristic:

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{\frac{K}{R_1 R_2 C_1 C_2}}{(j\omega)^2 + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{1-K}{R_2 C_1} \right) (j\omega) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\left((j\omega)^2 + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{1-K}{R_2 C_1} \right) (j\omega) + \frac{1}{R_1 R_2 C_1 C_2} \right) Y(\omega) = \frac{K}{R_1 R_2 C_1 C_2} X(\omega)$$

Inverse Fourier transforming each term:

$$(j\omega)^2 Y(\omega) + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{1-K}{R_2 C_1} \right) (j\omega) Y(\omega) + \frac{1}{R_1 R_2 C_1 C_2} Y(\omega) = \frac{K}{R_1 R_2 C_1 C_2} X(\omega)$$

$$\ddot{y}(t) + \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{1-K}{R_2 C_1} \right) \dot{y}(t) + \frac{1}{R_1 R_2 C_1 C_2} y(t) = \frac{K}{R_1 R_2 C_1 C_2} x(t)$$

Before plotting the magnitude and phase of $H(\omega)$ it is helpful to have an idea about what to expect. We can get that from low and high frequency asymptotes.

$$H(\omega) = \frac{K}{(j\omega)^2 R_1 R_2 C_1 C_2 + j\omega (C_1(R_1 + R_2) + R_1 C_2(1 - K)) + 1}$$

Low frequency (LF) asymptote:

$$\lim_{\omega \rightarrow 0} H(\omega) = H(\omega_{LF}) = \frac{K}{1} = K$$

High frequency (HF) asymptote:

$$\lim_{\omega \rightarrow \infty} H(\omega) = H(\omega_{HF}) = \frac{K}{(j\omega)^2 R_1 R_2 C_1 C_2} = -\frac{K}{R_1 R_2 C_1 C_2} \cdot \frac{1}{\omega^2} \rightarrow 0$$

The asymptotes tell us that the filter has constant gain at low frequencies and that the gain goes to zero for high frequency. So, we have a low pass filter.

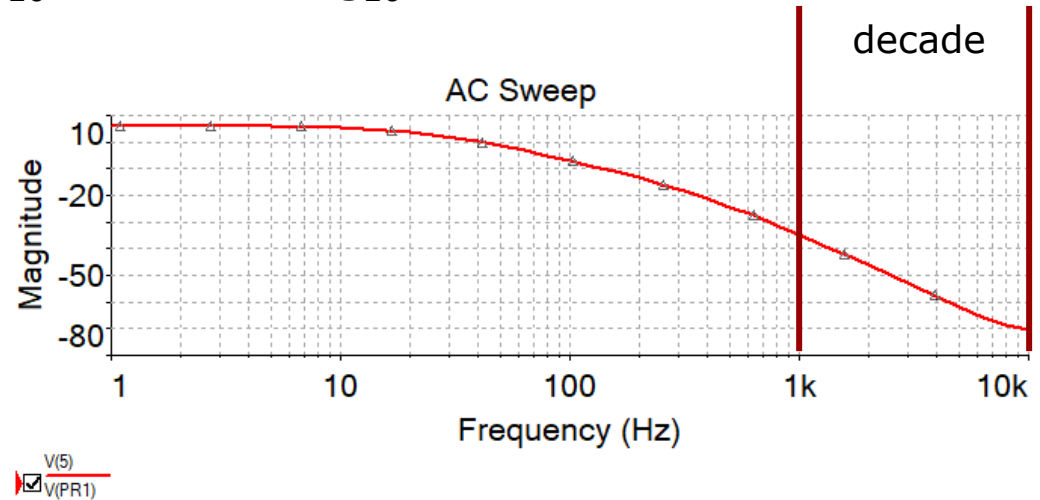
When interpreting the limit of an expression, we include only the term with dominant magnitude and consider the contributions from the other terms to be insignificant. So, for the denominator, we only include the 2nd order term for the high frequency asymptote, while to the low frequency asymptote the constant term will dominate.

Filter 10 – Lowpass filter (sol)

We can also predict the order of the filter. It is revealed by the slope of the declining asymptote, but we need to plot the amplitude spectrum in decibels. A first order filter has a slope of 20 dB per decade frequency change. A 2nd order filter a slope of 40 dB/decade, a 3rd order filter a slope of 60 dB/decade, and so on.

DC gain in decibels:

$$|H(\omega_{LF})|_{dB} = 20\text{dB} \cdot \log_{10} K = 20\text{dB} \cdot \log_{10} 2 = 6\text{dB}$$



HF slope:

$$\begin{aligned}
 |H(10 \cdot \omega_{HF})|_{dB} - |H(\omega_{HF})|_{dB} &= 20\text{dB} \cdot \log_{10} \frac{K}{R_1 R_2 C_1 C_2} \cdot \frac{1}{(10 \cdot \omega)^2} - 20\text{dB} \cdot \log_{10} \frac{K}{R_1 R_2 C_1 C_2} \cdot \frac{1}{(\omega)^2} \\
 &= 20\text{dB} \cdot \log_{10} \frac{(\omega)^2}{(10 \cdot \omega)^2} = 20\text{dB} \cdot \log_{10} \frac{1}{(10)^2} = 20\text{dB} \cdot \log_{10} 10^{-2} \\
 &= -2 \cdot 20\text{dB} \cdot \log_{10} 10 = -40\text{dB/decade}
 \end{aligned}$$

Phase characteristic:

$$H(\omega_{LF}) = \frac{K}{1} = K$$

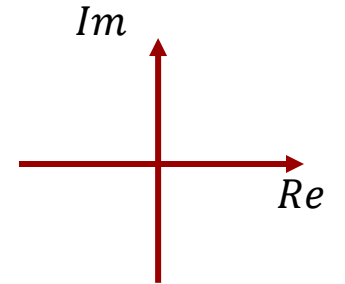
$$H(\omega_{HF}) = \frac{K}{(j\omega)^2 R_1 R_2 C_1 C_2} = -\frac{K}{R_1 R_2 C_1 C_2} \cdot \frac{1}{\omega^2}$$

Low frequency (LF) asymptote:

$$\angle H(\omega_{LF}) = \tan^{-1} \frac{0}{K} = 0^\circ$$

High frequency (HF) asymptote:

$$\angle H(\omega_{HF}) = \pm 180^\circ$$



Tricky cases of $\tan^{-1}()$:

$$z = x + jy$$

$$\theta = \tan^{-1} \frac{\text{Im}\{z\}}{\text{Re}\{z\}}$$

If $\text{Im}\{z\} = 0$, then if $\text{Re}\{z\} > 0 \Rightarrow \theta = 0$

If $\text{Im}\{z\} = 0$, then if $\text{Re}\{z\} < 0 \Rightarrow \theta = \pm 180^\circ$

If $\text{Re}\{z\} = 0$, then if $\text{Im}\{z\} > 0 \Rightarrow \theta = 90^\circ$

If $\text{Re}\{z\} = 0$, then if $\text{Im}\{z\} < 0 \Rightarrow \theta = -90^\circ$

A few commands to standardize plots.

Magnitude plots are in decibel with logarithmic frequency axis in Hz.

Phase plots are in degrees with logarithmic frequency axes in Hz.

Using display with *aligncolumns* will align the y-axes for better comparison of magnitude and phase plots.

There will be some annoying table borders. They can be removed using the interactive Table tools palette.

$$dB := \omega \rightarrow 20 \cdot \log_{10}(|H(\omega)|) :$$

$$angle := \omega \rightarrow \arg(H(\omega)) \cdot \frac{180}{\pi} :$$

Plotting solution

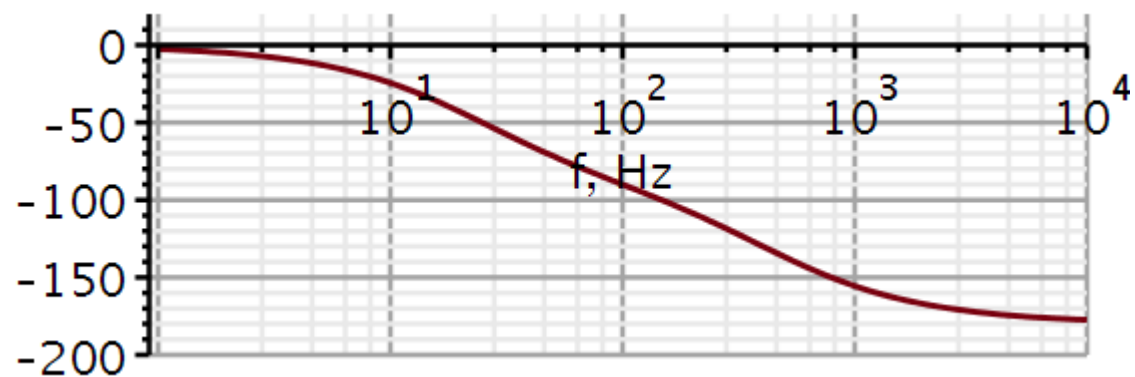
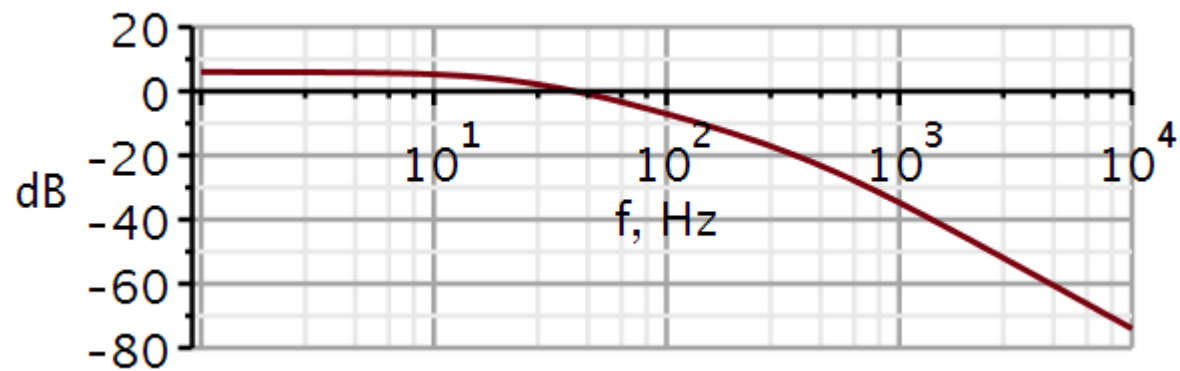
```
p1 := semilogplot(dB(2·π·f), f=1..1E5, -80..20, thickness
= 3, font = [Helvetica, roman, 18], axis[2] = [thickness
= 2.5], axis[1] = [thickness = 2.5], labels = ["f, Hz",
"dB"], labelfont = [Helvetica, 18], numpoints = 100,
gridlines, size = [600, 200]) :
```

```
p2 := semilogplot(angle(2·π·f), f=1..1E5, -200..200,
thickness = 3, font = [Helvetica, roman, 18], axis[2]
= [thickness = 2.5], axis[1] = [thickness = 2.5], labels
= ["f, Hz", " "], labelfont = [Helvetica, 18], numpoints
= 100, gridlines, size = [600, 200]) :
```

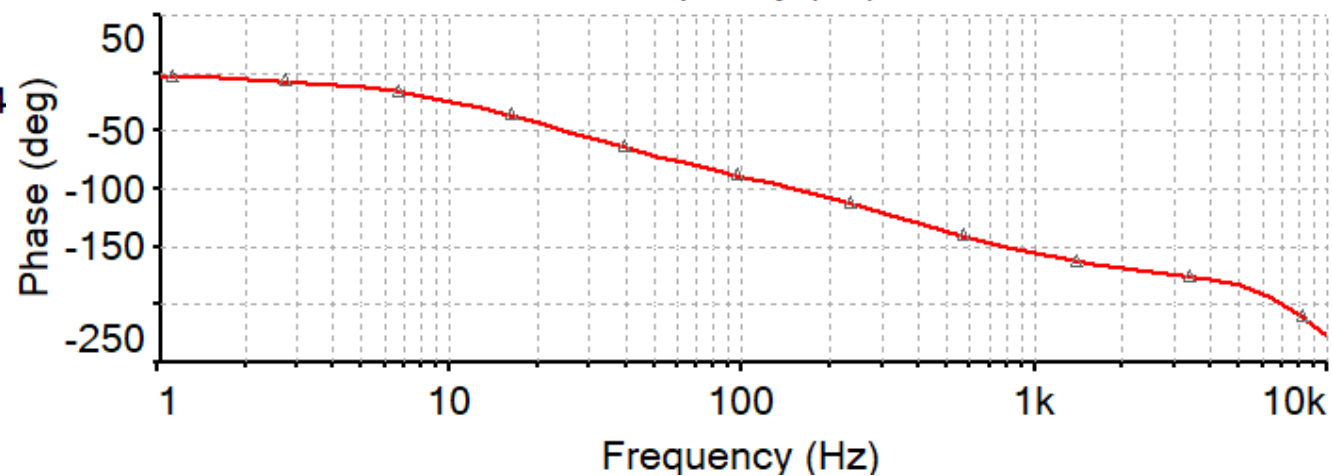
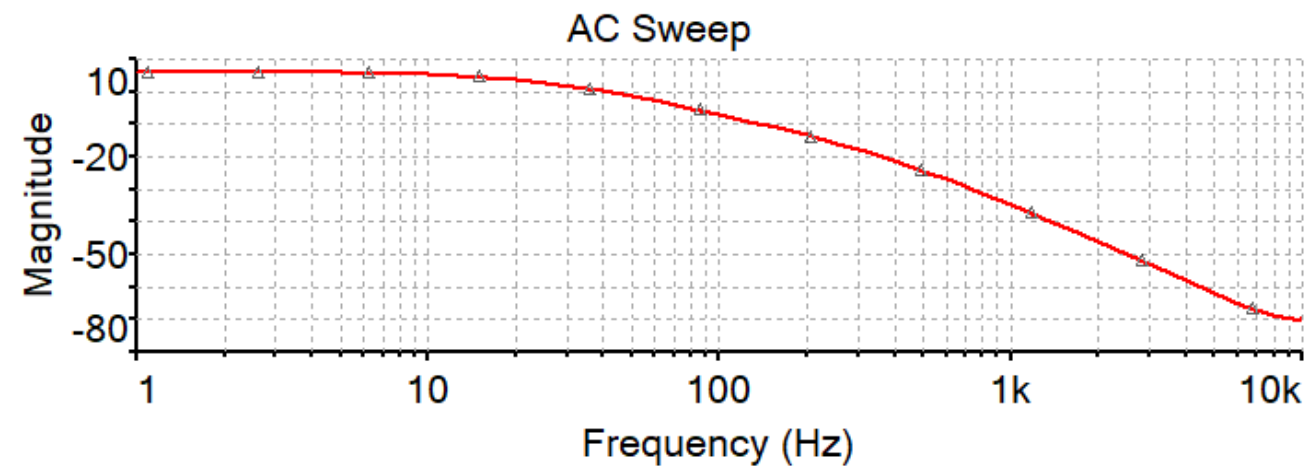
```
plotstack := Array(1..2, 1..1) :
plotstack[1, 1] := p1 :
plotstack[2, 1] := p2 :
display(plotstack, aligncolumns = [1])
```

Filter 10 – Lowpass filter (sol)

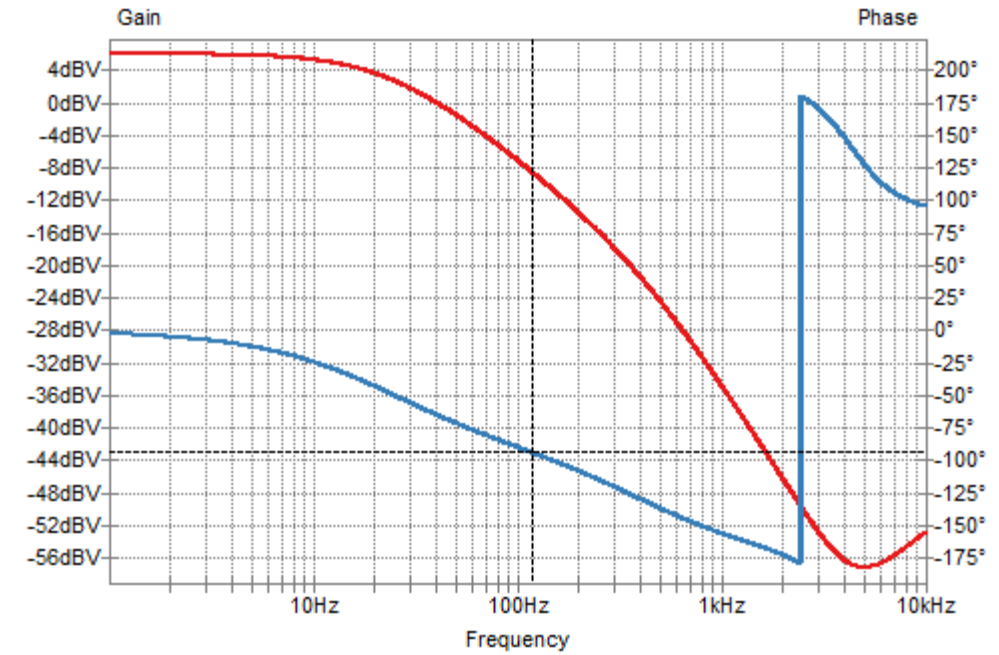
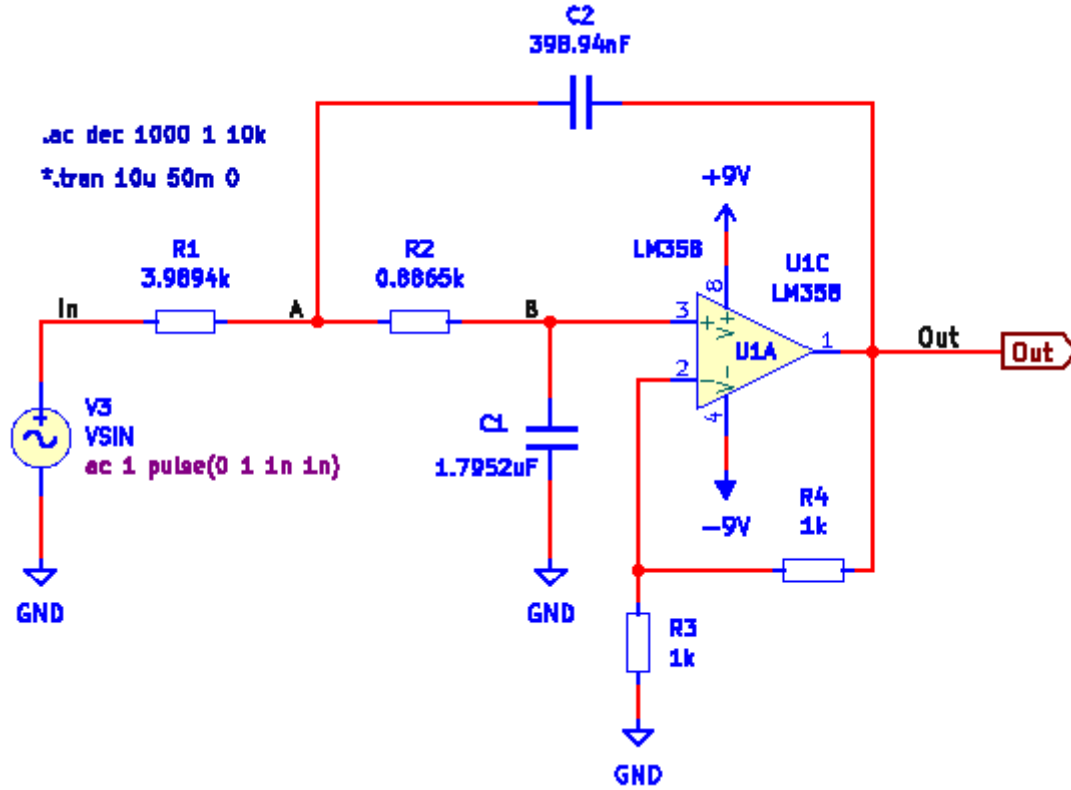
Maple



Multisim



Filter 10 – Lowpass filter (sol)



We can derive a general expression for the 3-dB cut-off frequency for a 2nd order lowpass filter.

$$|H(j\omega_{3dB})| = \frac{|b_0|}{|(j\omega_{3dB})^2 + a_1(j\omega_{3dB}) + a_0|} = \frac{1}{\sqrt{2}} \cdot |H(j\omega_{LF})|$$

$$|H(j\omega_{3dB})| = \frac{|b_0|}{|a_0| \sqrt{\left(1 - \frac{\omega_{3dB}^2}{a_0}\right)^2 + \left(\frac{a_1}{a_0} \omega_{3dB}\right)^2}} = \frac{1}{\sqrt{2}} \cdot \frac{|b_0|}{|a_0|}$$

$$\left(1 - \frac{\omega_{3dB}^2}{a_0}\right)^2 + \left(\frac{a_1}{a_0} \omega_{3dB}\right)^2 - 2 = 0$$

$$a_0^2 - 2a_0\omega_{3dB}^2 + \omega_{3dB}^4 + a_1^2\omega_{3dB}^2 - 2a_0^2 = 0$$

$$\omega_{3dB}^4 + (a_1^2 - 2a_0)\omega_{3dB}^2 - a_0^2 = 0$$

Filter 10 – Lowpass filter (sol)

$$\omega_{3dB}^4 + (a_1^2 - 2a_0)\omega_{3dB}^2 - a_0^2 = 0$$

$$y = \omega^2$$

$$y^2 + (a_1^2 - 2a_0)y - a_0^2 = 0$$

$$y = \frac{-(a_1^2 - 2a_0) \pm \sqrt{(a_1^2 - 2a_0)^2 - 4(-a_0^2)}}{2}$$

$$y = \frac{-(a_1^2 - 2a_0) \pm \sqrt{a_1^4 - 4a_0a_1^2 + 8a_0^2}}{2}$$

2nd order lowpass filter

$$a_0 = 1; a_1 = \sqrt{2}$$

Cut-off frequencies

$$H := \omega \rightarrow \frac{b0}{(j \cdot \omega)^2 + a1 \cdot (j \cdot \omega) + a0} :$$

$$\text{solve}(y^2 + (a1^2 - 2 \cdot a0) \cdot y - a0^2 = 0, y)$$

$$-\frac{a1^2}{2} + a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}, -\frac{a1^2}{2} + a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}$$

$$y1 := -\frac{a1^2}{2} + a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} :$$

$$y2 := -\frac{a1^2}{2} + a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} :$$

$$a0 := 1 ;; a1 := \sqrt{2} ;; b0 := 1 :$$

$$\text{evalf}(y1)$$

$$1.000000000$$

$$\text{evalf}(y2)$$

$$-1.000000000$$

$$x1 := \text{evalf}(\sqrt{y1})$$

$$x1 := 1.$$

Filter 10 – Lowpass filter (sol)

Same filter, but different components:

restart

$$y1 := -\frac{a1^2}{2} + a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} ;$$

$$y2 := -\frac{a1^2}{2} + a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} ;$$

$$C1 := 33 \cdot 10^{-9} ; C2 := 68 \cdot 10^{-9} ; R1 := 22000 ; R2 := 22000 ; K := 1 ;$$

$$a1 := \frac{1}{R1 \cdot C2} + \frac{1}{R2 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1}$$

$$a1 := \frac{250000}{187}$$

$$a0 := \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$a0 := \frac{62500000000}{67881}$$

$$b0 := \frac{K}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$b0 := \frac{62500000000}{67881}$$

$$x1 := \text{evalf}(\sqrt{y1})$$

$$x1 := 973.7593838$$

$$x1$$

$$973.7593838$$

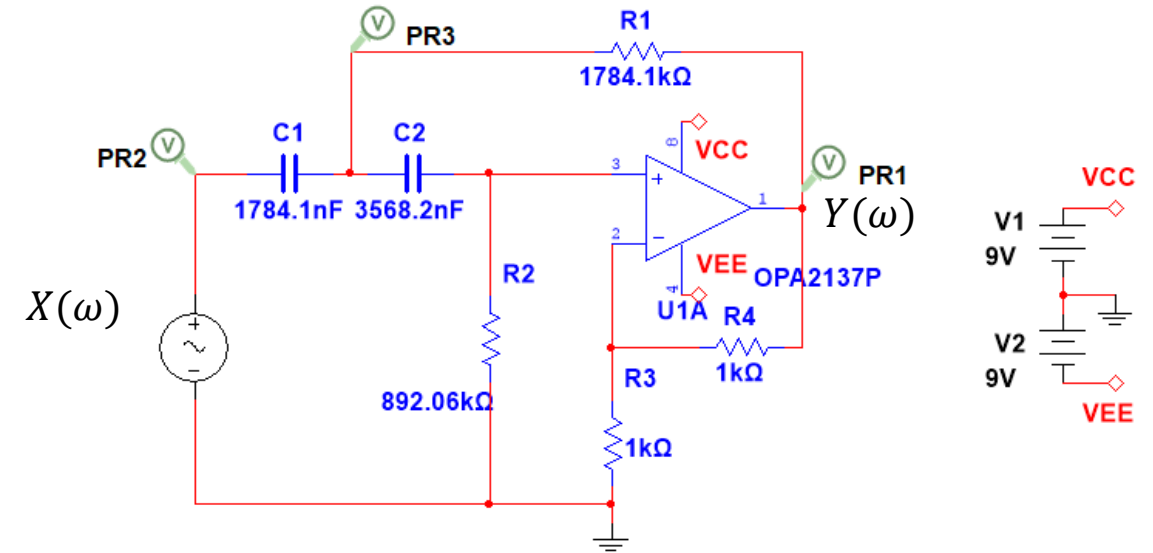
$$\frac{x1}{2 \cdot \pi}$$

$$154.9786193$$

Filter 11 – Highpass filter

A Sallen-Key filter is shown here. You are asked to analyse this circuit and:

1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.



C values
E6:
10
15
22
33
47
68

Filter 11 – Highpass filter (sol)

Nodal equations:

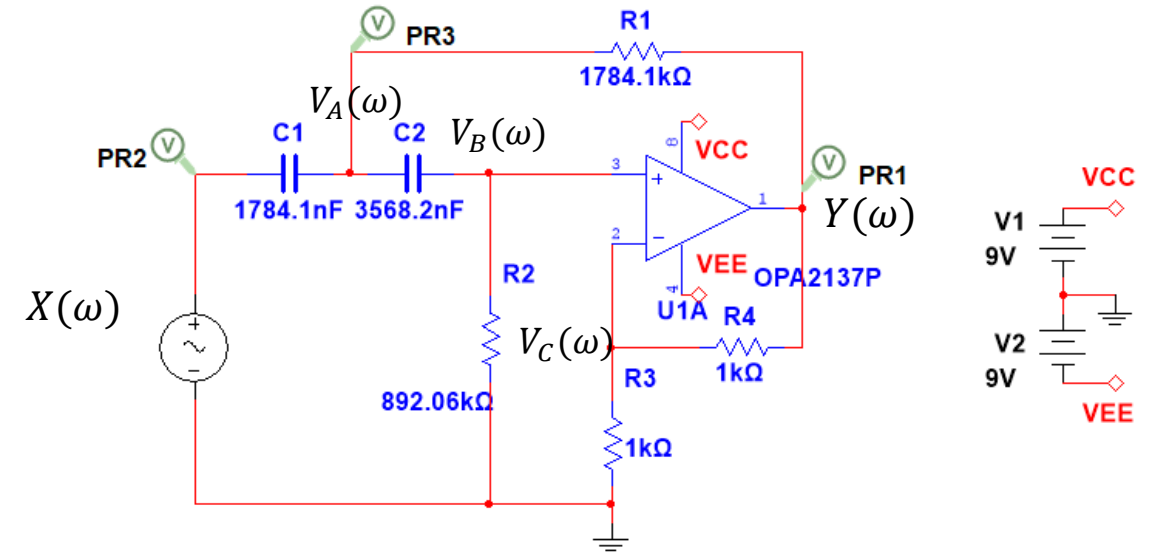
$$A: -\frac{X(\omega) - V_A(\omega)}{Z_{C_1}} + \frac{V_A(\omega) - Y(\omega)}{R_1} + \frac{V_A(\omega) - V_B(\omega)}{Z_{C_2}} = 0$$

$$B: -\frac{V_A(\omega) - V_B(\omega)}{Z_{C_2}} + \frac{V_B(\omega)}{R_2} = 0$$

$$C: V_C(\omega) = \frac{R_3}{R_3 + R_4} Y(\omega) = \frac{1}{K} Y(\omega)$$

Constraints:

$$V_B(\omega) = V_C(\omega) = \frac{1}{K} Y(\omega)$$



$$\frac{V_C(\omega)}{Y(\omega)} = \frac{R_3}{R_3 + R_4}$$

$$V_B(\omega) = V_C(\omega) = \frac{R_3}{R_3 + R_4} Y(\omega) = \frac{1}{K} Y(\omega)$$

$$Y(\omega) = \left(1 + \frac{R_4}{R_3}\right) V_B(\omega) = K V_B(\omega)$$

Filter 11 – Highpass filter (sol)

Nodal equations:

$$A: -\frac{X(\omega) - V_A(\omega)}{Z_{C_1}} + \frac{V_A(\omega) - Y(\omega)}{R_1} + \frac{V_A(\omega) - V_B(\omega)}{Z_{C_2}} = 0$$

$$B: -\frac{V_A(\omega) - V_B(\omega)}{Z_{C_2}} + \frac{V_B(\omega)}{R_2} = 0$$

$$C: V_C(\omega) = \frac{R_3}{R_3 + R_4} Y(\omega) = \frac{1}{K} Y(\omega)$$

Constraints:

$$V_B(\omega) = V_C(\omega) = \frac{1}{K} Y(\omega)$$

Constraints

$$V_B := \frac{1}{K} \cdot Y:$$

Equations

$$eqA1 := -\frac{(X - V_A)}{Z_{C1}} + \frac{(V_A - Y)}{R1} + \frac{(V_A - V_B)}{Z_{C2}} = 0:$$

$$eqB1 := -\frac{(V_A - V_B)}{Z_{C2}} + \frac{V_B}{R2} = 0:$$

Solve

$$solutions := simplify(solve(\{eqA1, eqB1\}, [V_A, Y], symbolic = true)):$$

$$assign(solutions)$$

$$simplify(Y)$$

$$-\frac{K R1 R2 X}{(-R1 + (K - 1) R2 - ZC2) ZC1 - R1 (R2 + ZC2)}$$

Impedances

$$ZC1 := \frac{1}{j \cdot \omega \cdot C1} :: ZC2 := \frac{1}{j \cdot \omega \cdot C2} ::$$

$$simplify(Y)$$

$$\frac{K R1 R2 X \omega^2 C2 C1}{-1 + C1 C2 R1 R2 \omega^2 + ((-1 R1 + (1 K - 1) R2) C2 - 1 C1 R1) \omega}$$

$$Y := \omega \rightarrow \frac{K \cdot R1 R2 \cdot (j \cdot \omega)^2 C1 C2}{C1 C2 R1 R2 (j \cdot \omega)^2 + j \cdot \omega \cdot (R1 \cdot (C1 + C2) + R2 \cdot C2 \cdot (1 - K)) + 1} \cdot X:$$

2. Reorganizing and inverse Fourier transforming yields the differential equation:

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{K(j\omega)^2}{(j\omega)^2 + j\omega \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_1 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\left((j\omega)^2 + j\omega \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_1 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2} \right) Y(\omega) = K(j\omega)^2 X(\omega)$$

$$(j\omega)^2 Y(\omega) + (j\omega) Y(\omega) \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_1 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2} Y(\omega) = K(j\omega)^2 X(\omega)$$

$$\ddot{y}(t) + \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_1 C_1} \right) \dot{y}(t) + \frac{1}{R_1 R_2 C_1 C_2} y(t) = K \ddot{x}(t)$$

Rewriting $H(\omega)$ manually

Transfer function

$$H := \omega \rightarrow \frac{K \cdot (j \cdot \omega)^2}{(j \cdot \omega)^2 + j \omega \left(\frac{1}{R2 \cdot C1} + \frac{1}{R2 \cdot C2} + \frac{(1 - K)}{R1 \cdot C1} \right) + \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}} :$$

Choice of components

$$C1 := 1784.1\text{E}-9 ;; C2 := 3568.2\text{E}-9 ;; R1 := 1784.1\text{E}3 ;; R2 := 892.06\text{E}3 ;; R3 := 1\text{E}3 ;; R4 := 1\text{E}3 :$$

$$K := 1 + \frac{R4}{R3} :$$

Calculated parameters

$$a1 := \frac{C1 + C2}{R2 \cdot C1 \cdot C2} + \frac{(1 - K)}{R1 \cdot C1}$$

$$a1 := 0.6283249517$$

$$a0 := \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$a0 := 0.09870027408$$

$$b2 = K$$

$$b2 = 2.000000000$$

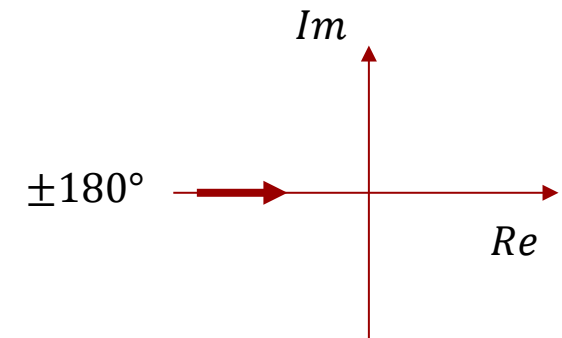
Coefficients agree with those calculated in Lecture 1.

3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope.

$$H(\omega) = \frac{(j\omega)^2 \cdot K}{(j\omega)^2 + a_1(j\omega) + a_0}$$

Low frequency asymptote

$$\lim_{\omega \rightarrow 0} H(\omega) = \lim_{\omega \rightarrow 0} \frac{(j\omega)^2 \cdot K}{a_0} = \lim_{\omega \rightarrow 0} \frac{-(\omega)^2 \cdot K}{a_0} = 0$$



High frequency asymptote

$$\lim_{\omega \rightarrow \infty} H(\omega) = \lim_{\omega \rightarrow \infty} \frac{(j\omega)^2 \cdot K}{(j\omega)^2} = K$$

The asymptotes reveal that the low frequencies are attenuated, and the high frequencies are amplified by K .

$$H(\omega) = \frac{K(j\omega)^2}{(j\omega)^2 + j\omega \left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{(1-K)}{R_1 C_1} \right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

The low-frequency asymptote is:

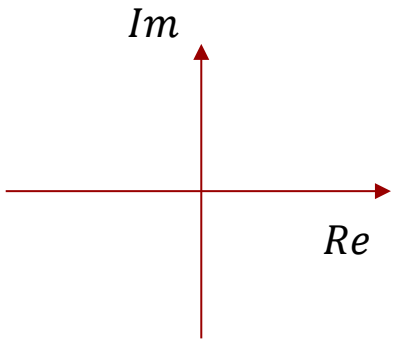
$$H_{LF}(\omega) = \frac{K(j\omega)^2}{\frac{1}{R_1 R_2 C_1 C_2}} = R_1 R_2 C_1 C_2 K (j\omega)^2$$

LF slope:

$$\begin{aligned} |H_{LF}(10 \cdot \omega)|_{dB} - |H_{LF}(\omega)|_{dB} &= 20\text{dB} \cdot \log_{10} R_1 R_2 C_1 C_2 K (10 \omega)^2 - 20\text{dB} \cdot \log_{10} R_1 R_2 C_1 C_2 K (\omega)^2 \\ &= 20\text{dB} \cdot \log_{10} \frac{(10 \omega)^2}{(\omega)^2} = 20\text{dB} \cdot \log_{10} 10^2 \\ &= 2 \cdot 20\text{dB} \cdot \log_{10} 10 = 40\text{dB/decade} \end{aligned}$$

$$\log A - \log B = \log \frac{A}{B}$$

4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.



$$H(\omega) = \frac{(j\omega)^2 \cdot K}{(j\omega)^2 + a_1(j\omega) + a_0}$$

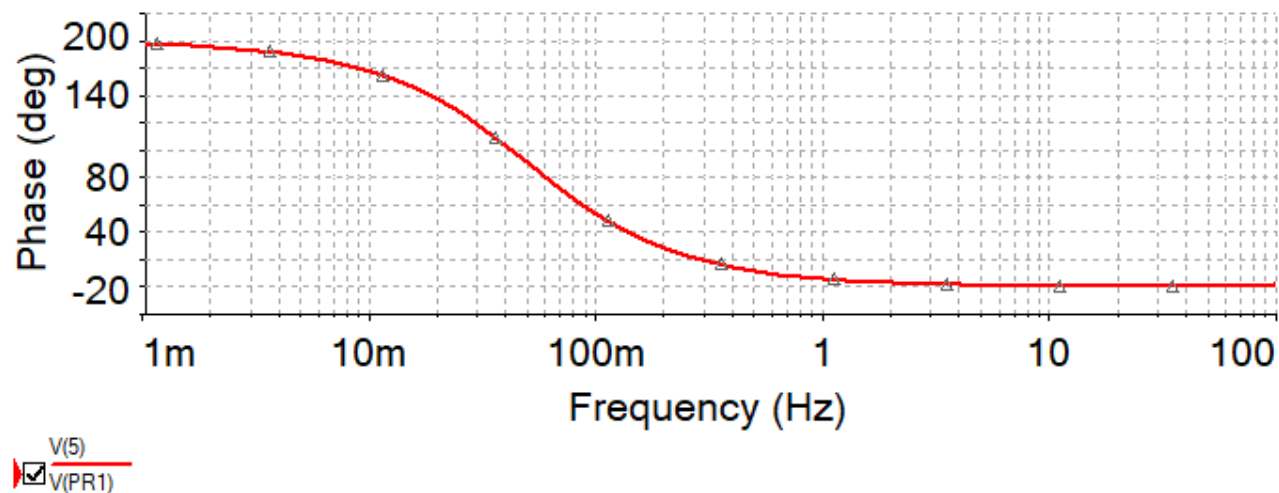
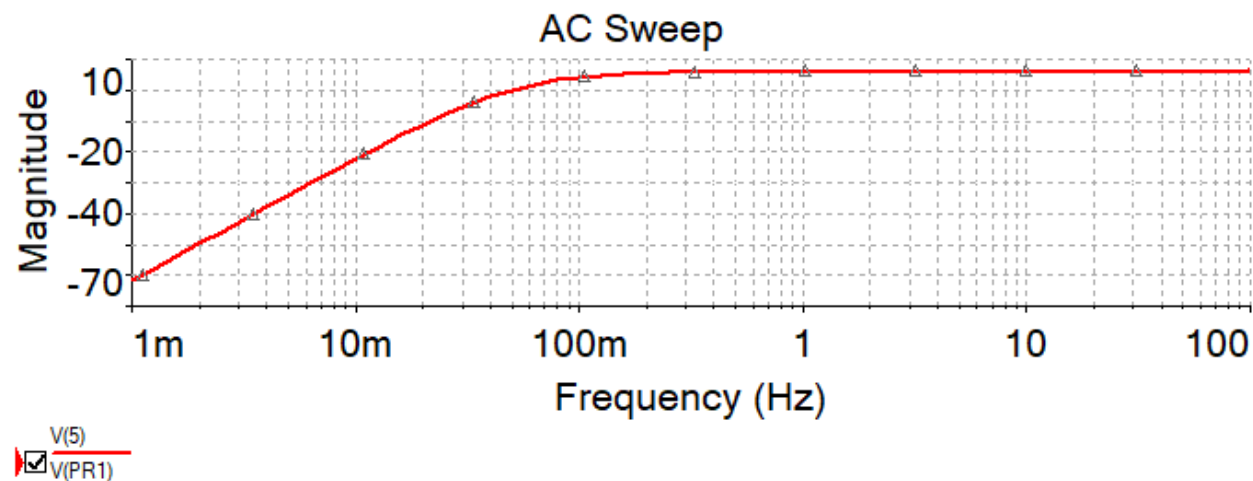
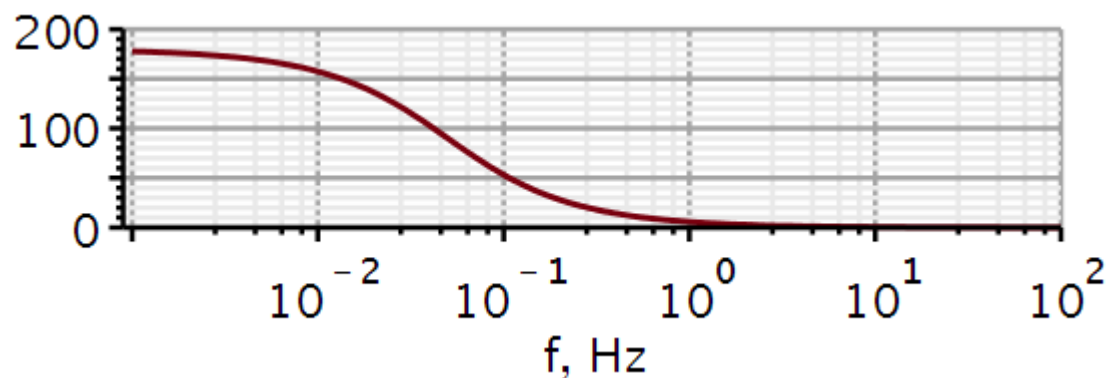
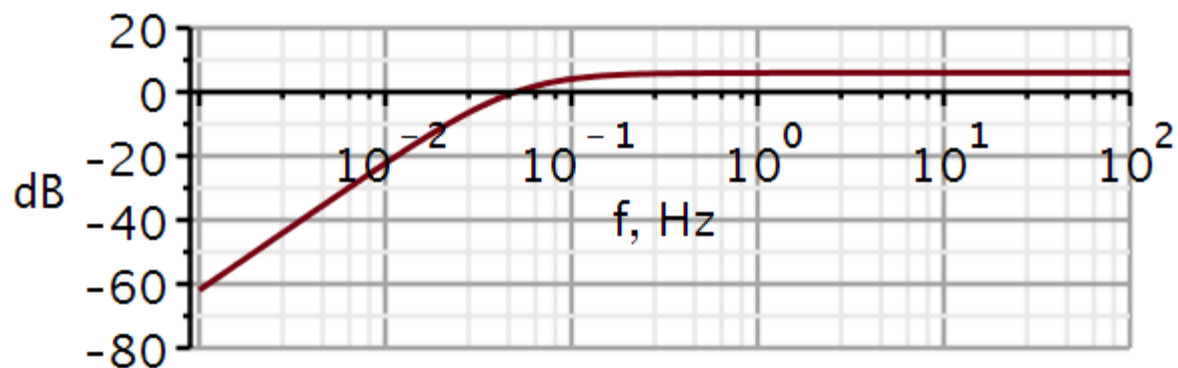
$$\angle H(\omega) = \underbrace{\angle P(\omega)}_{\text{numerator angle}} - \underbrace{\angle Q(\omega)}_{\text{denominator angle}}$$

$$\angle H(\omega) = \angle(j\omega)^2 - \angle((j\omega)^2 + a_1(j\omega) + a_0)$$

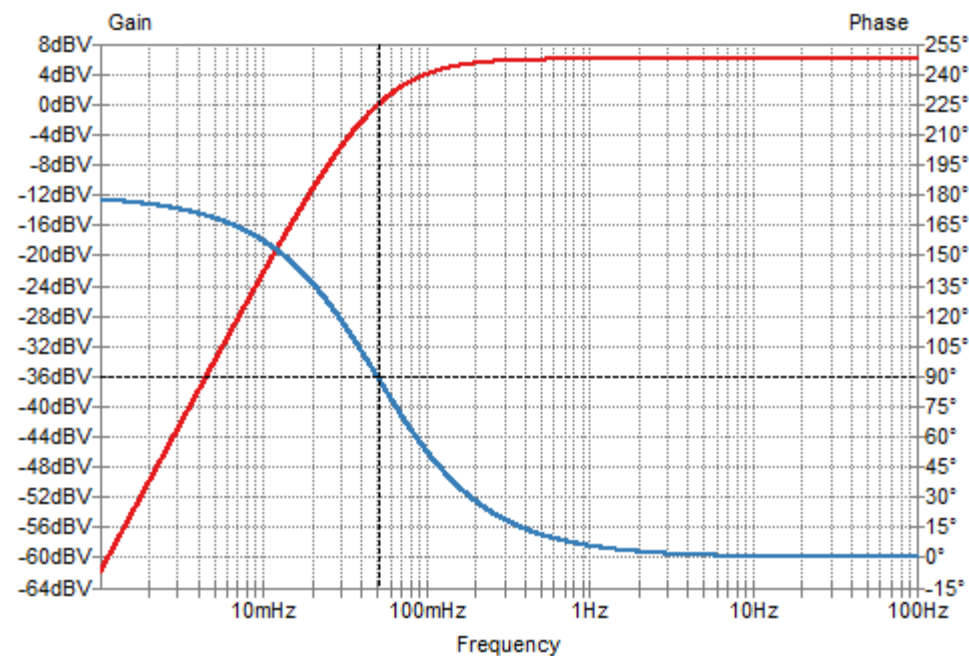
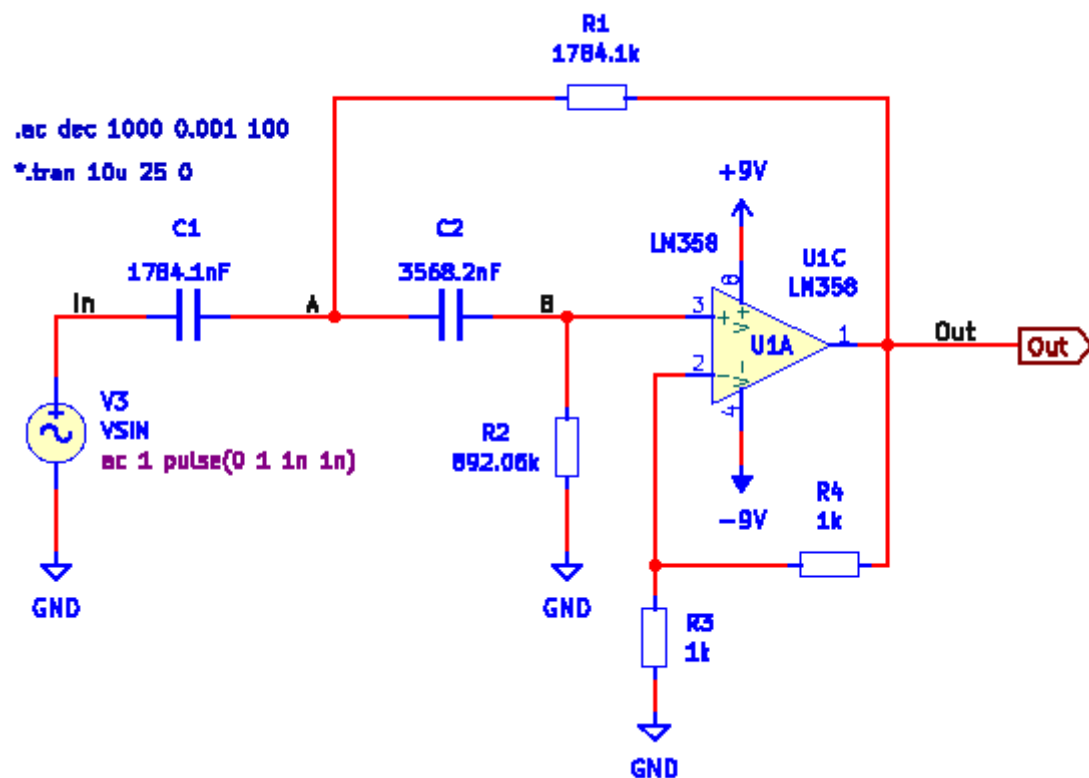
$$\angle H(0) = 180^\circ - 0^\circ = 180^\circ$$

$$\angle H(\infty) = 180^\circ - 180^\circ = 0^\circ$$

5. - 7. Plot the amplitude and phase characteristics in Maple.



Filter 11 – Highpass filter (sol)



We can derive a general expression for the 3-dB cut-off frequency for a 2nd order highpass filter.

$$|H(j\omega_{3dB})| = \frac{|b_2||j\omega_{3dB}|^2}{|(j\omega_{3dB})^2 + a_1(j\omega_{3dB}) + a_0|} = \frac{1}{\sqrt{2}} \cdot |H(j\omega_{HF})|$$

$$|H(j\omega_{3dB})| = \frac{|b_2||j\omega_{3dB}|^2}{\sqrt{(a_0 - \omega_{3dB}^2)^2 + (a_1\omega_{3dB})^2}} = \frac{|b_2||j\omega_{3dB}|^2}{\sqrt{2}|j\omega_{3dB}|^2}$$

$$(a_0 - \omega_{3dB}^2)^2 + (a_1\omega_{3dB})^2 = 2\omega_{3dB}^4$$

$$a_0^2 - 2a_0\omega_{3dB}^2 + \omega_{3dB}^4 + (a_1\omega_{3dB})^2 = 2\omega_{3dB}^4$$

$$-\omega_{3dB}^4 + (a_1^2 - 2a_0)\omega_{3dB}^2 + a_0^2 = 0$$

Filter 11 – Highpass filter (sol)

$$-\omega_{3dB}^4 + (a_1^2 - 2a_0)\omega_{3dB}^2 + a_0^2 = 0$$

$$y = \frac{-(a_1^2 - 2a_0) \pm \sqrt{(a_1^2 - 2a_0)^2 + 4a_0^2}}{-2}$$

$$x = \sqrt{y}$$

2nd order highpass

restart

$$\text{solve}(-y^2 + (a1^2 - 2 \cdot a0) \cdot y + a0^2 = 0, y)$$

$$\frac{a1^2}{2} - a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}, \frac{a1^2}{2} - a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}$$

$$y1 := \frac{a1^2}{2} - a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} :$$

$$y2 := \frac{a1^2}{2} - a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} :$$

$$a0 := 1 ;; a1 := \sqrt{2} ;; b2 := 1 :$$

evalf(y1)

1.000000000

evalf(y2)

-1.000000000

$$x1 := \text{evalf}(\sqrt{y1})$$

x1 := 1.

Filter 11 – Highpass filter (sol)

Highpass filter, not normalized

restart

solve($-y^2 + (a1^2 - 2 \cdot a0) \cdot y + a0^2 = 0, y$)

$$\frac{a1^2}{2} - a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}, \frac{a1^2}{2} - a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2}$$

$$y1 := \frac{a1^2}{2} - a0 + \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} ;$$

$$y2 := \frac{a1^2}{2} - a0 - \frac{\sqrt{a1^4 - 4 a0 a1^2 + 8 a0^2}}{2} ;$$

$$C1 := 10 \cdot 10^{-6} ; C2 := 10 \cdot 10^{-6} ; R1 := 470000 ; R2 := 220000 ; K := 1 ;$$

$$a1 := \frac{1}{R1 \cdot C1} + \frac{1}{R1 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1}$$

$$a1 := \frac{20}{47}$$

$$a0 := \frac{1}{R1 \cdot R2 \cdot C1 \cdot C2}$$

$$a0 := \frac{50}{517}$$

$$b2 := K$$

$$b2 := 1$$

$$x1 := \text{evalf}(\sqrt{y1})$$

$$x1 := 0.3012233701$$

$$x1$$

$$0.3012233701$$

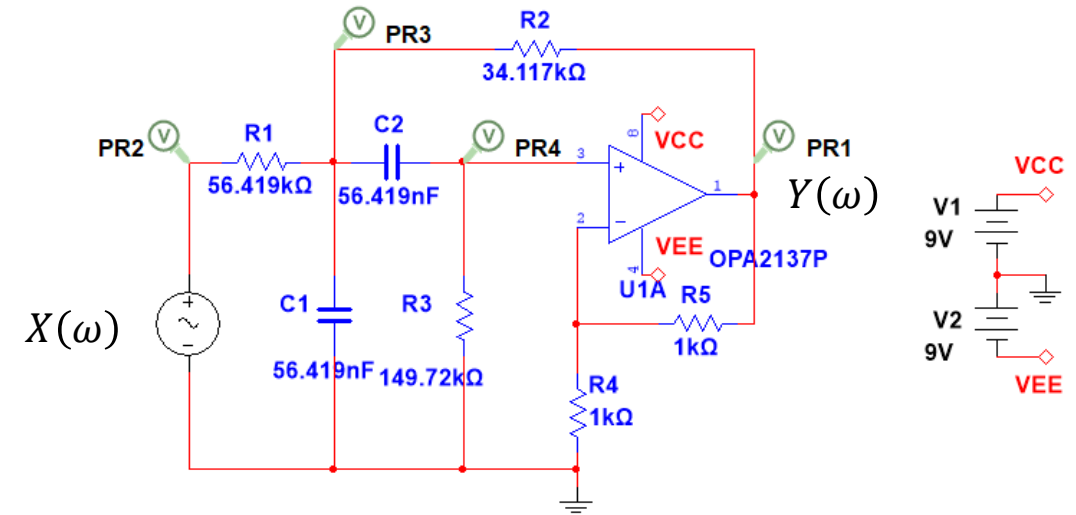
$$\frac{x1}{2 \cdot \pi}$$

$$0.04794118832$$

Filter 12 – Bandpass filter

A Sallen-Key filter is shown here. You are asked to analyse this circuit and:

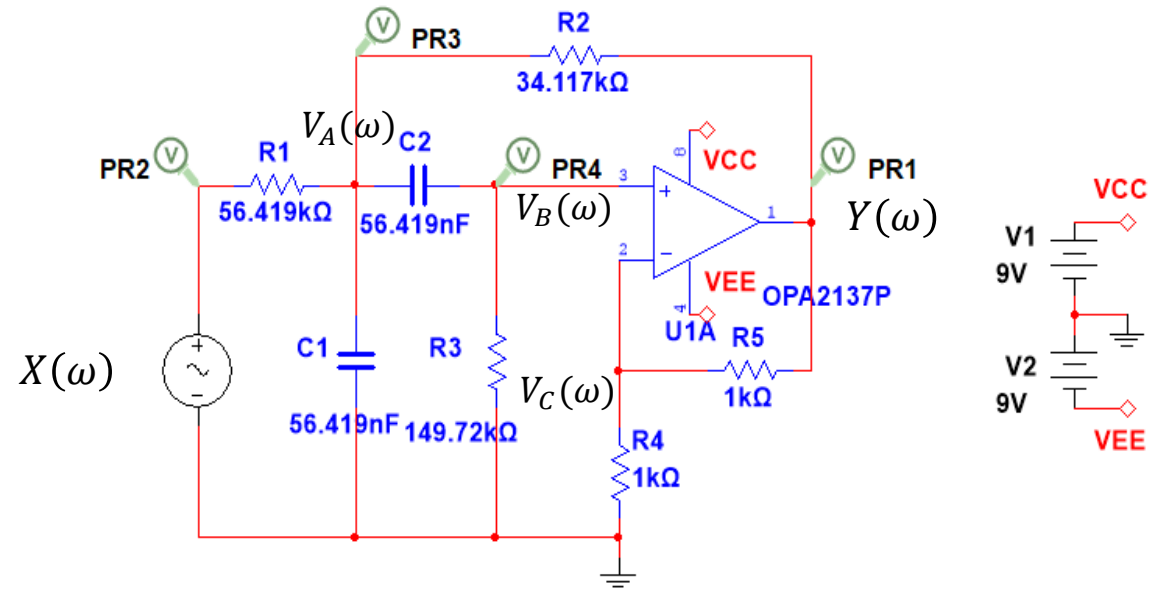
1. Derive the equation for the frequency characteristic $H(\omega)$. Use Maple/Python if possible. Verify the numerical values of the coefficients (a_1, a_0) with those calculated in Lecture 1.
2. Derive the differential equation.
3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope (dB/decade).
4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.
5. Plot the amplitude and phase characteristics in Maple/Python.
6. Draw the circuit in KiCad/Spice, run an AC sweep and plot the amplitude and phase characteristics.
7. Validate the mathematical analysis by comparing Maple/Python graphs with KiCad/Spice graphs.



C values
E6:
10
15
22
33
47
68

Filter 12 – Bandpass filter (sol)

Setting up the nodal equations in the frequency domain:



Constraints

$$V_B := \frac{1}{K} \cdot Y :$$

Equations

$$eqA1 := -\frac{(X - V_A)}{R1} + \frac{(V_A - Y)}{R2} + \frac{(V_A - V_B)}{ZC2} + \frac{V_A}{ZC1} = 0 :$$

$$eqB1 := -\frac{(V_A - V_B)}{ZC2} + \frac{V_B}{R3} = 0 :$$

Filter 12 – Bandpass filter (sol)

Solve

```
solutions := simplify(solve( {eqA1, eqB1}, [VA, Y], symbolic = true)) :
assign(solutions)
simplify(Y)
```

$$-\frac{K R_2 R_3 X ZC1}{((-R_2 + (K - 1) R_3 - ZC2) R_1 - R_2 (R_3 + ZC2)) ZC1 - R_1 R_2 (R_3 + ZC2)}$$

Impedances

$$ZC1 := \frac{1}{j \cdot \omega \cdot C1} ; ZC2 := \frac{1}{j \cdot \omega \cdot C2} ;$$

simplify(Y)

$$\frac{-I K R_2 R_3 X \omega C2}{C1 C2 R1 R2 R3 \omega^2 + (((-I R_2 + (I K - I) R_3) R_1 - I R_2 R_3) C2 - I C1 R1 R2) \omega - R1 - R2}$$

After some manual rewriting, we get the equation in polynomial form:

$$H(\omega) = \frac{\frac{K}{R_1 C_1} (j\omega)}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_1} + \frac{1}{R_3 C_2} + \frac{(1 - K)}{R_2 C_1} \right) + \frac{R_1 + R_2}{R_1 R_2 R_3 C_1 C_2}}$$

Filter 12 – Bandpass filter (sol)

2. Coefficients agree with those calculated in Lecture 1.

Transfer function

$$H := \omega \rightarrow \frac{\frac{K \cdot j \cdot \omega}{R1 \cdot C1}}{(j \cdot \omega)^2 + \left(\frac{1}{R1 \cdot C1} + \frac{1}{R3 \cdot C1} + \frac{1}{R3 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1} \right) (j \cdot \omega) + \frac{(R1 + R2)}{R1 \cdot R2 \cdot R3 \cdot C1 \cdot C2}} :$$

Choice of components

$$\begin{aligned} C1 &:= 56.419\text{E}-9 ; C2 := 56.419\text{E}-9 ; R1 := 56.419\text{E}3 ; R2 := 34.117\text{E}3 ; R3 := 149.72\text{E}3 ; R4 := 1\text{E}3 ; \\ R5 &:= 1\text{E}3 ; \\ K &:= 1 + \frac{R5}{R4} ; \end{aligned}$$

Calculated parameters

$$a1 := \frac{1}{R1 \cdot C1} + \frac{1}{R3 \cdot C1} + \frac{1}{R3 \cdot C2} + \frac{(1 - K)}{R2 \cdot C1}$$

$$a1 := 31.4059709$$

$$a0 := \frac{(R1 + R2)}{R1 \cdot R2 \cdot R3 \cdot C1 \cdot C2}$$

$$a0 := 98694.85113$$

$$b1 := \frac{K}{R1 \cdot C1}$$

$$b1 := 628.3176031$$

3. Derive the equation for the amplitude characteristic $|H(\omega)|$ and determine Pass band gain and Stop band asymptote slope.

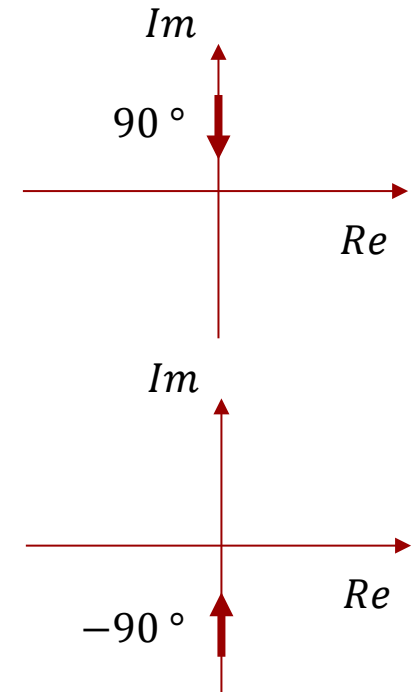
$$H(\omega) = \frac{j \omega \cdot b_1}{(j \omega)^2 + a_1(j \omega) + a_0}$$

Low frequency asymptote

$$\lim_{\omega \rightarrow 0} H(\omega) = \lim_{\omega \rightarrow 0} \frac{j \omega \cdot b_1}{a_0} = 0$$

High frequency asymptote

$$\lim_{\omega \rightarrow \infty} H(\omega) = \lim_{\omega \rightarrow \infty} \frac{j \omega \cdot b_1}{(j \omega)^2} = \lim_{\omega \rightarrow \infty} \frac{b_1}{j \omega} = \lim_{\omega \rightarrow \infty} \frac{-j \cdot b_1}{\omega} = 0$$



The asymptotes reveal that both the low and high frequencies are attenuated. It must be a bandpass filter.

Filter 12 – Bandpass filter (sol)

$$H(\omega) = \frac{\frac{K}{R_1 C_1} (j\omega)}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_1} + \frac{1}{R_3 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{R_1 + R_2}{R_1 R_2 R_3 C_1 C_2}}$$

The low-frequency asymptote is:

$$H_{LF}(\omega) = \frac{\frac{K}{R_1 C_1} (j\omega)}{\frac{R_1 + R_2}{R_1 R_2 R_3 C_1 C_2}} = \frac{K R_2 R_3 C_2}{R_1 + R_2} (j\omega)$$

LF slope:

$$\begin{aligned} |H_{LF}(10 \cdot \omega)|_{dB} - |H_{LF}(\omega)|_{dB} &= 20\text{dB} \cdot \log_{10} \frac{K R_2 R_3 C_2}{R_1 + R_2} (10 \omega) - 20\text{dB} \cdot \log_{10} \frac{K R_2 R_3 C_2}{R_1 + R_2} (\omega) \\ &= 20\text{dB} \cdot \log_{10} \frac{(10 \omega)^1}{(\omega)^1} = 20\text{dB} \cdot \log_{10} 10^1 \\ &= 1 \cdot 20\text{dB} \cdot \log_{10} 10 = 20\text{dB/decade} \end{aligned}$$

$$\log A - \log B = \log \frac{A}{B}$$

Filter 12 – Bandpass filter (sol)

$$H(\omega) = \frac{\frac{K}{R_1 C_1} (j\omega)}{(j\omega)^2 + j\omega \left(\frac{1}{R_1 C_1} + \frac{1}{R_3 C_1} + \frac{1}{R_3 C_2} + \frac{(1-K)}{R_2 C_1} \right) + \frac{R_1 + R_2}{R_1 R_2 R_3 C_1 C_2}}$$

The high-frequency asymptote is:

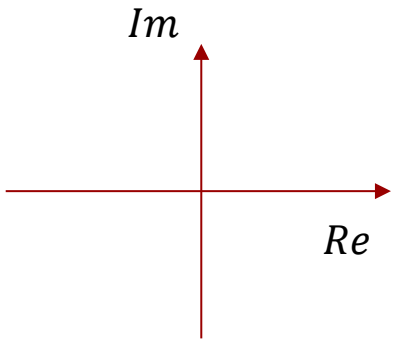
$$H_{HF}(\omega) = \frac{\frac{K}{R_1 C_1} (j\omega)}{(j\omega)^2} = \frac{K}{R_1 C_1 (j\omega)}$$

HF slope:

$$\begin{aligned} |H_{HF}(10 \cdot \omega)|_{dB} - |H_{HF}(\omega)|_{dB} &= 20\text{dB} \cdot \log_{10} \frac{K}{R_1 C_1 (10 \omega)} - 20\text{dB} \cdot \log_{10} \frac{K}{R_1 C_1 (\omega)} \\ &= 20\text{dB} \cdot \log_{10} \frac{(\omega)^1}{(10 \omega)^1} = 20\text{dB} \cdot \log_{10} \frac{1}{(10)^1} = 20\text{dB} \cdot \log_{10} 10^{-1} \\ &= -1 \cdot 20\text{dB} \cdot \log_{10} 10 = -20\text{dB/decade} \end{aligned}$$

$$\log A - \log B = \log \frac{A}{B}$$

4. Derive the equation for phase angle $\angle H(\omega)$ and predict phase angles at low and high frequencies.



$$H(\omega) = \frac{(j\omega)b_1}{(j\omega)^2 + a_1(j\omega) + a_0}$$

$$\angle H(\omega) = \underbrace{\angle P(\omega)}_{\text{numerator angle}} - \underbrace{\angle Q(\omega)}_{\text{denominator angle}}$$

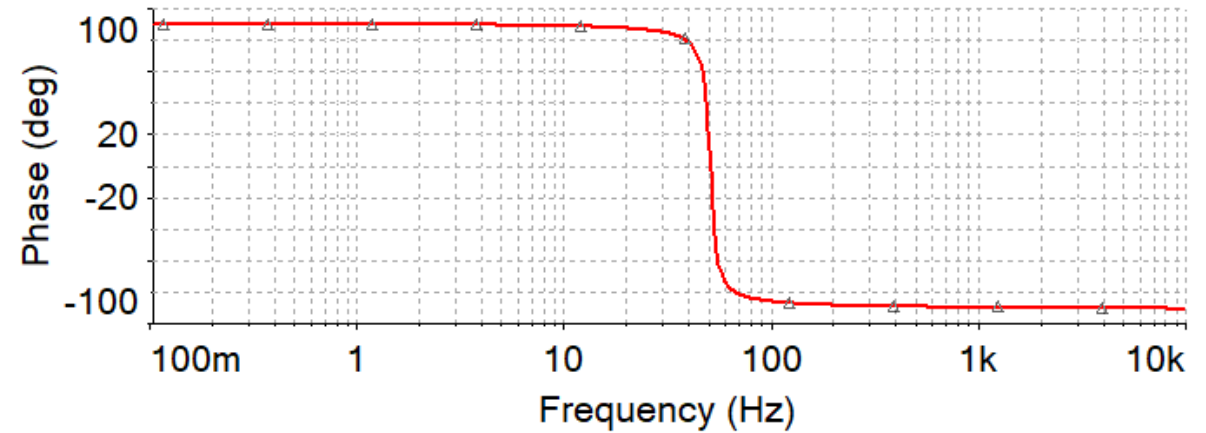
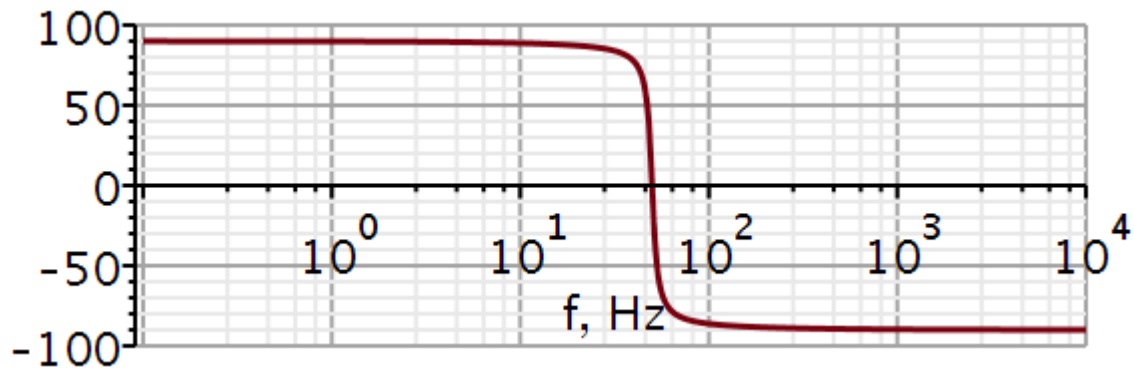
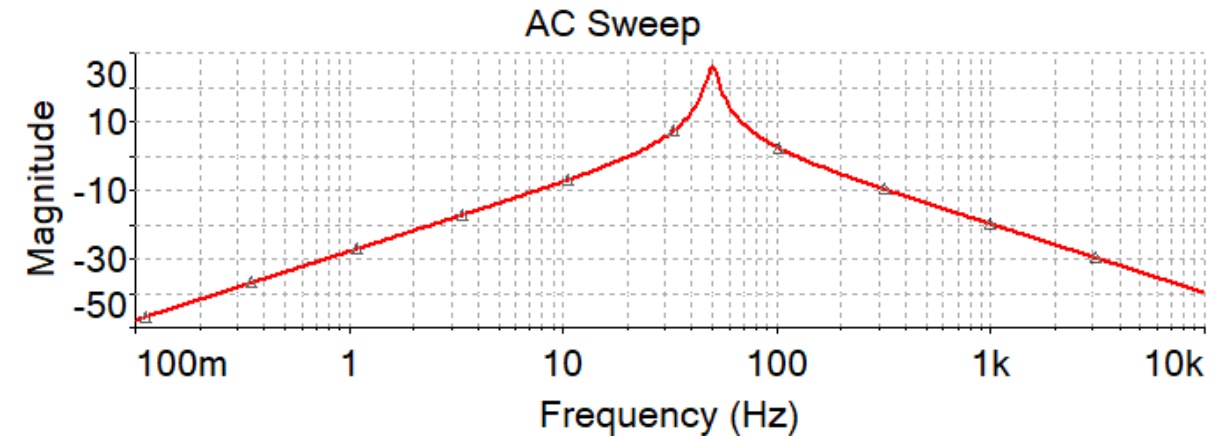
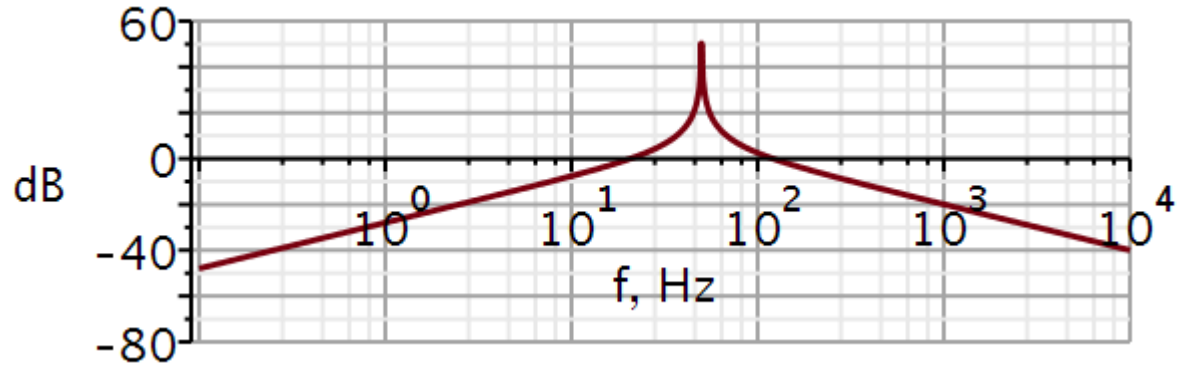
$$\angle H(\omega) = \angle(j\omega)b_1 - \angle((j\omega)^2 + a_1(j\omega) + a_0)$$

$$\angle H(0) = 90^\circ - 0^\circ = 90^\circ$$

$$\angle H(\infty) = 90^\circ - 180^\circ = -90^\circ$$

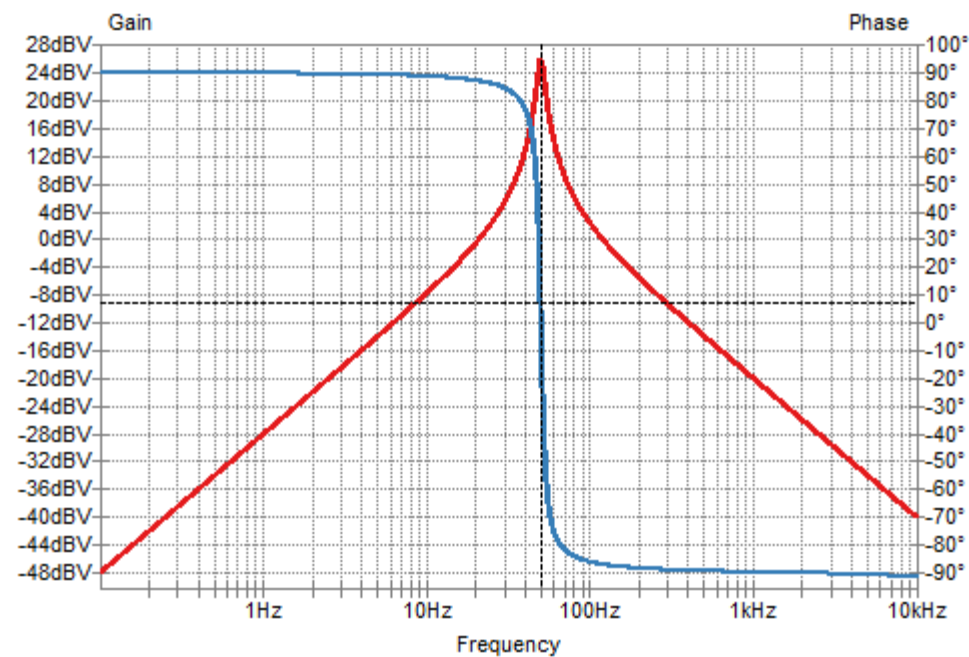
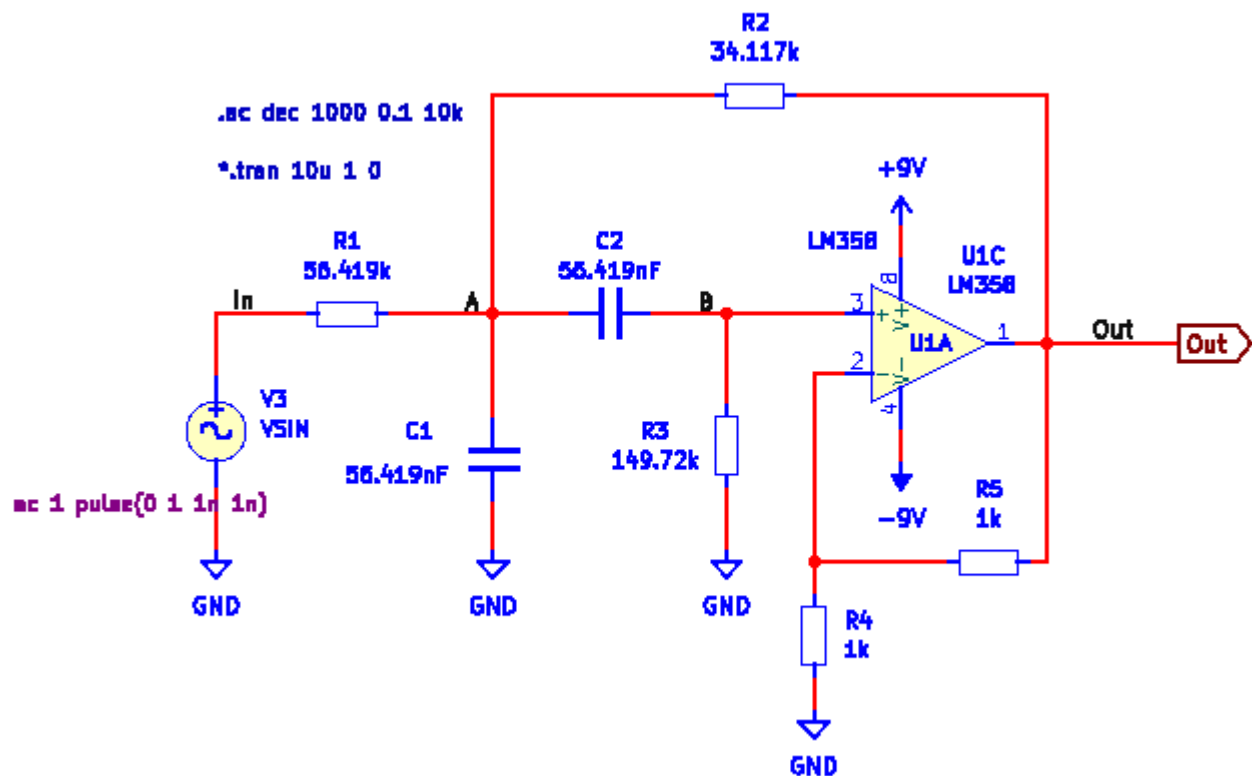
Filter 12 – Bandpass filter (sol)

5. – 7. Plot and compare.



See next slide

Filter 12 – Bandpass filter (sol)



Filter 12 – Bandpass filter (sol)

$$H := \omega \rightarrow \frac{\frac{K \cdot j \cdot \omega}{R1 \cdot C1}}{(j \cdot \omega)^2 + \left(\frac{1}{R1 \cdot C1} + \frac{1}{R3 \cdot C1} + \frac{1}{R3 \cdot C2} + \frac{(1-K)}{R2 \cdot C1} \right) (j \cdot \omega) + \frac{(R1 + R2)}{R1 \cdot R2 \cdot R3 \cdot C1 \cdot C2}} :$$

$$dB := \omega \rightarrow 20 \cdot \log_{10}(|H(\omega)|) :$$

$$angle := \omega \rightarrow \text{argument}(H(\omega)) \cdot \frac{180}{\pi} :$$

$$\theta := \omega \rightarrow 90 - \frac{180}{\pi} \arctan \left(\left(\frac{1}{R1 \cdot C1} + \frac{1}{R3 \cdot C1} + \frac{1}{R3 \cdot C2} + \frac{(1-K)}{R2 \cdot C1} \right) \cdot \omega, \frac{(R1 + R2)}{R1 \cdot R2 \cdot R3 \cdot C1 \cdot C2} - \omega^2 \right) :$$

Imaginary argument

Real argument

So far, the phase function `angle()` has given satisfying results when plotting the phase. For unknown reasons `angle()` gives a very sharp phase jump from 90 to -90 degrees for the bandpass filter. Then I tried to use a function $\theta()$ shown above, it gives a smoother phase transition in good agreement with what KiCad shows.

`arctan()` can take two arguments, the imaginary part first, and the real part second. In this way it can figure out, in what quadrant the complex number is located.